# Berliner's games, 5th World Correspondence Championship

Generated by Stockfish 16

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# Chapter 1

# Abramson Moisey vs Berliner Hans J (USA)

## 1.1 Game Information

| | |
|---|---|
| **Event** | Ch World (corr.) |
| **Site** | corr. |
| **Date** | 1965.??.?? |
| **Round** | 5 |
| **White** | Abramson Moisey |
| **Black** | Berliner Hans J (USA) |
| **Result** | 0-1 |
| **Opening** | A41 – Unknown |

## 1.2 Player Statistics

### 1.2.1 Abramson Moisey (White)

- Total moves: 36
- Accuracy: 86.5%
- Average centipawn loss: 29.0
- Best/Excellent moves: 8 / 6
- Good moves: 9
- Inaccuracies: 6
- Mistakes: 7
- Blunders: 0

### 1.2.2 Berliner Hans J (USA) (Black)

- Total moves: 36
- Accuracy: 92.7%

- Average centipawn loss: 15.1
- Best/Excellent moves: 12 / 6
- Good moves: 11
- Inaccuracies: 6
- Mistakes: 1
- Blunders: 0

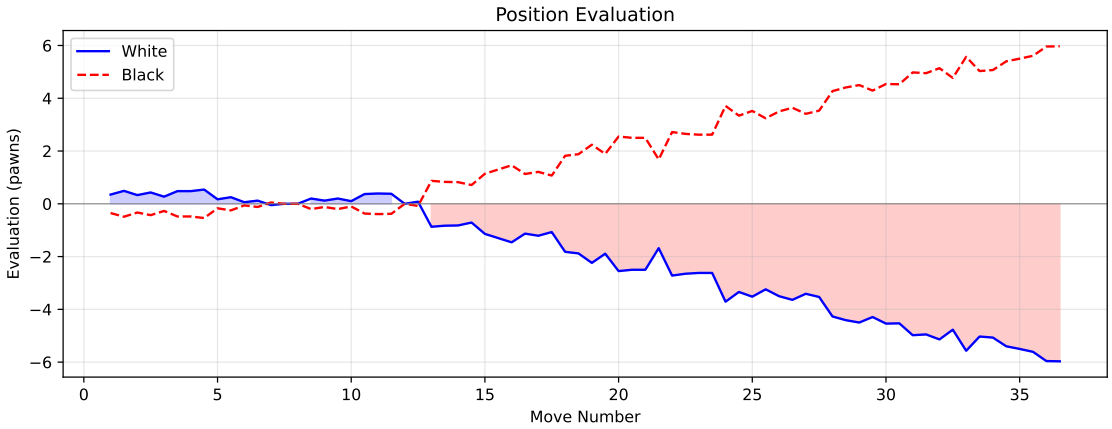
## 1.3 Positional Analysis

This section shows how key positional factors evolved throughout the game.

### 1.3.1 Position Evaluation Over Time

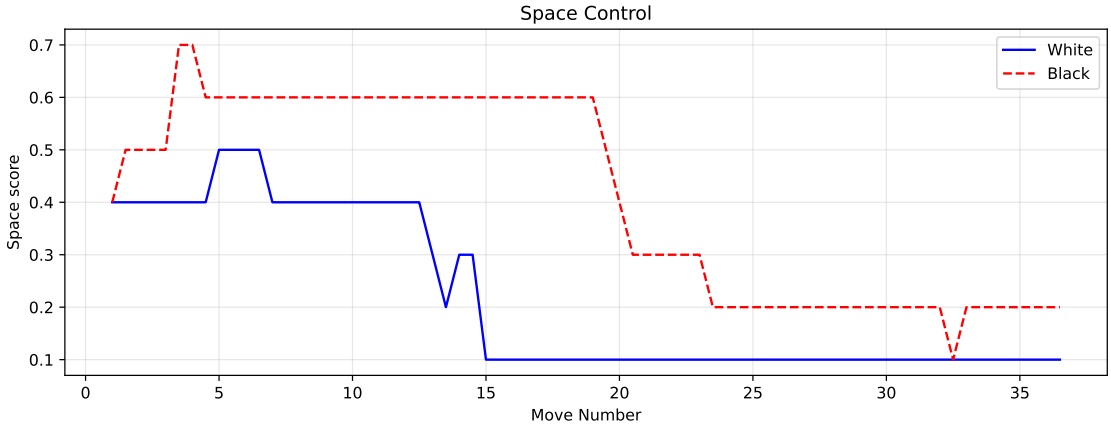
**Game Character Classification.** Evaluation range: minimum $m_1 = -5.97$, maximum $m_2 = +0.54$, spread $d = m_2 - m_1 = 6.51$.

This game is classified as **chaotic** (normal). A chaotic game with moderate swings.



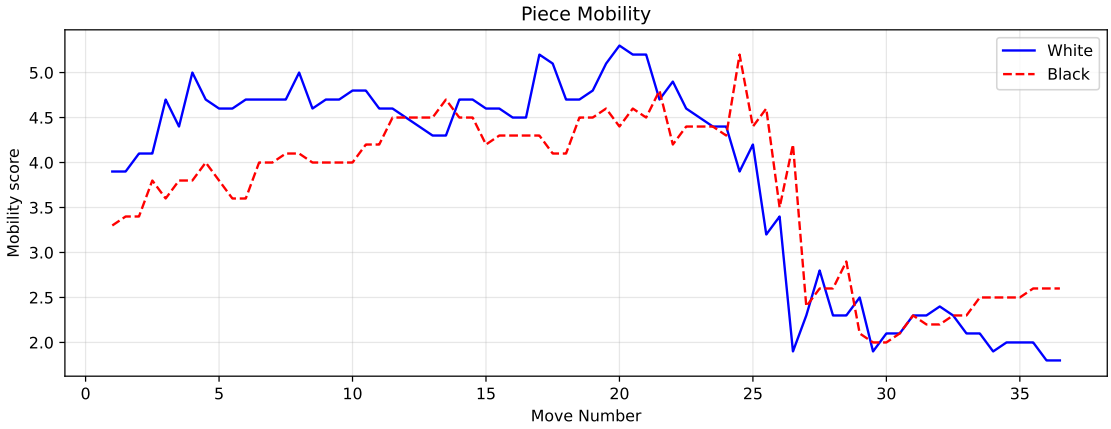

### 1.3.2 Space Control

- White average: 0.22 (range: 0.10 to 0.50)
- Black average: 0.41 (range: 0.10 to 0.70)



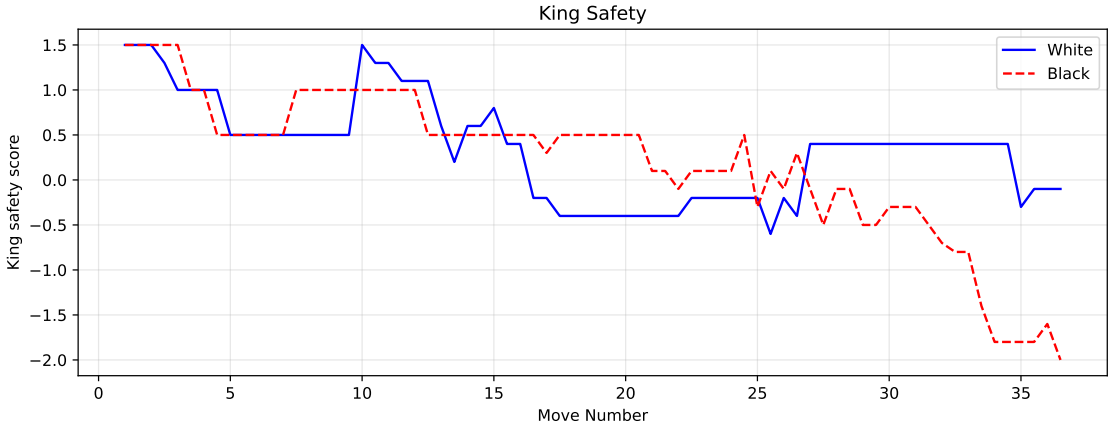

### 1.3.3 Mobility (Piece Activity)

- White average: 3.87 (range: 1.80 to 5.30)
- Black average: 3.68 (range: 2.00 to 5.20)



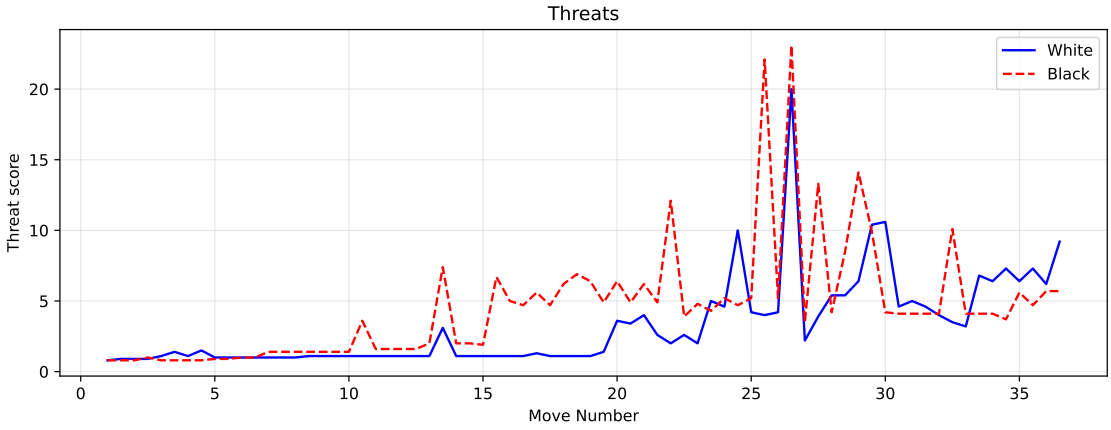

### 1.3.4 King Safety

- White average: 0.34
- Black average: 0.20




### 1.3.5 Threats

- White average: 3.26 (range: 0.80 to 20.00)
- Black average: 4.60 (range: 0.80 to 23.10)
- Average advantage: -1.35 (Black maintained more threats)




## 1.4 Annotated Game

1. Nf3 g6 2. c4 Bg7 3. d4 d6 4. Nc3 e5 5. e3?! Nd7 6. a3?! Ngf6 7. b4 O-O 8. Bb2 c6?! 9. Be2! Re8 10. O-O e4 11. Nd2! Nf8 12. Re1?! h5 13. f4? *White loses 95cp. Consider instead: a4, d5, b5.*

13...exf3! 14. Bxf3! N8h7 15. e4?! Ng4! 16. Nf1 Qh4?! 17. Qd2 Ng5 18. Re2? *White loses 89cp. Consider instead: Bxg4.*

18...Be6! 19. Qd3 c5?! 20. bxc5? *White loses 72cp. Consider instead: Nd5, Nb5.*

20...dxc5! 21. Nd5 cxd4? *Black loses 78cp. Consider instead: b5.*

22. Bxg4? *White loses 100cp. Consider instead: Bxd4.*

22...Qxg4! 23. Rae1! b5 24. c5? *White loses 68cp. Consider instead: cxb5.*

24...Bxd5?! 25. exd5 Rxe2! 26. Qxe2! Qxe2! 27. Rxe2 d3! 28. Rd2? *White loses 83cp. Consider instead: Re5, Rf2.*

28...Ne4! 29. Bxg7 Nxd2 30. Bd4 Ne4 31. c6 Kf8 32. Be5?! f6?! 33. Bf4? *White loses 79cp. Consider instead: Bc7.*

33...Ke7?! 34. Ne3 d2! 35. Kf1 Nc3! 36. d6+?! Ke6

0-1

## 1.5 Critical Positions

This section highlights critical moments where the evaluation shifted substantially—either due to errors or missed opportunities. Each diagram shows the position *after* the move was played.

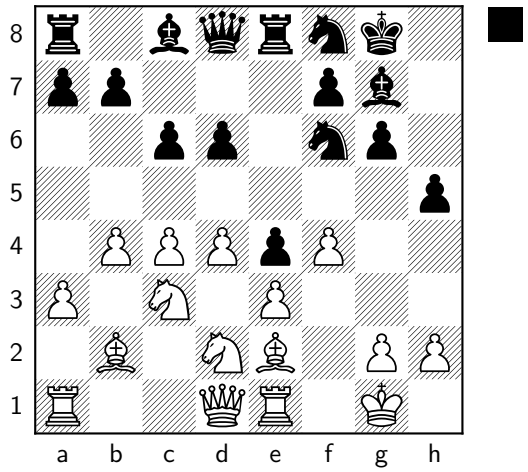
- **Instead of [move]**: The move(s) that should have been played instead.
- **Best continuation**: The optimal sequence of moves from the diagrammed position.

Positions marked with ★ represent the biggest evaluation swings in the game. The positional metrics table summarizes the spatial control, piece activity, king safety, and tactical threats for each side.

### Position 1: After 13. f4 ★

*Biggest evaluation swing(s) (95cp)*

Evaluation: -0.87



| Metric | White | Black |
|---|---|---|
| Space | 0.30 | 0.60 |
| Mobility | 4.30 | 4.50 |
| King Safety | 0.60 | 0.50 |
| Threats | 1.10 | 2.00 |

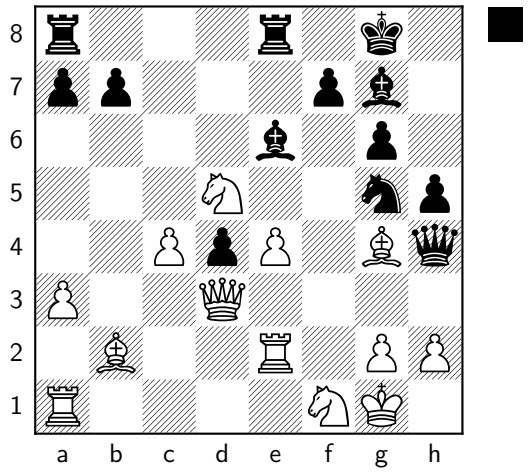
Instead of f4: a4, d5, b5

Best continuation: exf3 Bxf3 Be6

### Position 2: After 22. Bxg4

*Critical moment*

Evaluation: -2.72



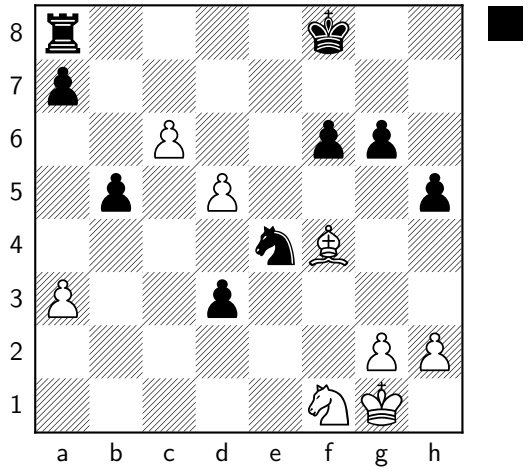
| Metric | White | Black |
|---|---|---|
| Space | 0.10 | 0.30 |
| Mobility | 4.90 | 4.20 |
| King Safety | -0.40 | -0.10 |
| Threats | 2.00 | 12.10 |

Instead of Bxg4: Bxd4 Best continuation: Qxg4 Ree1 b5

### Position 3: After 33. Bf4 ★

*Biggest evaluation swing(s) (80cp)*

Evaluation: -5.57



| Metric | White | Black |
|---|---|---|
| Space | 0.10 | 0.20 |
| Mobility | 2.10 | 2.30 |
| King Safety | 0.40 | -0.80 |
| Threats | 3.20 | 4.10 |

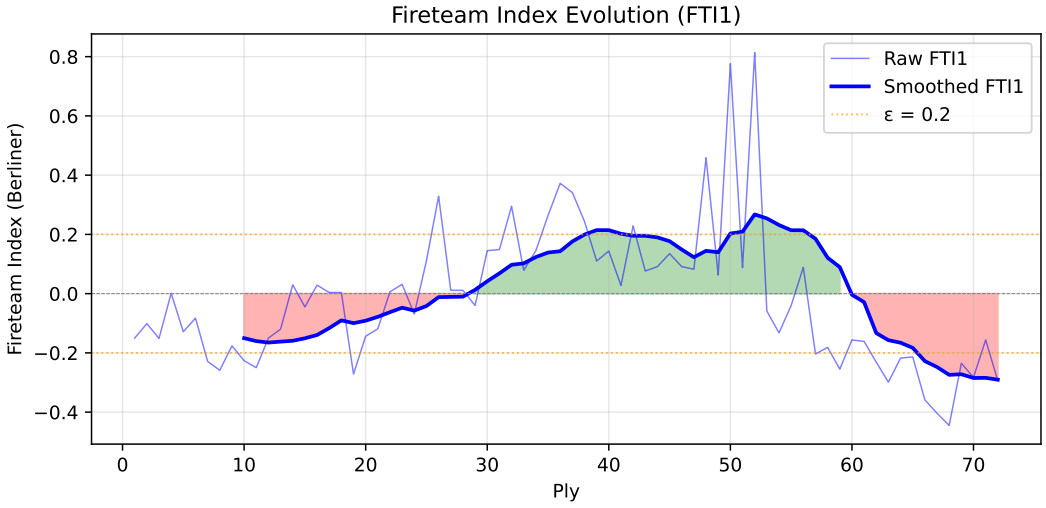
Instead of Bf4: Bc7 Best continuation: Rc8 g4 Nc3

## 1.6 Fireteam Index Prediction

Tracking: **Berliner** (Black)

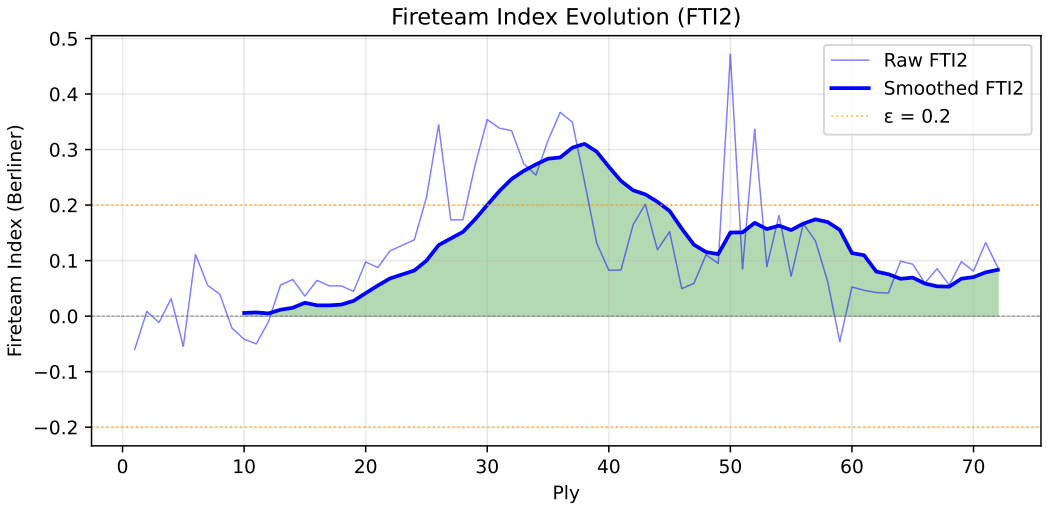
### FTI1: Harmonious $(\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4})$

| Algorithm | Prediction | Max Streak / Confidence |
|---|---|---|
| Per-ply | DRAW | 3 plies |
| Windowed | DRAW | 2 plies |
| Balance | DRAW | $B_n$=-0.05, C=0.04 |



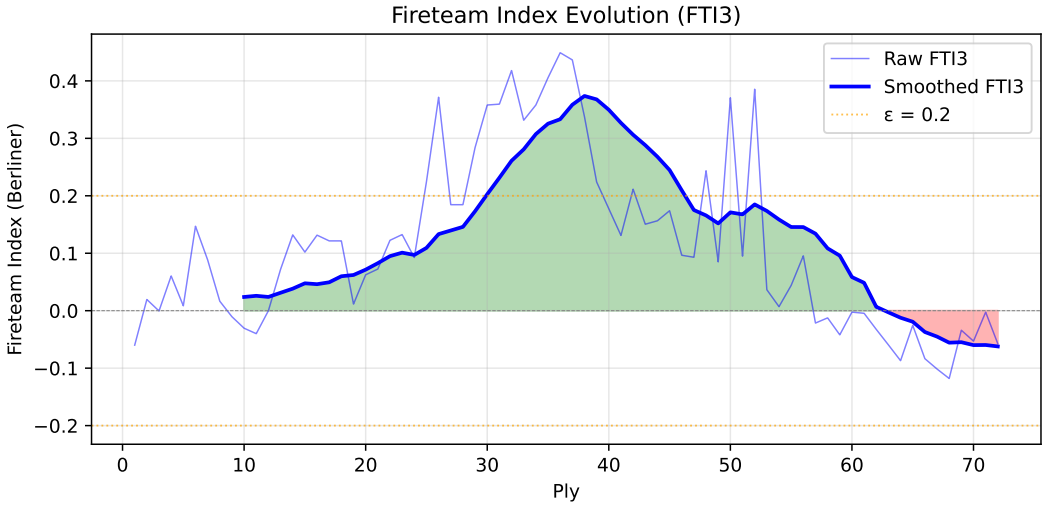

### FTI2: Tactical $(0.6, 0.1, 0, 0.3)$

| Algorithm | Prediction | Max Streak / Confidence |
|---|---|---|
| Per-ply | DRAW | 9 plies |
| Windowed | DRAW | 8 plies |
| Balance | WIN | $B_n$=+0.27, C=0.27 |



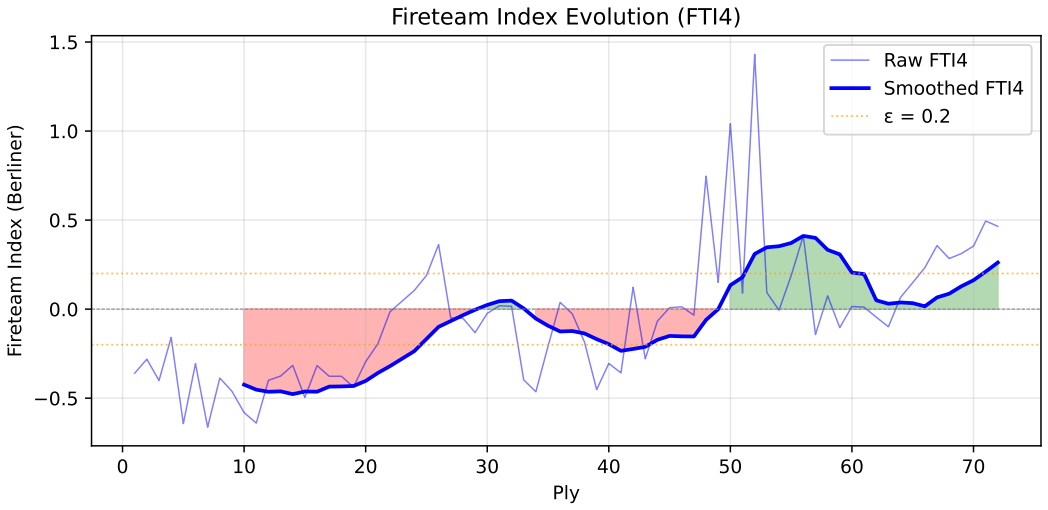

**FTI3: Strategic** $(0.7, 0.1, 0.1, 0.1)$

| Algorithm | Prediction | Max Streak / Confidence |
|---|---|---|
| Per-ply | WIN | 10 plies |
| Windowed | WIN | 13 plies |
| Balance | WIN | $B_n$=+0.30, C=0.30 |



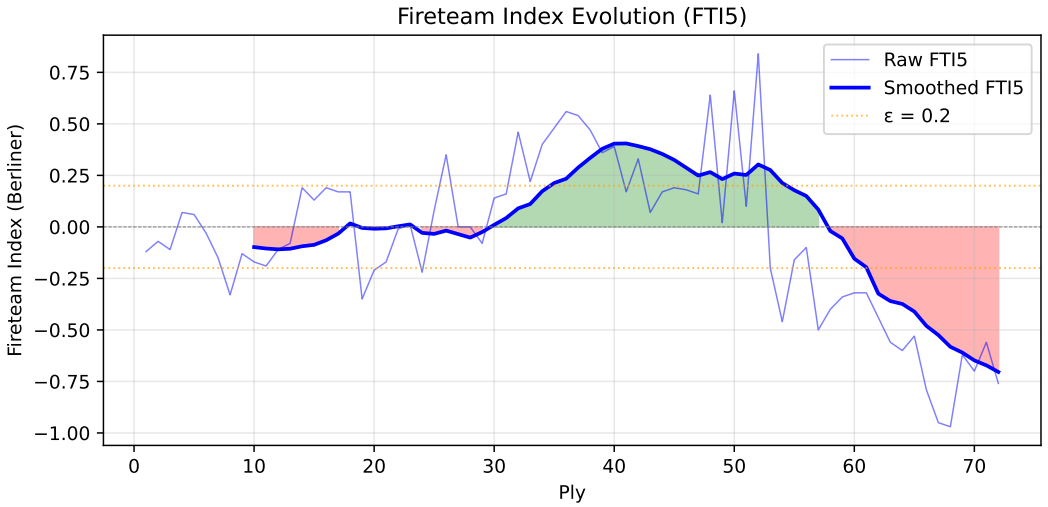

**FTI4: Dynamic** $(0.2, 0.6, 0, 0.2)$

| Algorithm | Prediction | Max Streak / Confidence |
|---|---|---|
| Per-ply | DRAW | 6 plies |
| Windowed | DRAW | 8 plies |
| Balance | DRAW | $B_n$=+0.02, C=0.01 |




**FTI5: Solid** $(0.3, 0.2, 0.5, 0)$

| Algorithm | Prediction | Max Streak / Confidence |
|---|---|---|
| Per-ply | DRAW | 7 plies |
| Windowed | WIN | 10 plies |
| Balance | DRAW | $B_n$=-0.12, C=0.10 |




Actual result: 0-1 (Berliner's outcome: WIN)

FTI1: Per-ply × Windowed × Balance ×
FTI2: Per-ply × Windowed × Balance ✓
FTI3: Per-ply ✓ Windowed ✓ Balance ✓
FTI4: Per-ply × Windowed × Balance ×
FTI5: Per-ply × Windowed ✓ Balance ×

## 1.7 Analysis Information

- Engine: Stockfish 16
- Depth: 20
- Analysis time: 247.4 seconds

# Chapter 2

# Roman Zinovievich Altshuler vs Hans Berliner

## 2.1 Game Information

| | |
|---|---|
| **Event** | 5th Correspondence World Championship |
| **Site** | ICCF |
| **Date** | 1965.07.01 |
| **Round** | 10 |
| **White** | Roman Zinovievich Altshuler |
| **Black** | Hans Berliner |
| **Result** | 0-1 |
| **Opening** | B04 – Unknown |

## 2.2 Player Statistics

### 2.2.1 Roman Zinovievich Altshuler (White)

- Total moves: 25
- Accuracy: 88.7%
- Average centipawn loss: 24.0
- Best/Excellent moves: 9 / 4
- Good moves: 6
- Inaccuracies: 0
- Mistakes: 5
- Blunders: 1

### 2.2.2 Hans Berliner (Black)

- Total moves: 25
- Accuracy: 93.5%

- Average centipawn loss: 13.5
- Best/Excellent moves: 9 / 5
- Good moves: 7
- Inaccuracies: 2
- Mistakes: 2
- Blunders: 0

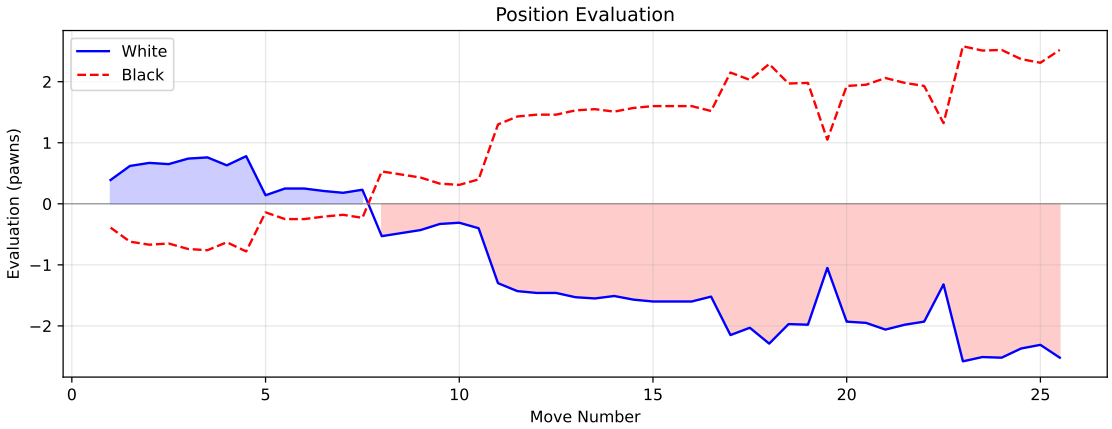
## 2.3 Positional Analysis

This section shows how key positional factors evolved throughout the game.

### 2.3.1 Position Evaluation Over Time

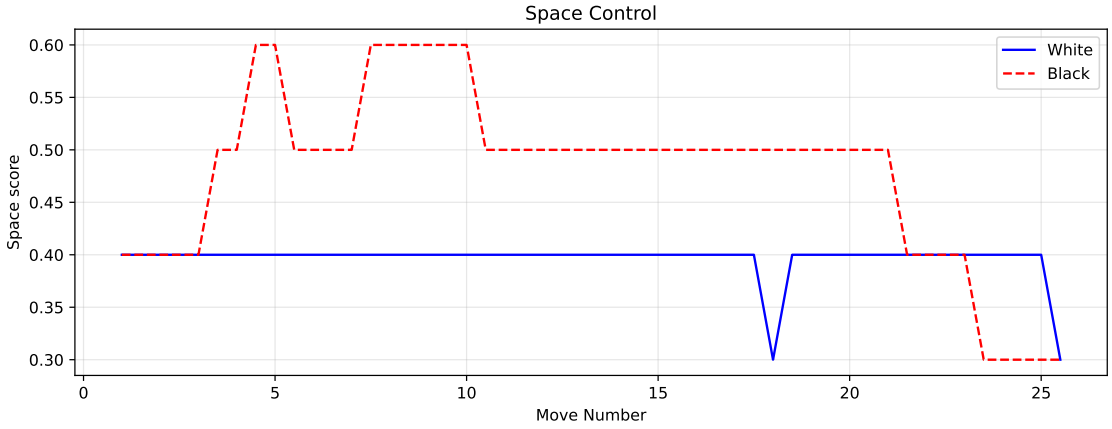
**Game Character Classification.** Evaluation range: minimum $m_1 = -2.58$, maximum $m_2 = +0.78$, spread $d = m_2 - m_1 = 3.36$.

This game is classified as **tactical** (normal). A tactical game with moderate swings.



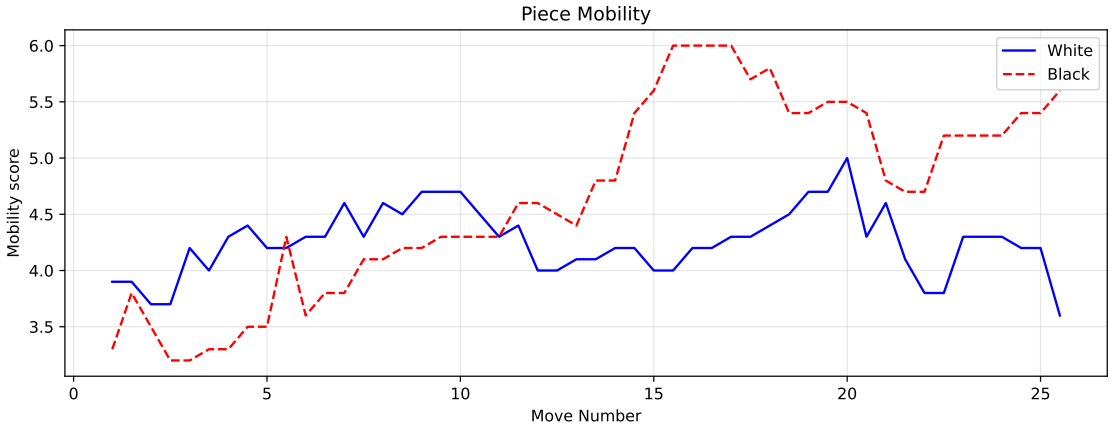

### 2.3.2 Space Control

- White average: 0.40 (range: 0.30 to 0.40)
- Black average: 0.48 (range: 0.30 to 0.60)



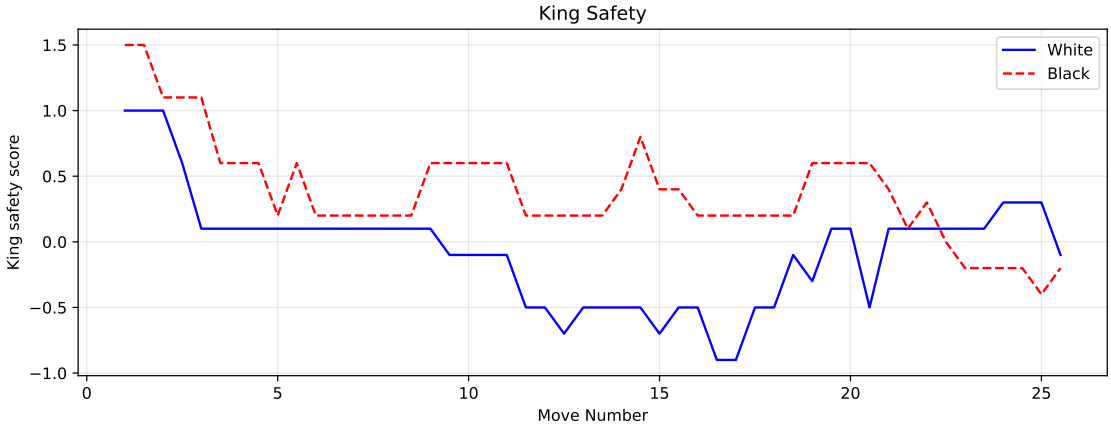

### 2.3.3 Mobility (Piece Activity)

- White average: 4.26 (range: 3.60 to 5.00)
- Black average: 4.65 (range: 3.20 to 6.00)



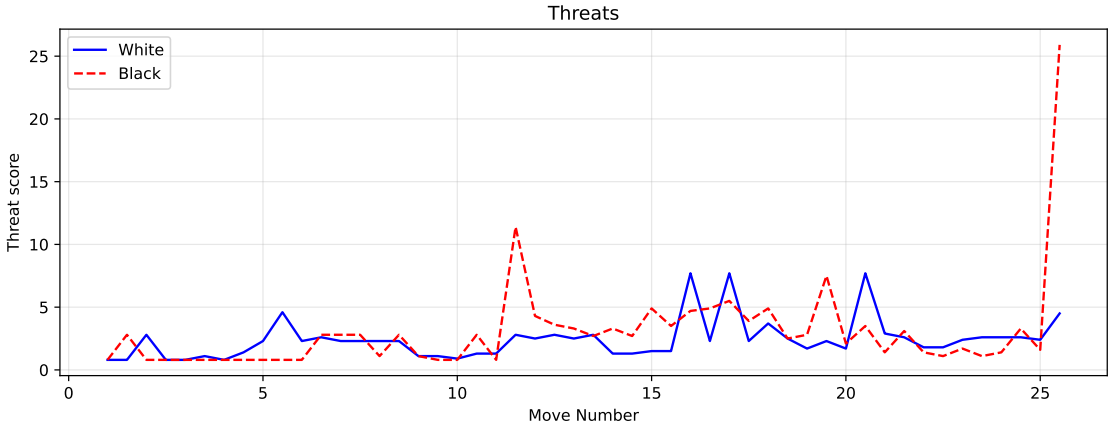

### 2.3.4 King Safety

- White average: -0.06
- Black average: 0.38




### 2.3.5 Threats

- White average: 2.42 (range: 0.80 to 7.70)
- Black average: 3.07 (range: 0.80 to 25.90)
- Average advantage: -0.65 (Black maintained more threats)




## 2.4 Annotated Game

1. e4 Nf6?! 2. e5 Nd5 3. d4 d6 4. Nf3 g6 5. Ng5? *White loses 66cp. Consider instead: Bc4, c4, exd6.*

5...dxe5 6. dxe5 Bg7 7. Bc4 c6 8. Qe2? *White loses 70cp. Consider instead: e6, O-O, f4.*

8...h6! 9. Nf3! Bg4! 10. Nbd2! b5! 11. Bb3? *White loses 85cp. Consider instead: Bxd5.*

11...Nf4! 12. Qf1 Qa5! 13. c3! Nd7! 14. Bc2 Nxe5 15. Nd4! Qc7 16. Ne4 Nc4 17. h3? *White loses 63cp. Consider instead: f3, Bxf4.*

17...Bd7 18. g3 Ne6?! 19. Nf3! Ne5? *Black loses 96cp. Consider instead: c5, Nd6, O-O.*

20. Qe2? *White loses 97cp. Consider instead: Nxe5.*

20...Nxf3+! 21. Qxf3! f5 22. Nd2! O-O? *Black loses 63cp. Consider instead: Qe5+, Rb8, Ng5.*

23. Nb3?? *White loses 132cp. Consider instead: Qe2.*

23...c5! 24. O-O! c4 25. Nd4! Nxd4!

0-1

## 2.5 Critical Positions

This section highlights critical moments where the evaluation shifted substantially—either due to errors or missed opportunities. Each diagram shows the position *after* the move was played.

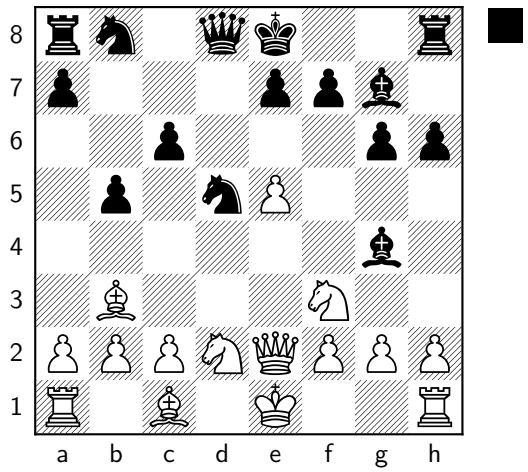
- **Instead of [move]**: The move(s) that should have been played instead.
- **Best continuation**: The optimal sequence of moves from the diagrammed position.

Positions marked with ★ represent the biggest evaluation swings in the game. The positional metrics table summarizes the spatial control, piece activity, king safety, and tactical threats for each side.

### Position 1: After 11. Bb3 ★

*Biggest evaluation swing(s) (90cp)*

Evaluation: -1.30



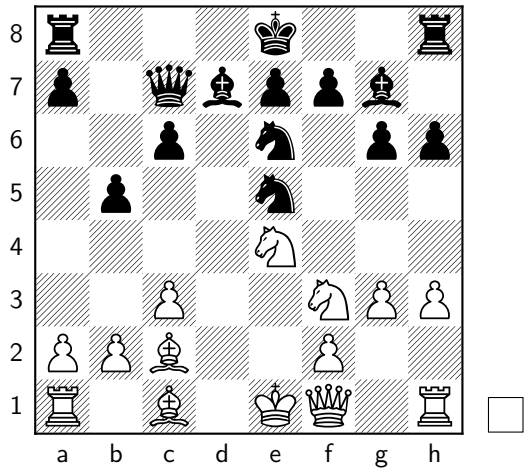
| Metric | White | Black |
|---|---|---|
| Space | 0.40 | 0.50 |
| Mobility | 4.30 | 4.30 |
| King Safety | -0.10 | 0.60 |
| Threats | 1.30 | 0.80 |

Instead of Bb3: Bxd5 Best continuation: Nf4 Qf1 Qa5

## Position 2: After 19...Ne5 ★

*Biggest evaluation swing(s) (93cp)*

Evaluation: -1.05



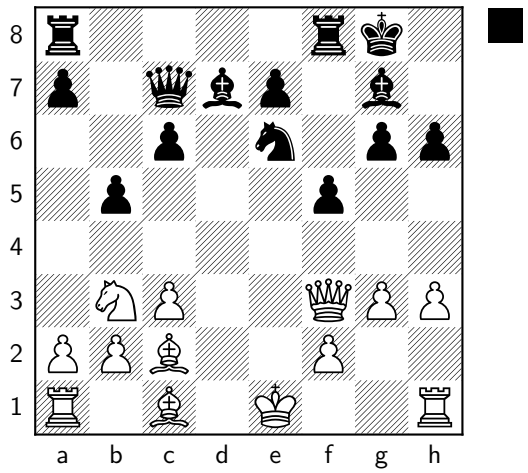
| Metric | White | Black |
|---|---|---|
| Space | 0.40 | 0.50 |
| Mobility | 4.70 | 5.50 |
| King Safety | 0.10 | 0.60 |
| Threats | 2.30 | 7.50 |

Instead of Ne5: c5, Nd6, O-O Best continuation: Nxe5 Qxe5 Be3

## Position 3: After 23. Nb3

*White blunders (-126cp)*

Evaluation: -2.58



| Metric | White | Black |
|---|---|---|
| Space | 0.40 | 0.40 |
| Mobility | 4.30 | 5.20 |
| King Safety | 0.10 | -0.20 |
| Threats | 2.40 | 1.70 |

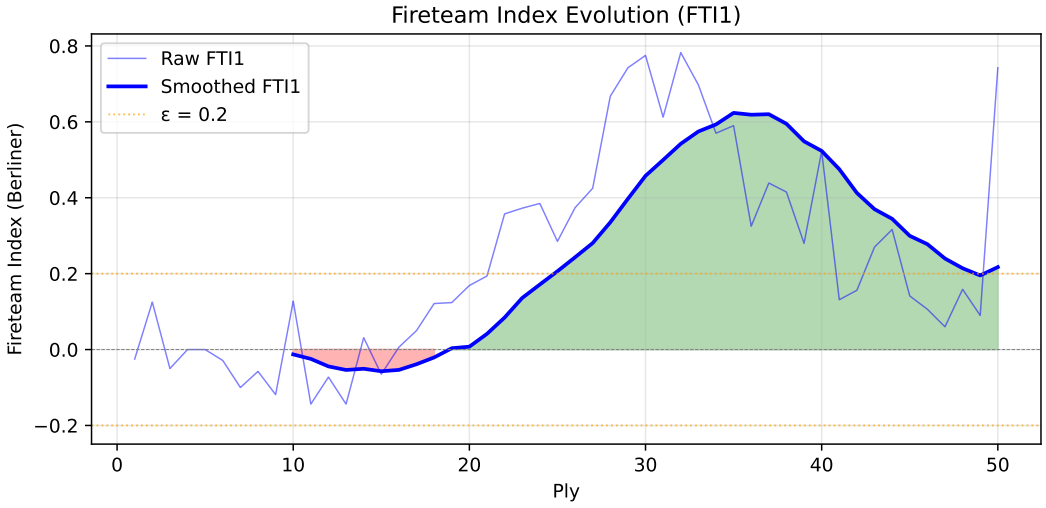
Instead of Nb3: Qe2 Best continuation: c5 O-O a5

## 2.6 Fireteam Index Prediction

Tracking: **Berliner** (Black)

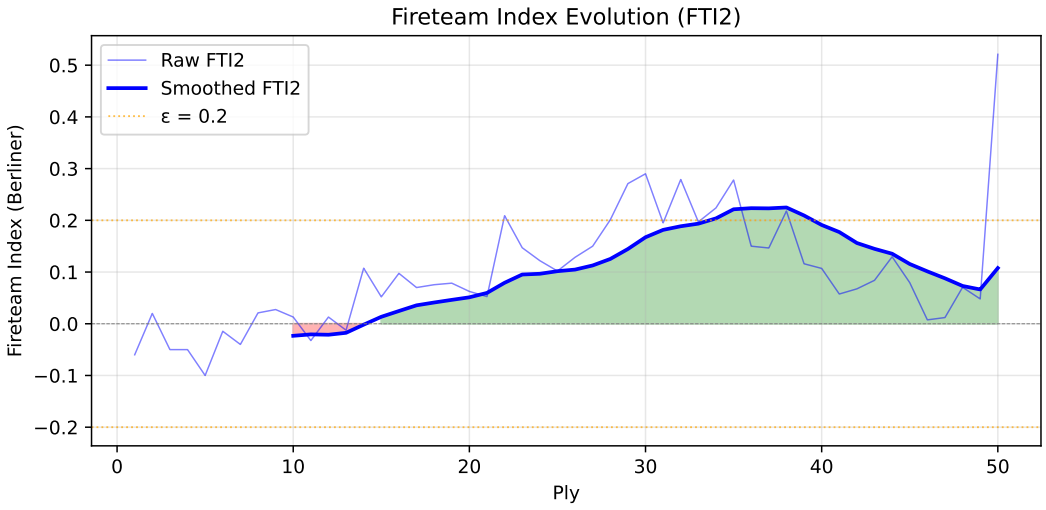
**FTI1: Harmonious** $(\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4})$

| Algorithm | Prediction | Max Streak / Confidence |
|---|---|---|
| Per-ply | WIN | 19 plies |
| Windowed | WIN | 20 plies |
| Balance | WIN | $B_n$=+0.65, C=0.65 |



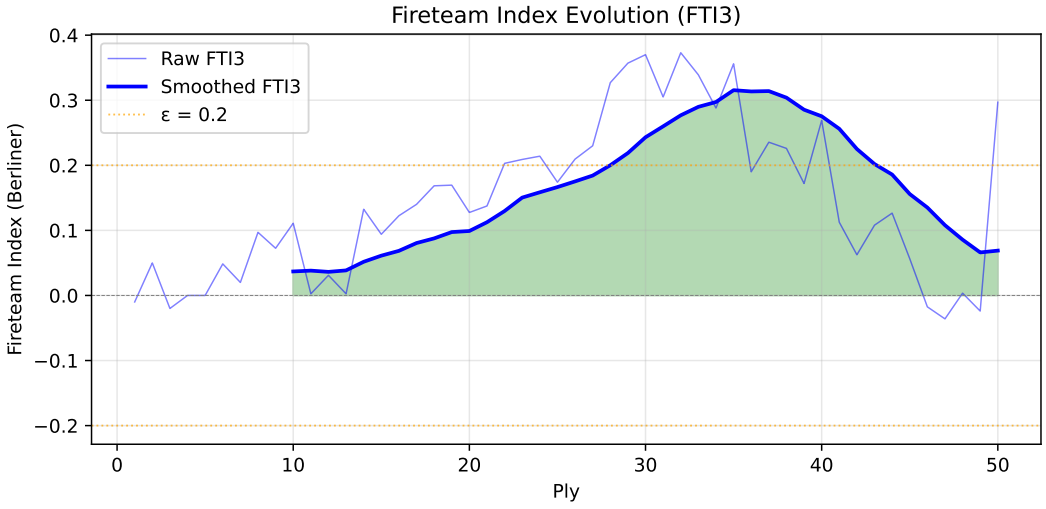

**FTI2: Tactical** $(0.6, 0.1, 0, 0.3)$

| Algorithm | Prediction | Max Streak / Confidence |
|---|---|---|
| Per-ply | DRAW | 2 plies |
| Windowed | DRAW | 0 plies |
| Balance | WIN | $B_n$=+0.26, C=0.26 |



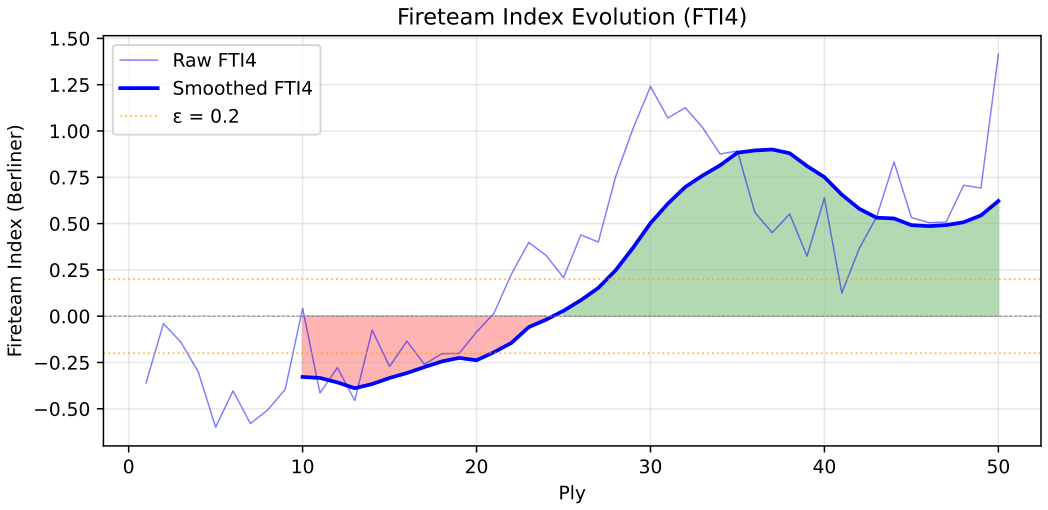

**FTI3: Strategic** $(0.7, 0.1, 0.1, 0.1)$

| Algorithm | Prediction | Max Streak / Confidence |
|---|---|---|
| Per-ply | DRAW | 8 plies |
| Windowed | WIN | 11 plies |
| Balance | WIN | $B_n$=+0.50, C=0.50 |



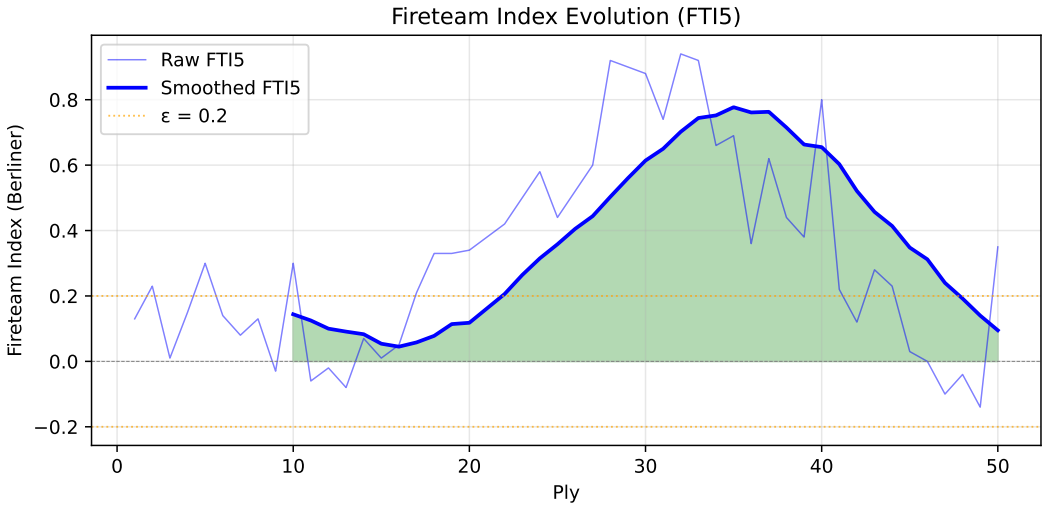

**FTI4: Dynamic** $(0.2, 0.6, 0, 0.2)$

| Algorithm | Prediction | Max Streak / Confidence |
|---|---|---|
| Per-ply | WIN | 15 plies |
| Windowed | WIN | 22 plies |
| Balance | WIN | $B_n$=+0.74, C=0.63 |




**FTI5: Solid** $(0.3, 0.2, 0.5, 0)$

| Algorithm | Prediction | Max Streak / Confidence |
|---|---|---|
| Per-ply | WIN | 23 plies |
| Windowed | WIN | 24 plies |
| Balance | WIN | $B_n$=+0.82, C=0.82 |




Actual result: 0-1 (Berliner's outcome: WIN)

FTI1: Per-ply ✓ Windowed ✓ Balance ✓
FTI2: Per-ply × Windowed × Balance ✓
FTI3: Per-ply × Windowed ✓ Balance ✓
FTI4: Per-ply ✓ Windowed ✓ Balance ✓
FTI5: Per-ply ✓ Windowed ✓ Balance ✓

## 2.7 Analysis Information

- Engine: Stockfish 16
- Depth: 20
- Analysis time: 168.5 seconds

# Chapter 3

# Berliner Hans J (USA) vs Borisenko Georgy K (UZB)

## 3.1 Game Information

| | |
|---|---|
| **Event** | Ch World (corr.) |
| **Site** | corr. |
| **Date** | 1965.??.?? |
| **Round** | 16 |
| **White** | Berliner Hans J (USA) |
| **Black** | Borisenko Georgy K (UZB) (2354) |
| **Result** | 1-0 |
| **Opening** | D86 – Unknown |

## 3.2 Player Statistics

### 3.2.1 Berliner Hans J (USA) (White)

- Total moves: 17
- Accuracy: 96.2%
- Average centipawn loss: 7.7
- Best/Excellent moves: 6 / 2
- Good moves: 8
- Inaccuracies: 1
- Mistakes: 0
- Blunders: 0

### 3.2.2 Borisenko Georgy K (UZB) (Black)

- Total moves: 16
- Accuracy: 91.3%

- Average centipawn loss: 18.1
- Best/Excellent moves: 4 / 1
- Good moves: 7
- Inaccuracies: 2
- Mistakes: 2
- Blunders: 0

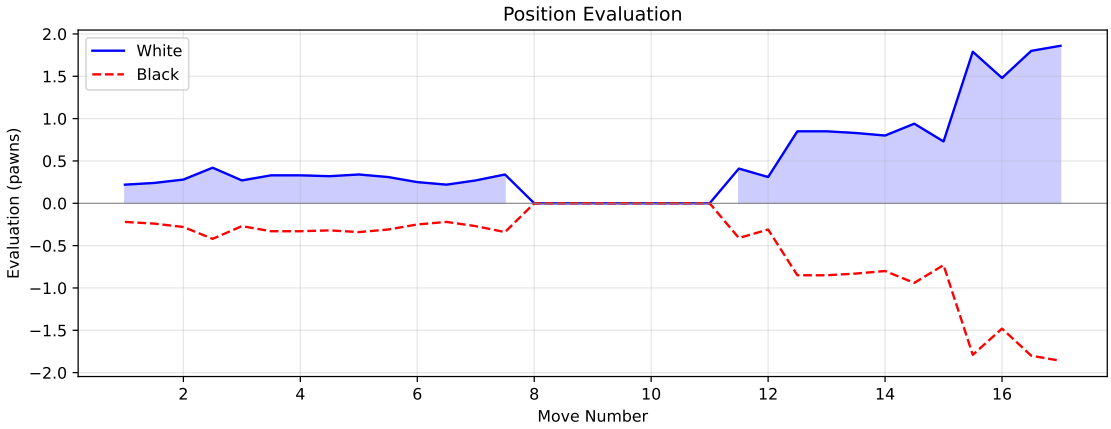
## 3.3 Positional Analysis

This section shows how key positional factors evolved throughout the game.

### 3.3.1 Position Evaluation Over Time

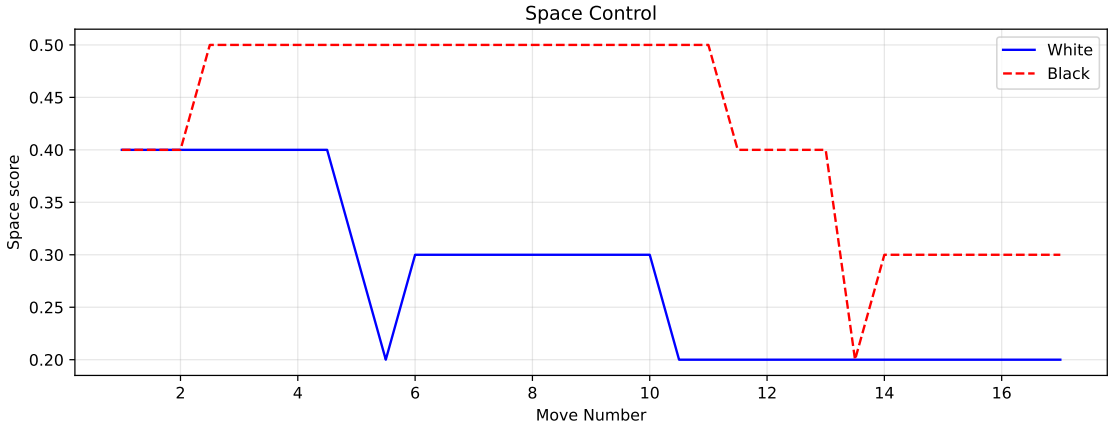
**Game Character Classification.** Evaluation range: minimum $m_1 = +0.00$, maximum $m_2 = +1.86$, spread $d = m_2 - m_1 = 1.86$.

This game is classified as **tense** (one-sided). A tense, one-sided game. White maintained the advantage throughout.




### 3.3.2 Space Control

- White average: 0.28 (range: 0.20 to 0.40)
- Black average: 0.43 (range: 0.20 to 0.50)



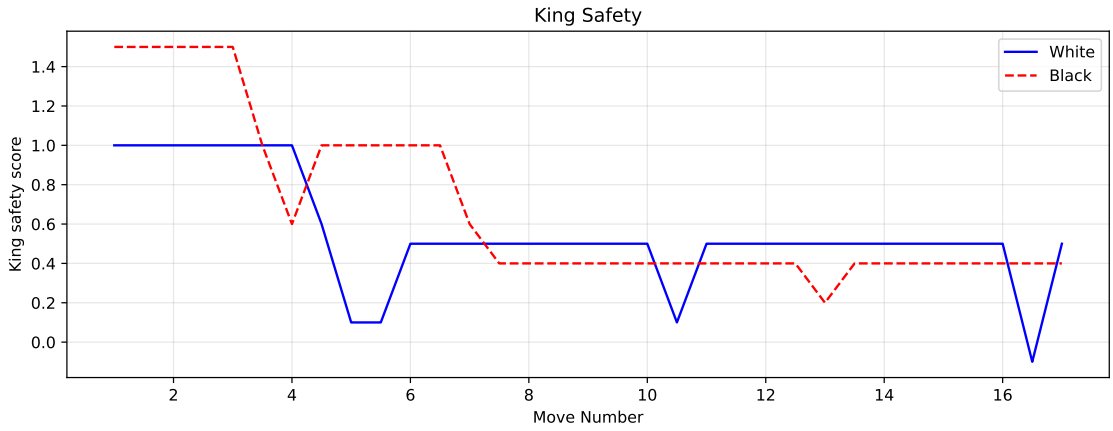

### 3.3.3 Mobility (Piece Activity)

- White average: 4.52 (range: 3.80 to 5.10)
- Black average: 3.86 (range: 3.20 to 4.40)



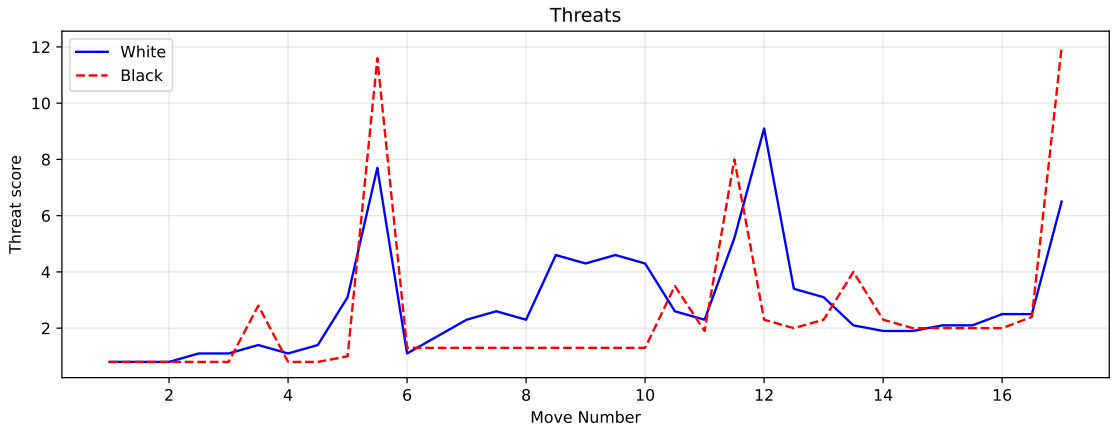

### 3.3.4 King Safety

- White average: 0.55
- Black average: 0.68




### 3.3.5 Threats

- White average: 2.86 (range: 0.80 to 9.10)
- Black average: 2.47 (range: 0.80 to 12.00)
- Average advantage: 0.39 (White maintained more threats)




## 3.4 Annotated Game

1. d4 Nf6 2. c4 g6 3. Nc3 d5 4. cxd5 Nxd5 5. e4 Nxc3 6. bxc3 Bg7 7. Bc4 O-O 8. h4?! c5! 9. Ne2! Nc6 10. Be3! cxd4! 11. cxd4! b5?! 12. Bd5! Bd7? *Black loses 62cp. Consider instead: Qa5+, Bb7, Qd7.*

13. h5! e6! 14. Bb3 Na5 15. Qd2 Nb7? *Black loses 112cp. Consider instead: Nxb3, Rc8, b4.*

16. Rc1 Rc8?! 17. Rxc8!

1-0

## 3.5 Critical Positions

This section highlights critical moments where the evaluation shifted substantially—either due to errors or missed opportunities. Each diagram shows the position *after* the move was played.

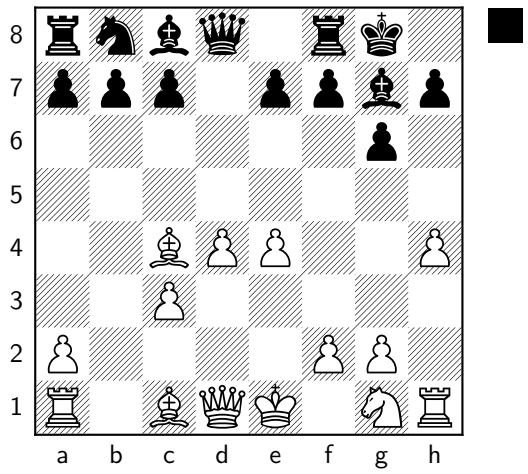
- **Instead of [move]**: The move(s) that should have been played instead.
- **Best continuation**: The optimal sequence of moves from the diagrammed position.

Positions marked with ★ represent the biggest evaluation swings in the game. The positional metrics table summarizes the spatial control, piece activity, king safety, and tactical threats for each side.

### Position 1: After 8. h4 ★

*Biggest evaluation swing(s) (34cp)*

Evaluation: +0.00



| Metric | White | Black |
|---|---|---|
| Space | 0.30 | 0.50 |
| Mobility | 4.70 | 3.50 |
| King Safety | 0.50 | 0.40 |
| Threats | 2.30 | 1.30 |

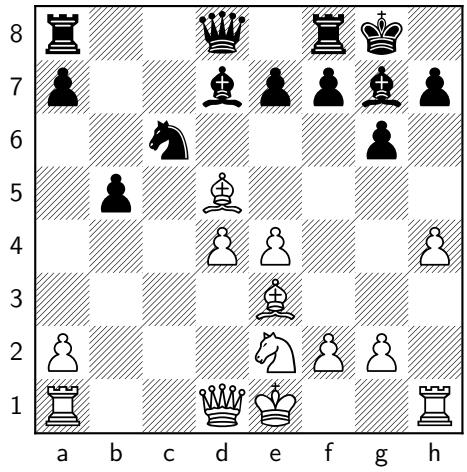
Instead of h4: Ne2, Be3, Nf3

Best continuation: c5 Ne2 cxd4

### Position 2: After 12...Bd7 ★

*Biggest evaluation swing(s) (54cp)*

Evaluation: +0.85



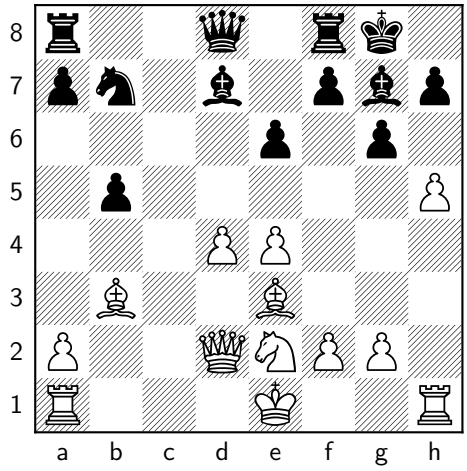
| Metric | White | Black |
|---|---|---|
| Space | 0.20 | 0.40 |
| Mobility | 4.70 | 4.40 |
| King Safety | 0.50 | 0.40 |
| Threats | 3.40 | 2.00 |

Instead of Bd7: Qa5+, Bb7, Qd7 Best continuation: h5 e6 Bb3

### Position 3: After 15...Nb7

*Critical moment*

Evaluation: +1.79



| Metric | White | Black |
|---|---|---|
| Space | 0.20 | 0.30 |
| Mobility | 4.60 | 3.90 |
| King Safety | 0.50 | 0.40 |
| Threats | 2.10 | 2.00 |

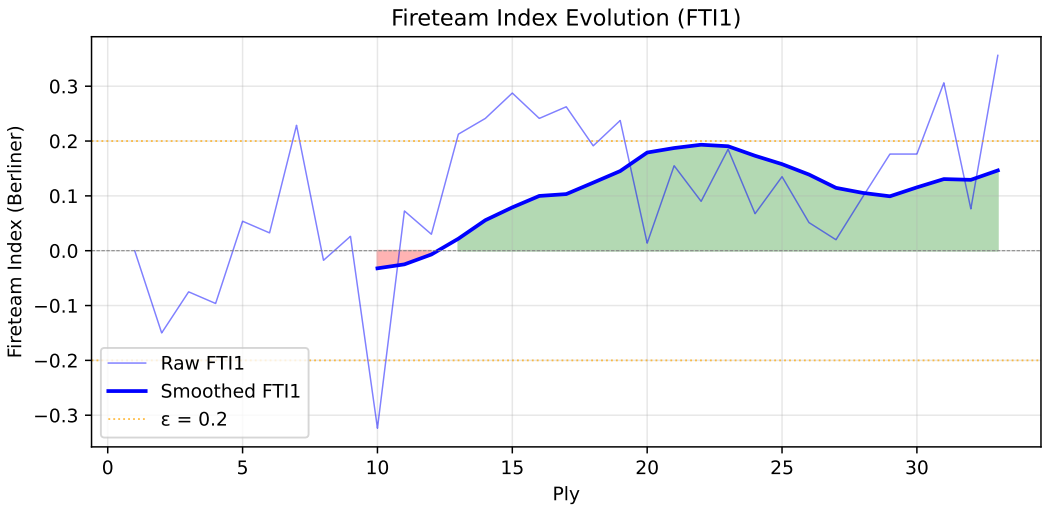
Instead of Nb7: Nxb3, Rc8, b4 Best continuation: e5 Na5 Bg5

## 3.6 Fireteam Index Prediction

Tracking: **Berliner** (White)

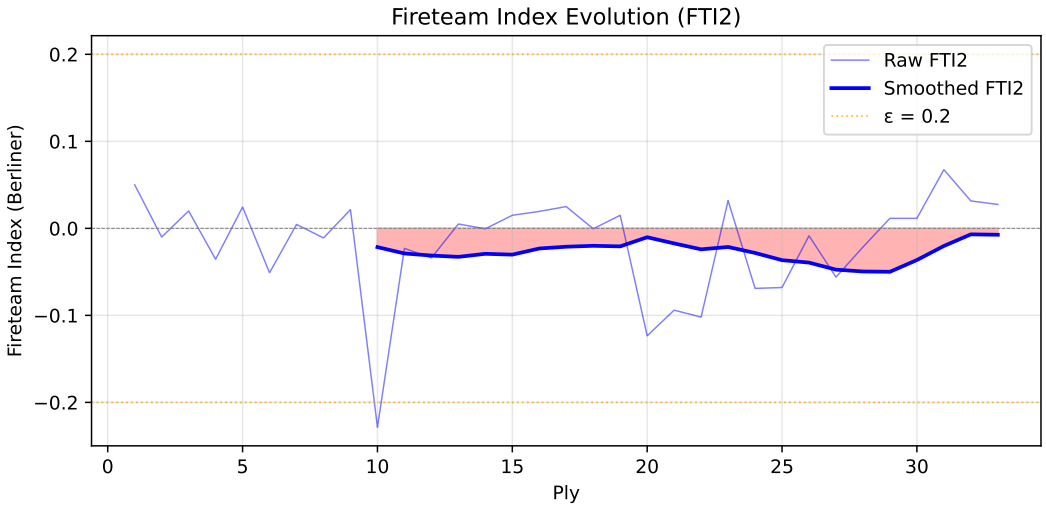
**FTI1: Harmonious** $(\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4})$

| Algorithm | Prediction | Max Streak / Confidence |
|---|---|---|
| Per-ply | DRAW | 1 plies |
| Windowed | DRAW | 0 plies |
| Balance | WIN | $B_n$=+0.24, C=0.24 |



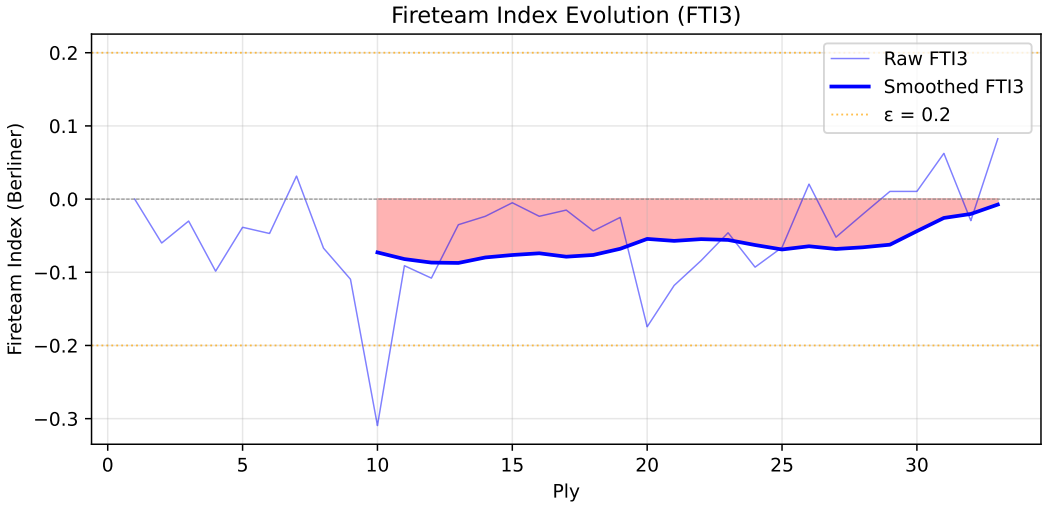

**FTI2: Tactical** $(0.6, 0.1, 0, 0.3)$

| Algorithm | Prediction | Max Streak / Confidence |
|---|---|---|
| Per-ply | DRAW | 0 plies |
| Windowed | DRAW | 0 plies |
| Balance | DRAW | $B_n$=+0.00, C=0.00 |



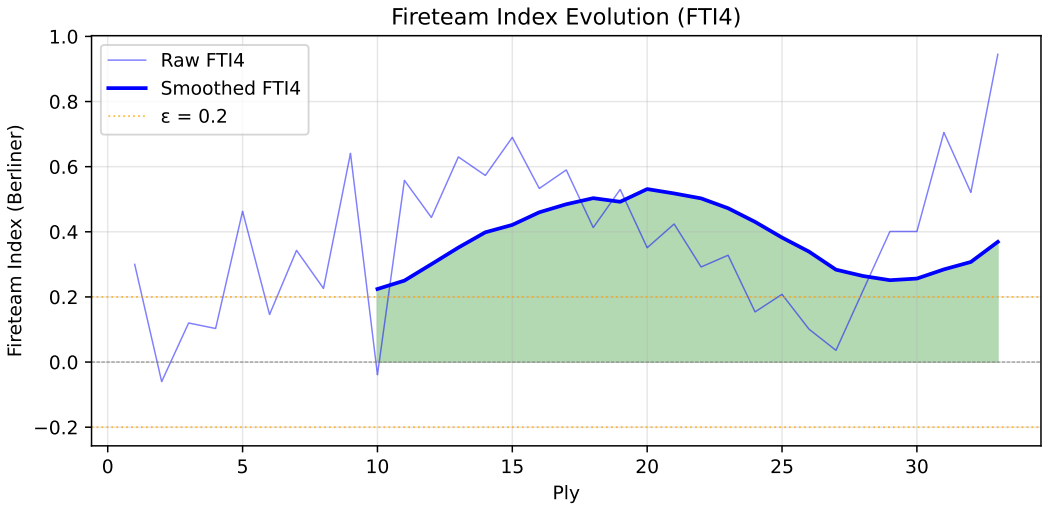

**FTI3: Strategic** $(0.7, 0.1, 0.1, 0.1)$

| Algorithm | Prediction | Max Streak / Confidence |
|---|---|---|
| Per-ply | DRAW | 0 plies |
| Windowed | DRAW | 0 plies |
| Balance | DRAW | $B_n$=+0.00, C=0.00 |



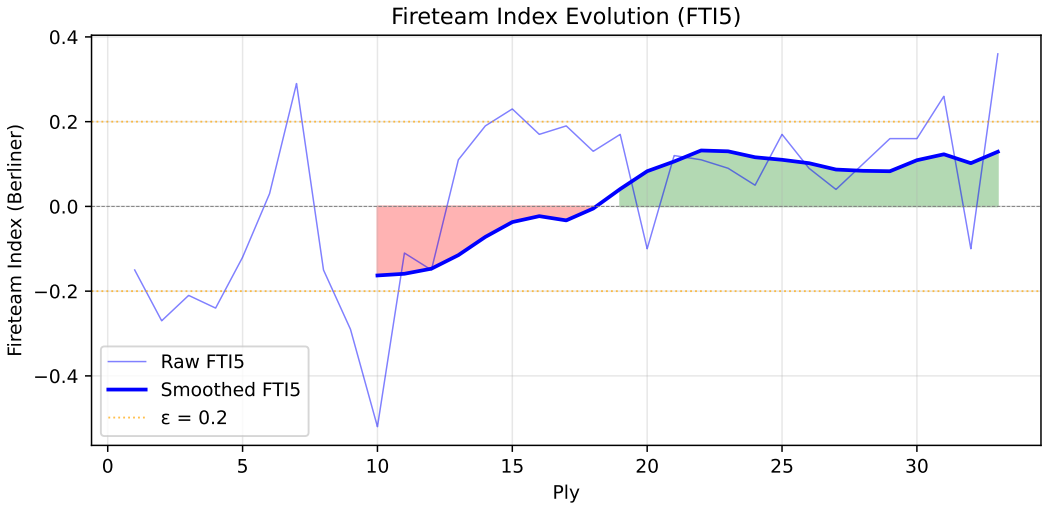

**FTI4: Dynamic** $(0.2, 0.6, 0, 0.2)$

| Algorithm | Prediction | Max Streak / Confidence |
|---|---|---|
| Per-ply | DRAW | 7 plies |
| Windowed | WIN | 17 plies |
| Balance | WIN | $B_n$=+0.82, C=0.82 |




**FTI5: Solid** $(0.3, 0.2, 0.5, 0)$

| Algorithm | Prediction | Max Streak / Confidence |
|---|---|---|
| Per-ply | DRAW | 1 plies |
| Windowed | DRAW | 0 plies |
| Balance | DRAW | $B_n$=+0.12, C=0.12 |




Actual result: 1-0 (Berliner's outcome: WIN)

FTI1: Per-ply × Windowed × Balance ✓
FTI2: Per-ply × Windowed × Balance ×
FTI3: Per-ply × Windowed × Balance ×
FTI4: Per-ply × Windowed ✓ Balance ✓
FTI5: Per-ply × Windowed × Balance ×

## 3.7 Analysis Information

- Engine: Stockfish 16
- Depth: 20
- Analysis time: 113.6 seconds

# Chapter 4

# Heinz De Carbonnel vs Hans Berliner

## 4.1 Game Information

| | |
|---|---|
| **Event** | 5th Correspondence World Championship |
| **Site** | ICCF |
| **Date** | 1965.07.01 |
| **Round** | 4 |
| **White** | Heinz De Carbonnel |
| **Black** | Hans Berliner |
| **Result** | 0-1 |
| **Opening** | E80 – Unknown |

## 4.2 Player Statistics

### 4.2.1 Heinz De Carbonnel (White)

- Total moves: 46
- Accuracy: 77.4%
- Average centipawn loss: 51.1
- Best/Excellent moves: 18 / 4
- Good moves: 11
- Inaccuracies: 8
- Mistakes: 3
- Blunders: 2

### 4.2.2 Hans Berliner (Black)

- Total moves: 46
- Accuracy: 93.5%
- Average centipawn loss: 13.5

- Best/Excellent moves: 23 / 7
- Good moves: 8
- Inaccuracies: 7
- Mistakes: 0
- Blunders: 1

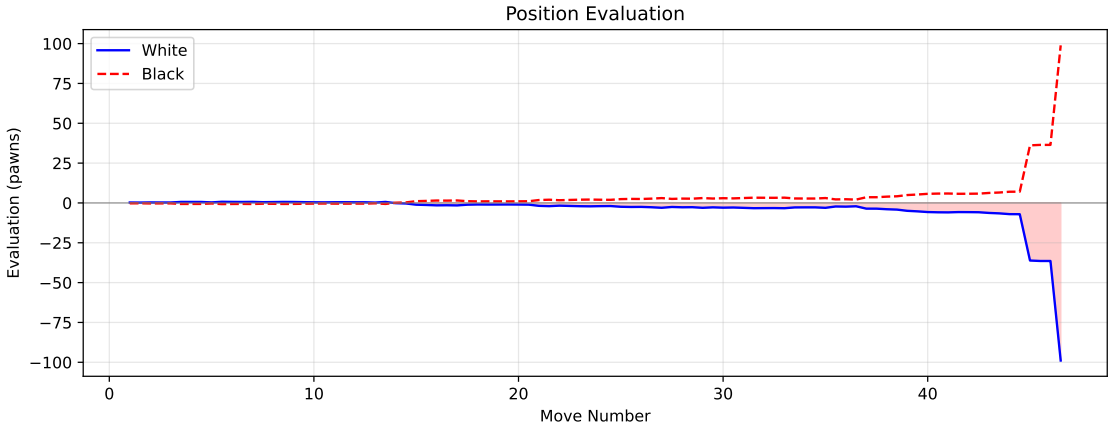
## 4.3 Positional Analysis

This section shows how key positional factors evolved throughout the game.

### 4.3.1 Position Evaluation Over Time

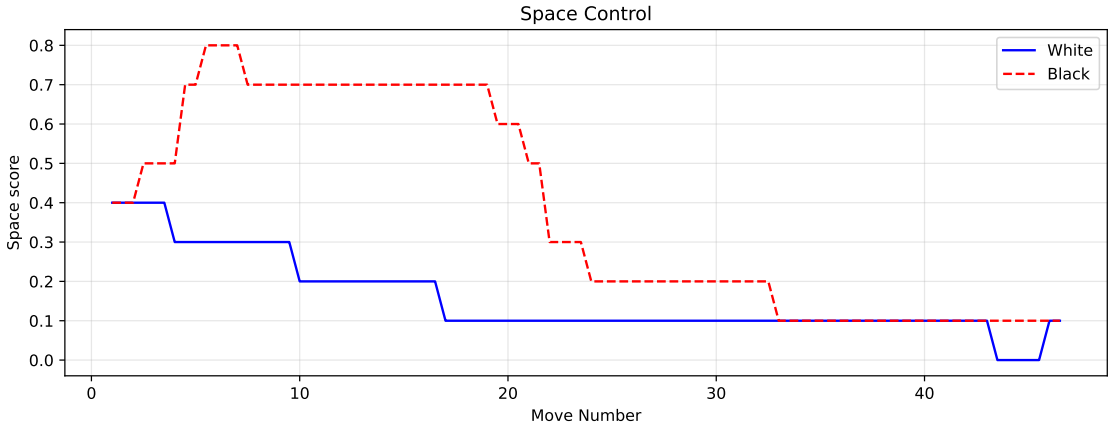
**Game Character Classification.** Evaluation range: minimum $m_1 = -15.00$, maximum $m_2 = +0.73$, spread $d = m_2 - m_1 = 15.73$.

This game is classified as **chaotic** (normal). A chaotic game with moderate swings.



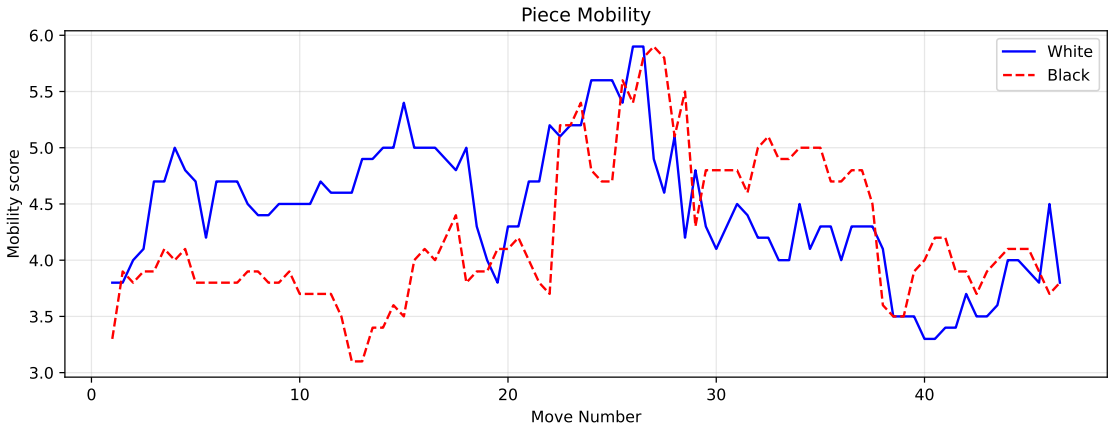

### 4.3.2 Space Control

- White average: 0.16 (range: 0.00 to 0.40)
- Black average: 0.38 (range: 0.10 to 0.80)



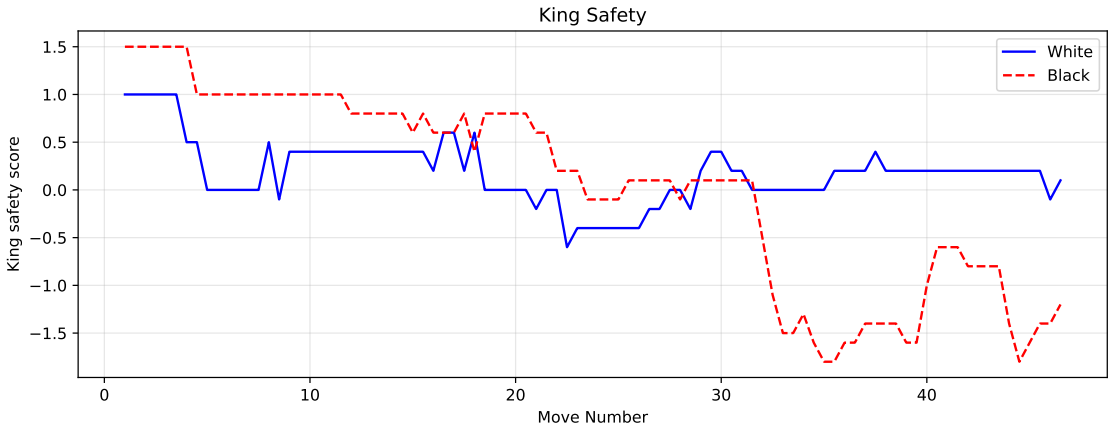
### 4.3.3 Mobility (Piece Activity)

- White average: 4.45 (range: 3.30 to 5.90)
- Black average: 4.23 (range: 3.10 to 5.90)



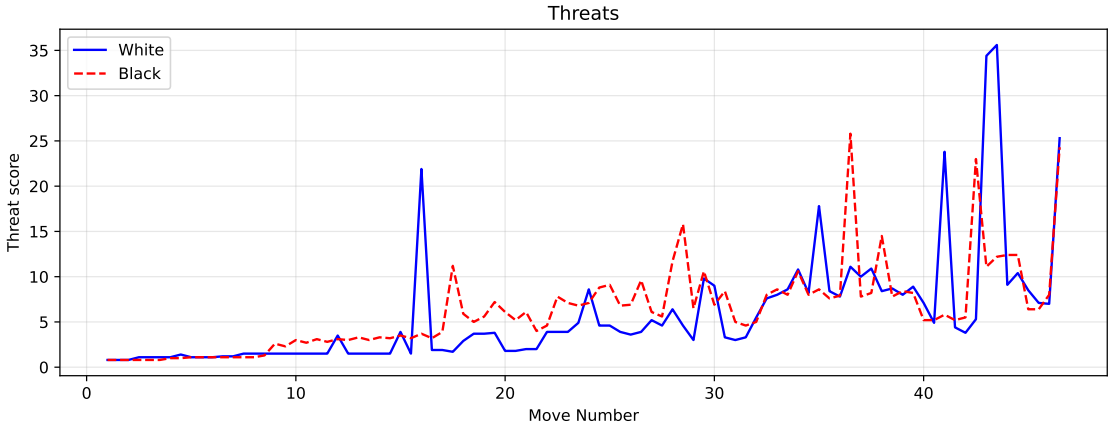

### 4.3.4 King Safety

- White average: 0.18
- Black average: 0.04




### 4.3.5 Threats

- White average: 5.72 (range: 0.80 to 35.60)
- Black average: 6.27 (range: 0.80 to 25.80)
- Average advantage: -0.55 (Black maintained more threats)




## 4.4 Annotated Game

1. d4 Nf6 2. c4 g6 3. Nc3 Bg7?! 4. e4 d6 5. f3?! c6?! 6. Be3 a6 7. Qd2 b5! 8. O-O-O Qa5 9. Kb1! Nbd7! 10. g4 Rb8 11. h4! h5! 12. g5! Nh7! 13. Rh2 Nhf8?! 14. Rc1?! Nb6 15. cxb5? *White loses 74cp. Consider instead: d5, c5, Nd1.*

15...axb5! 16. Nd1?! Qa8! 17. f4! Nc4?! 18. Bxc4! bxc4! 19. Rf2 c5! 20. Nc3! Ne6! 21. dxc5? *White loses 85cp. Consider instead: Nge2, Nf3.*

21...Bb7! 22. cxd6 Bxe4+! 23. Ka1 O-O! 24. dxe7 Rfe8! 25. Qd1?! Rxe7! 26. Nf3! Rd8! 27. Nd2?! Nd4?! 28. Bxd4 Bxd4 29. Ndxe4 Bxf2 30. Qc2! Be3 31. Rf1! Rd3 32. Nf6+! Kg7 33. f5! Re5?! 34. fxg6! fxg6 35. Qh2 Qb8?? *Black loses 148cp. Consider instead: Bd4, Re7.*

36. Rb1! Rb5?! 37. Qh1?? *White loses 121cp. Consider instead: Qg2, Qc2.*

37...Bd4 38. Nxb5 Qxb5 39. Qh2? *White loses 71cp. Consider instead: Qe1.*

39...Qb7! 40. Ne8+?! Kh8! 41. Nd6 Qd5 42. Qf4! Rf3! 43. Qe4?! c3! 44. Qe8+?! Kh7! 45. Qe7+?? *White loses 1500cp. Consider instead: Re1.*

45...Bg7! 46. bxc3! Qxd6!

0-1

## 4.5 Critical Positions

This section highlights critical moments where the evaluation shifted substantially—either due to errors or missed opportunities. Each diagram shows the position *after* the move was played.

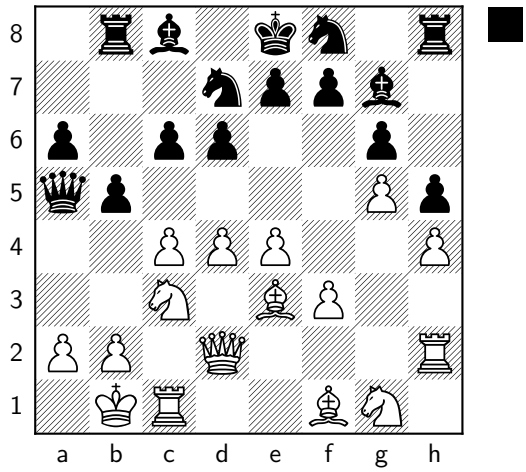
- **Instead of [move]**: The move(s) that should have been played instead.
- **Best continuation**: The optimal sequence of moves from the diagrammed position.

Positions marked with ★ represent the biggest evaluation swings in the game. The positional metrics table summarizes the spatial control, piece activity, king safety, and tactical threats for each side.

## Position 1: After 14. Rc1 ★

*Biggest evaluation swing(s) (79cp)*

Evaluation: -0.15



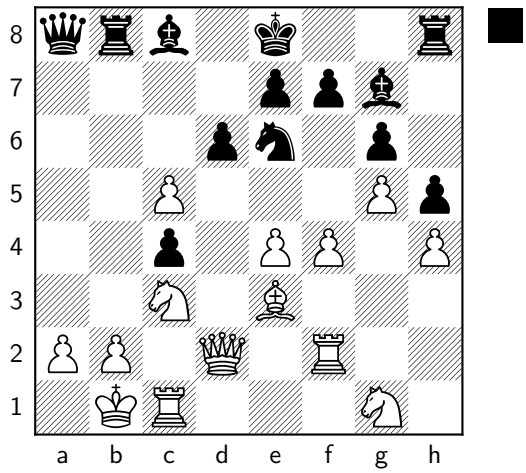
| Metric | White | Black |
|---|---|---|
| Space | 0.20 | 0.70 |
| Mobility | 5.00 | 3.40 |
| King Safety | 0.40 | 0.80 |
| Threats | 1.50 | 3.30 |

Instead of Rc1: Rf2, f4, e5 Best continuation: Nb6 d5 c5

## Position 2: After 21. dxc5 ★

*Biggest evaluation swing(s) (75cp)*

Evaluation: -1.82



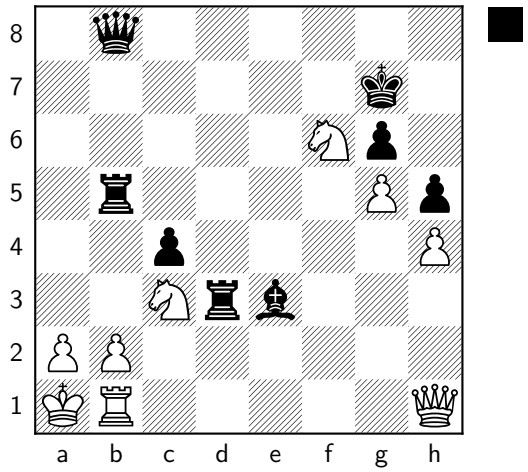
| Metric | White | Black |
|---|---|---|
| Space | 0.10 | 0.50 |
| Mobility | 4.70 | 4.00 |
| King Safety | -0.20 | 0.60 |
| Threats | 2.00 | 6.10 |

Instead of dxc5: Nge2, Nf3 Best continuation: Bb7 Ka1 Bxe4

### Position 3: After 37. Qh1

*White blunders (-149cp)*

Evaluation: -3.58



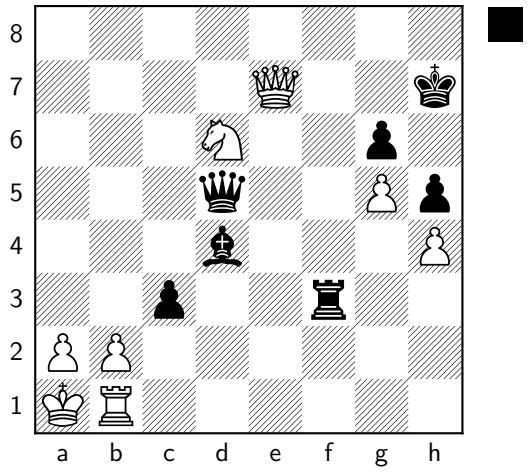
| Metric | White | Black |
|---|---|---|
| Space | 0.10 | 0.10 |
| Mobility | 4.30 | 4.80 |
| King Safety | 0.20 | -1.40 |
| Threats | 10.00 | 7.80 |

Instead of Qh1: Qg2, Qc2 Best continuation: Bd4 Nxb5 Qxb5

### Position 4: After 45. Qe7+

*White blunders (-2907cp)*

Evaluation: -36.14



| Metric | White | Black |
|---|---|---|
| Space | 0.00 | 0.10 |
| Mobility | 3.90 | 4.10 |
| King Safety | 0.20 | -1.60 |
| Threats | 8.50 | 6.40 |

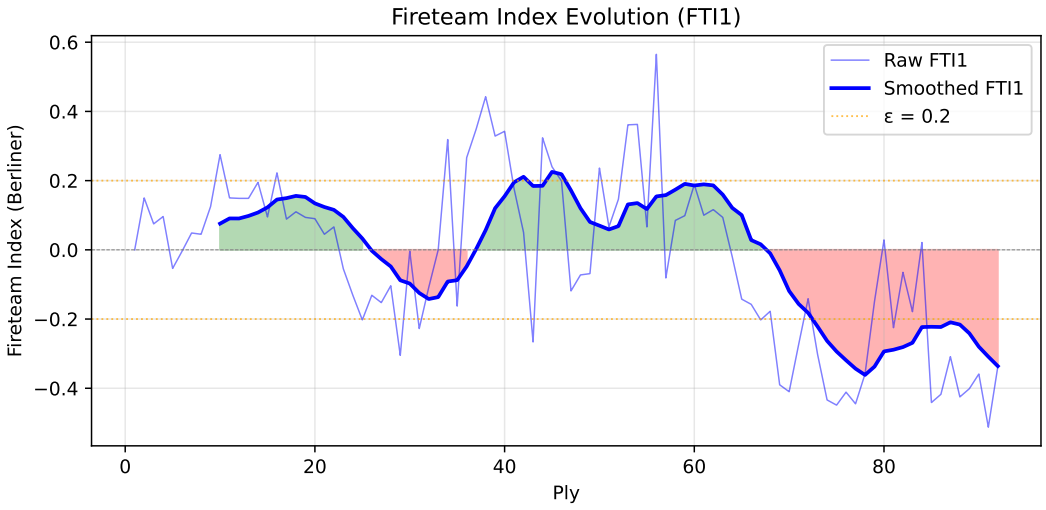
Instead of Qe7+: Re1 Best continuation: Bg7 bxc3 Qxd6

## 4.6 Fireteam Index Prediction

Tracking: **Berliner** (Black)

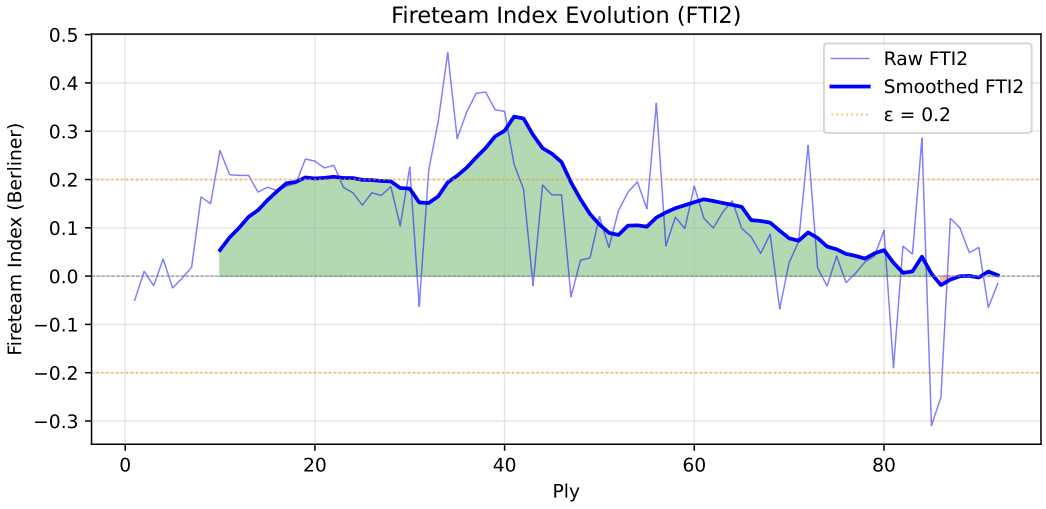
**FTI1: Harmonious** $(\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4})$

| **Algorithm** | **Prediction** | **Max Streak / Confidence** |
|---|---|---|
| Per-ply | DRAW | 5 plies |
| Windowed | DRAW | 0 plies |
| Balance | DRAW | $B_n$=-0.14, C=0.11 |



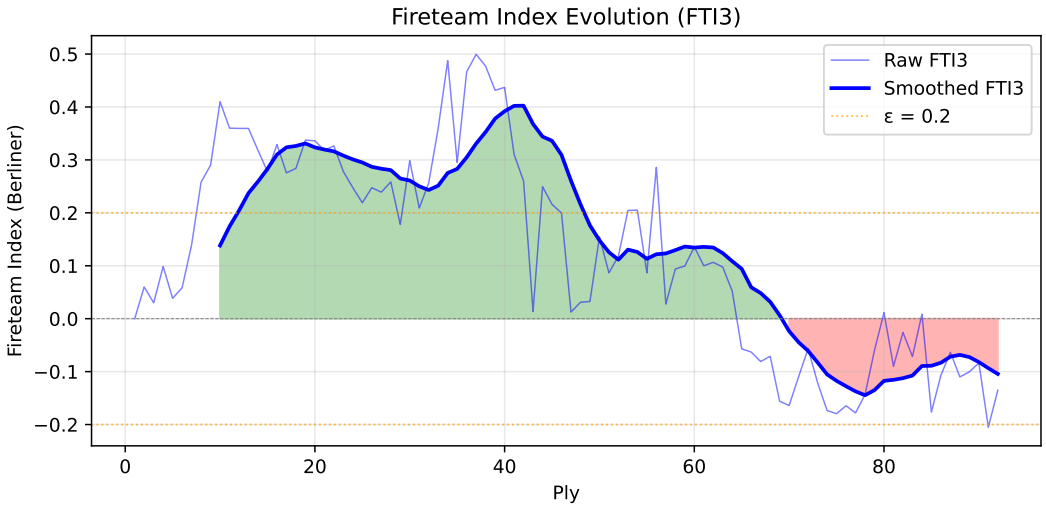

**FTI2: Tactical** $(0.6, 0.1, 0, 0.3)$

| Algorithm | Prediction | Max Streak / Confidence |
|---|---|---|
| Per-ply | DRAW | 8 plies |
| Windowed | DRAW | 8 plies |
| Balance | WIN | $B_n$=+0.21, C=0.20 |



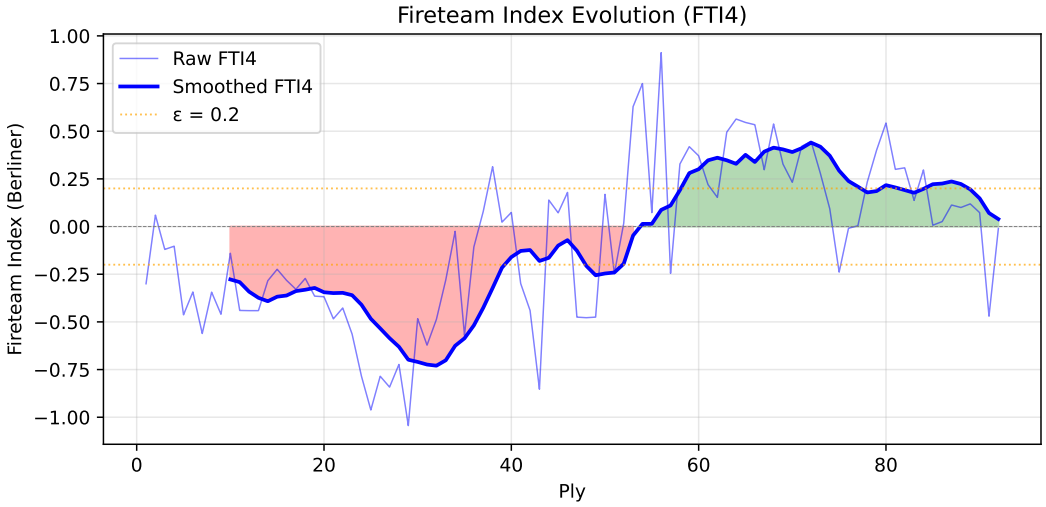

**FTI3: Strategic** $(0.7, 0.1, 0.1, 0.1)$

| Algorithm | Prediction | Max Streak / Confidence |
|---|---|---|
| Per-ply | WIN | 11 plies |
| Windowed | WIN | 15 plies |
| Balance | WIN | $B_n$=+0.38, C=0.36 |



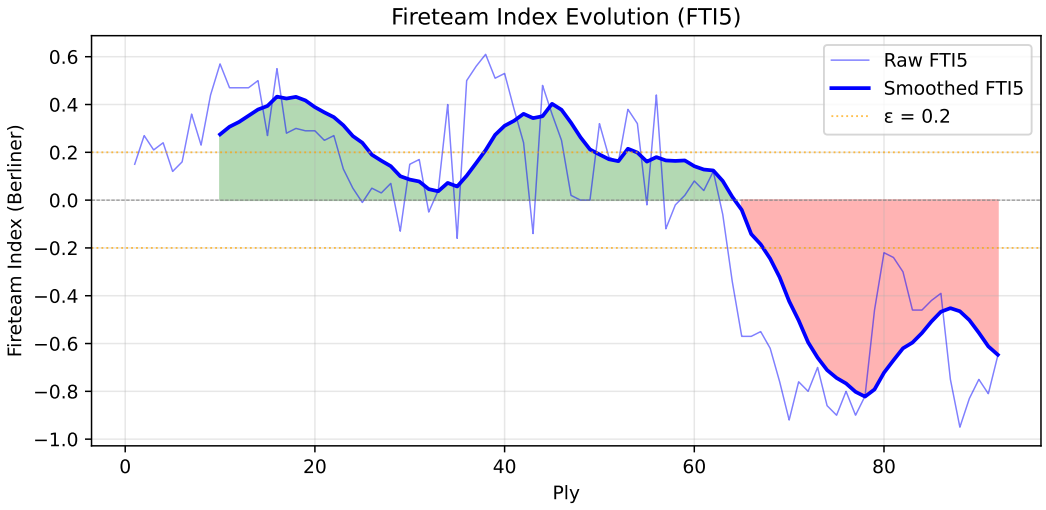

**FTI4: Dynamic** $(0.2, 0.6, 0, 0.2)$

| Algorithm | Prediction | Max Streak / Confidence |
|---|---|---|
| Per-ply | DRAW | 7 plies |
| Windowed | WIN | 17 plies |
| Balance | DRAW | $B_n$=-0.04, C=0.02 |




**FTI5: Solid** $(0.3, 0.2, 0.5, 0)$

| Algorithm | Prediction | Max Streak / Confidence |
|---|---|---|
| Per-ply | DRAW | 6 plies |
| Windowed | WIN | 10 plies |
| Balance | DRAW | $B_n$=-0.11, C=0.10 |




Actual result: 0-1 (Berliner's outcome: WIN)
FTI1: Per-ply × Windowed × Balance ×
FTI2: Per-ply × Windowed × Balance ✓
FTI3: Per-ply ✓ Windowed ✓ Balance ✓
FTI4: Per-ply × Windowed ✓ Balance ×
FTI5: Per-ply × Windowed ✓ Balance ×

## 4.7 Analysis Information

- Engine: Stockfish 16
- Depth: 20
- Analysis time: 329.2 seconds

# Chapter 5

# Lucius Endzelins vs Hans Berliner

## 5.1 Game Information

| | |
|---|---|
| **Event** | 5th Correspondence World Championship |
| **Site** | ICCF |
| **Date** | 1965.07.01 |
| **Round** | 6 |
| **White** | Lucius Endzelins |
| **Black** | Hans Berliner |
| **Result** | 1/2-1/2 |
| **Opening** | E98 – Unknown |

## 5.2 Player Statistics

### 5.2.1 Lucius Endzelins (White)

- Total moves: 21
- Accuracy: 95.2%
- Average centipawn loss: 9.8
- Best/Excellent moves: 6 / 5
- Good moves: 8
- Inaccuracies: 2
- Mistakes: 0
- Blunders: 0

### 5.2.2 Hans Berliner (Black)

- Total moves: 21
- Accuracy: 95.6%
- Average centipawn loss: 9.1

- Best/Excellent moves: 7 / 4
- Good moves: 9
- Inaccuracies: 1
- Mistakes: 0
- Blunders: 0

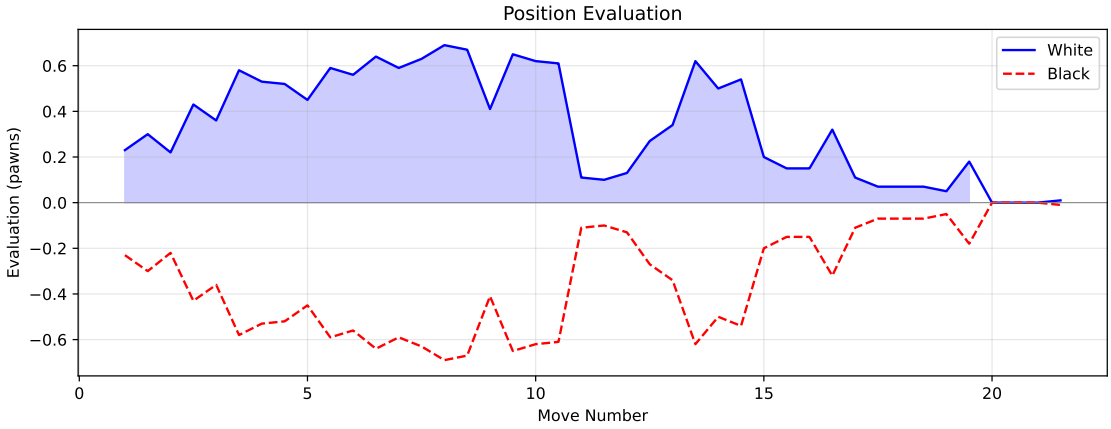
## 5.3 Positional Analysis

This section shows how key positional factors evolved throughout the game.

### 5.3.1 Position Evaluation Over Time

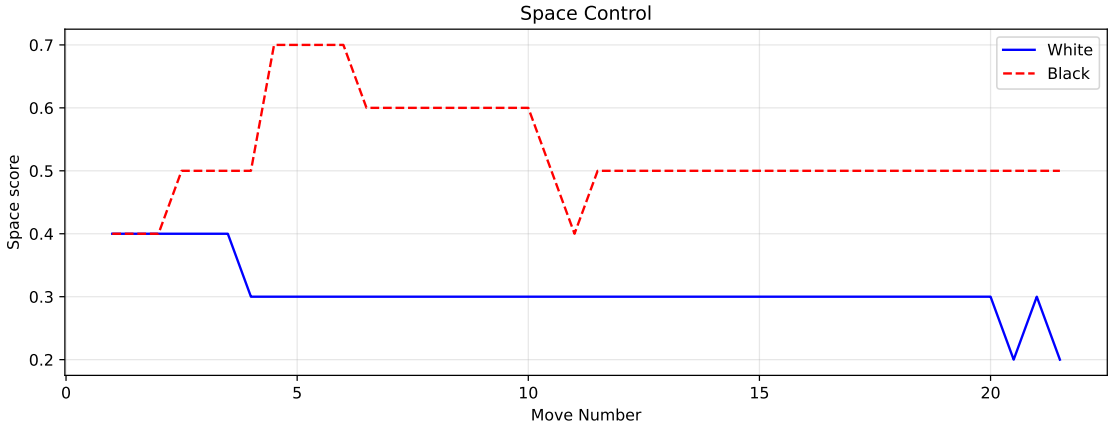
**Game Character Classification.** Evaluation range: minimum $m_1 = +0.00$, maximum $m_2 = +0.69$, spread $d = m_2 - m_1 = 0.69$.

This game is classified as **balanced** (one-sided). A balanced, one-sided game. White maintained the advantage throughout.



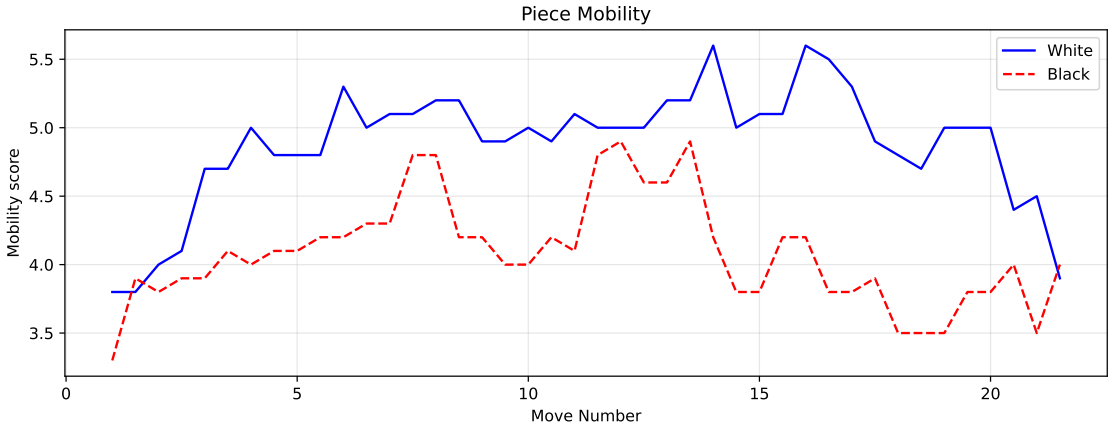

### 5.3.2 Space Control

- White average: 0.31 (range: 0.20 to 0.40)
- Black average: 0.53 (range: 0.40 to 0.70)



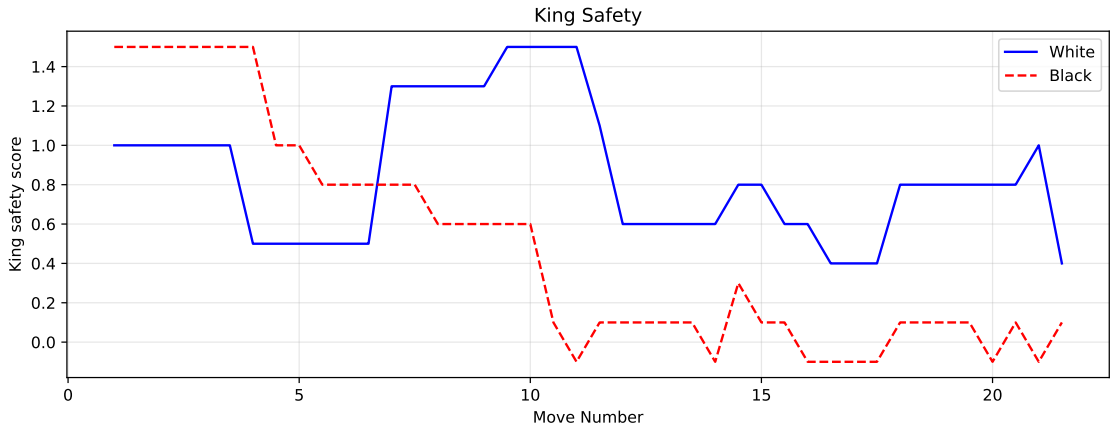

### 5.3.3 Mobility (Piece Activity)

- White average: 4.88 (range: 3.80 to 5.60)
- Black average: 4.08 (range: 3.30 to 4.90)



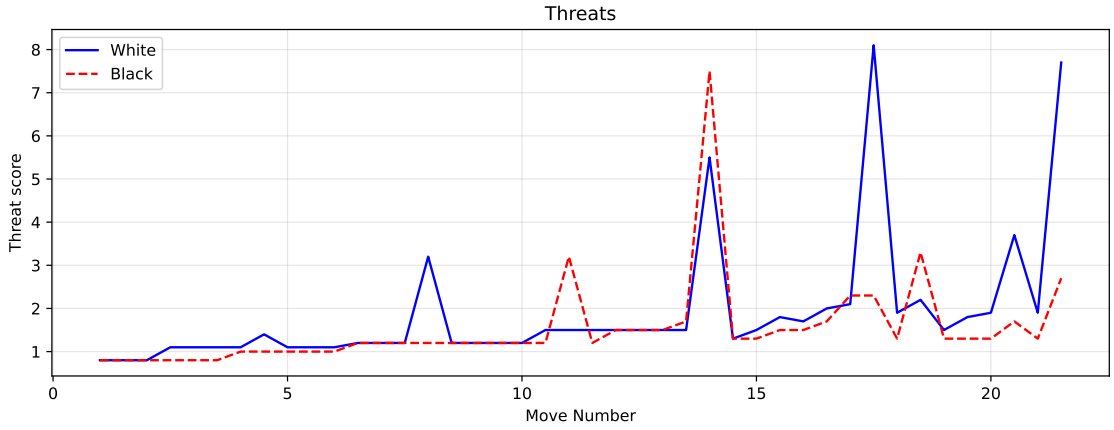

### 5.3.4 King Safety

- White average: 0.85
- Black average: 0.49




### 5.3.5 Threats

- White average: 1.91 (range: 0.80 to 8.10)
- Black average: 1.54 (range: 0.80 to 7.50)
- Average advantage: 0.37 (White maintained more threats)




## 5.4 Annotated Game

1. d4 Nf6 2. c4 g6 3. Nc3 Bg7 4. e4 d6 5. Be2 O-O 6. Nf3 e5 7. O-O Nc6 8. d5! Ne7! 9. Ne1 Nd7 10. Nd3 f5 11. exf5?! Nxf5 12. f3! Nf6 13. Nf2! Bh6?! 14. Bxh6 Nxh6 15. Nfe4?! Nf5 16. Qd2 Nd4 17. Nb5 Nxe2+! 18. Qxe2 a6 19. Nbc3! Bf5 20. Qe3 Bxe4! 21. Nxe4! Nxe4!

1/2-1/2

## 5.5 Critical Positions

This section highlights critical moments where the evaluation shifted substantially—either due to errors or missed opportunities. Each diagram shows the position *after* the move was played.

- **Instead of [move]**: The move(s) that should have been played instead.

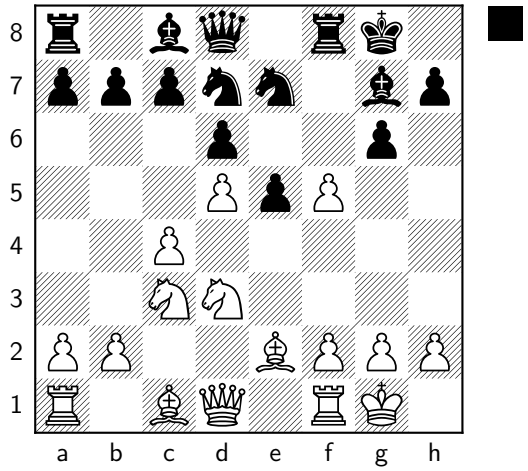
- **Best continuation**: The optimal sequence of moves from the diagrammed position.

Positions marked with ★ represent the biggest evaluation swings in the game. The positional metrics table summarizes the spatial control, piece activity, king safety, and tactical threats for each side.

## Position 1: After 11. exf5 ★

*Biggest evaluation swing(s) (50cp)*

Evaluation: +0.11



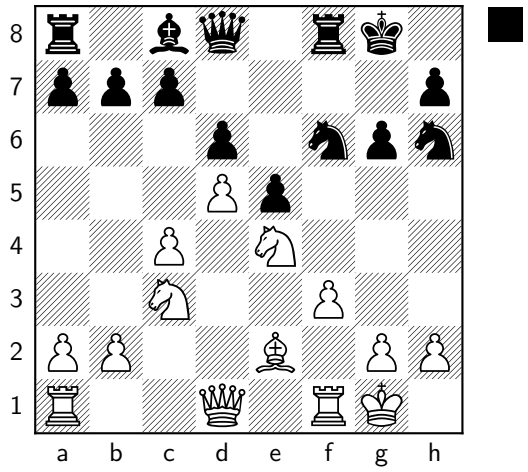
| Metric | White | Black |
|---|---|---|
| Space | 0.30 | 0.40 |
| Mobility | 5.10 | 4.10 |
| King Safety | 1.50 | -0.10 |
| Threats | 1.50 | 3.20 |

Instead of exf5: f3, b4, Kh1 Best continuation: Nxf5 f3 b6

## Position 2: After 15. Nfe4 ★

*Biggest evaluation swing(s) (34cp)*

Evaluation: +0.20



| Metric | White | Black |
|---|---|---|
| Space | 0.30 | 0.50 |
| Mobility | 5.10 | 3.80 |
| King Safety | 0.80 | 0.10 |
| Threats | 1.50 | 1.30 |

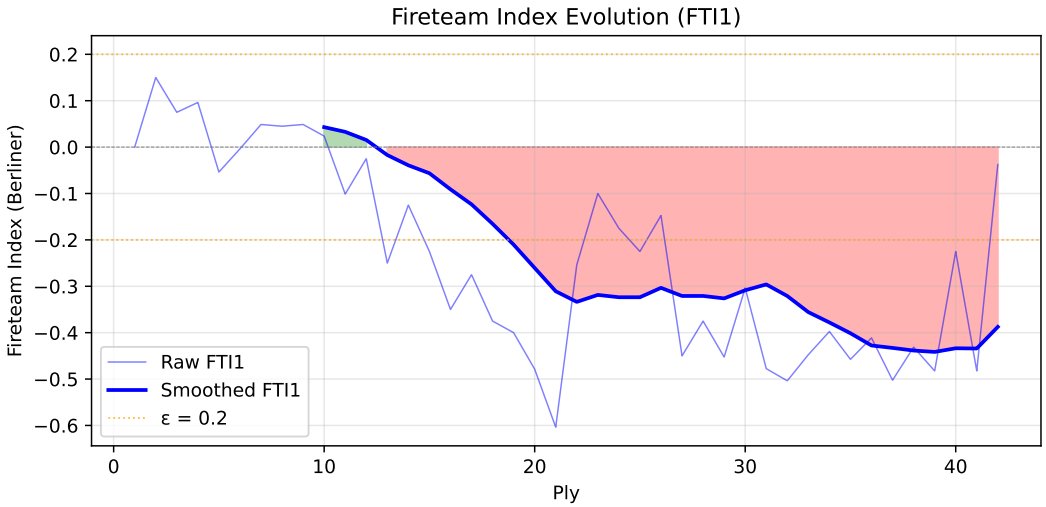
Instead of Nfe4: g4, Qd2, Bd3 Best continuation: Nxe4 Nxe4 Nf5

## 5.6 Fireteam Index Prediction

Tracking: **Berliner** (Black)

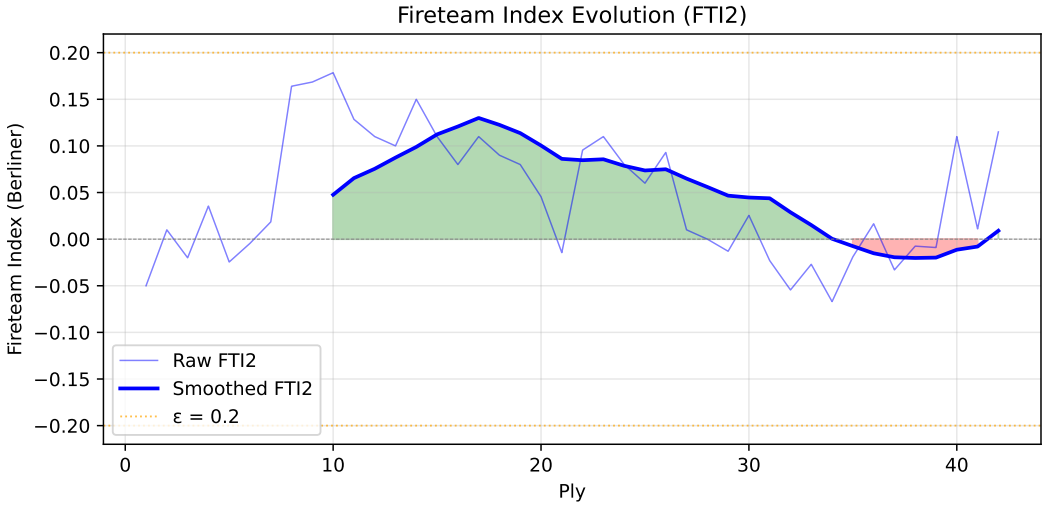
**FTI1: Harmonious** $(\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4})$

| **Algorithm** | **Prediction** | **Max Streak / Confidence** |
|---|---|---|
| Per-ply | DRAW | 0 plies |
| Windowed | DRAW | 0 plies |
| Balance | LOSS | $B_n$=-0.85, C=0.85 |



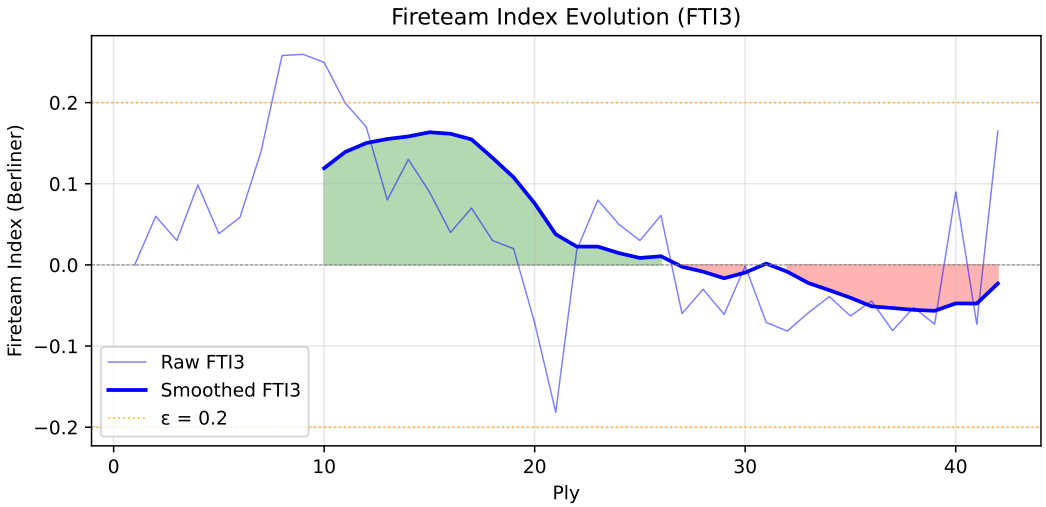

**FTI2: Tactical** $(0.6, 0.1, 0, 0.3)$

| Algorithm | Prediction | Max Streak / Confidence |
|---|---|---|
| Per-ply | DRAW | 0 plies |
| Windowed | DRAW | 0 plies |
| Balance | DRAW | $B_n$=+0.00, C=0.00 |



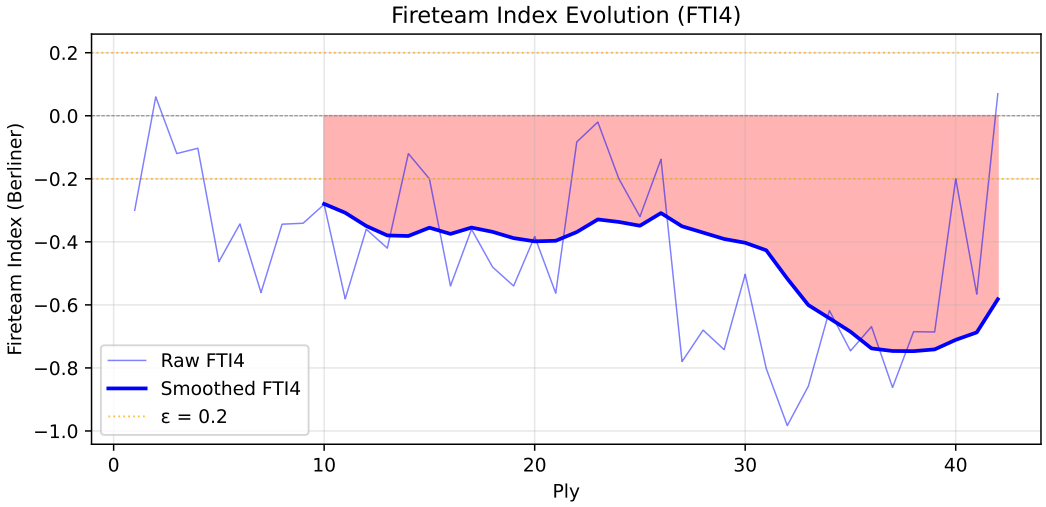

**FTI3: Strategic** $(0.7, 0.1, 0.1, 0.1)$

| Algorithm | Prediction | Max Streak / Confidence |
|---|---|---|
| Per-ply | DRAW | 0 plies |
| Windowed | DRAW | 0 plies |
| Balance | DRAW | $B_n$=+0.00, C=0.00 |



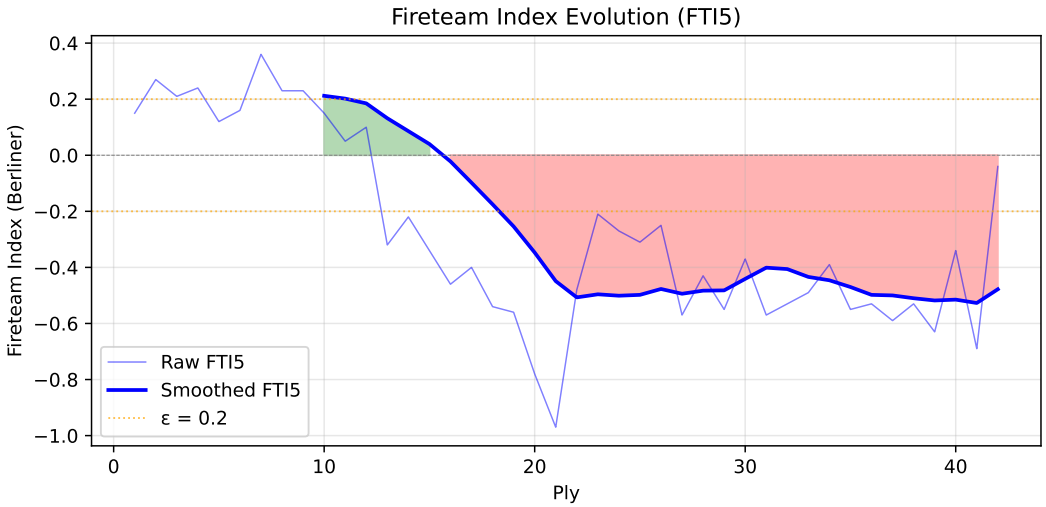

**FTI4: Dynamic** $(0.2, 0.6, 0, 0.2)$

| Algorithm | Prediction | Max Streak / Confidence |
|---|---|---|
| Per-ply | DRAW | 0 plies |
| Windowed | DRAW | 0 plies |
| Balance | LOSS | $B_n$=-0.77, C=0.77 |




**FTI5: Solid** $(0.3, 0.2, 0.5, 0)$

| Algorithm | Prediction | Max Streak / Confidence |
|---|---|---|
| Per-ply | DRAW | 0 plies |
| Windowed | DRAW | 0 plies |
| Balance | LOSS | $B_n$=-0.96, C=0.96 |




Actual result: 1/2-1/2 (Berliner's outcome: DRAW)
FTI1: Per-ply ✓ Windowed ✓ Balance ×
FTI2: Per-ply ✓ Windowed ✓ Balance ✓
FTI3: Per-ply ✓ Windowed ✓ Balance ✓
FTI4: Per-ply ✓ Windowed ✓ Balance ×
FTI5: Per-ply ✓ Windowed ✓ Balance ×

## 5.7 Analysis Information

- Engine: Stockfish 16
- Depth: 20
- Analysis time: 135.1 seconds

# Chapter 6

# Arne Ericson vs Hans Berliner

## 6.1 Game Information

| | |
|---|---|
| **Event** | 5th Correspondence World Championship |
| **Site** | ICCF |
| **Date** | 1965.07.01 |
| **Round** | 14 |
| **White** | Arne Ericson |
| **Black** | Hans Berliner |
| **Result** | 0-1 |
| **Opening** | B04 – Unknown |

## 6.2 Player Statistics

### 6.2.1 Arne Ericson (White)

- Total moves: 52
- Accuracy: 91.1%
- Average centipawn loss: 18.5
- Best/Excellent moves: 15 / 13
- Good moves: 13
- Inaccuracies: 7
- Mistakes: 3
- Blunders: 1

### 6.2.2 Hans Berliner (Black)

- Total moves: 52
- Accuracy: 94.6%
- Average centipawn loss: 11.1

- Best/Excellent moves: 24 / 12
- Good moves: 11
- Inaccuracies: 4
- Mistakes: 1
- Blunders: 0

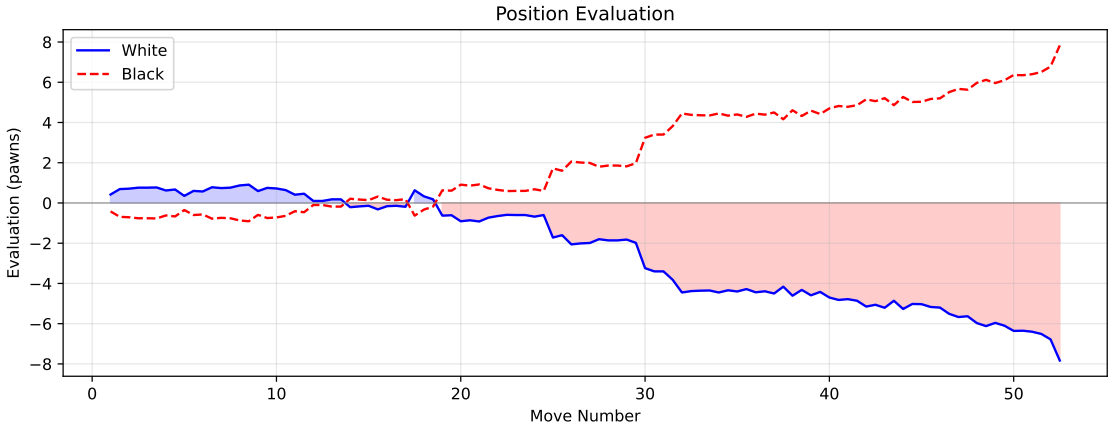
## 6.3 Positional Analysis

This section shows how key positional factors evolved throughout the game.

### 6.3.1 Position Evaluation Over Time

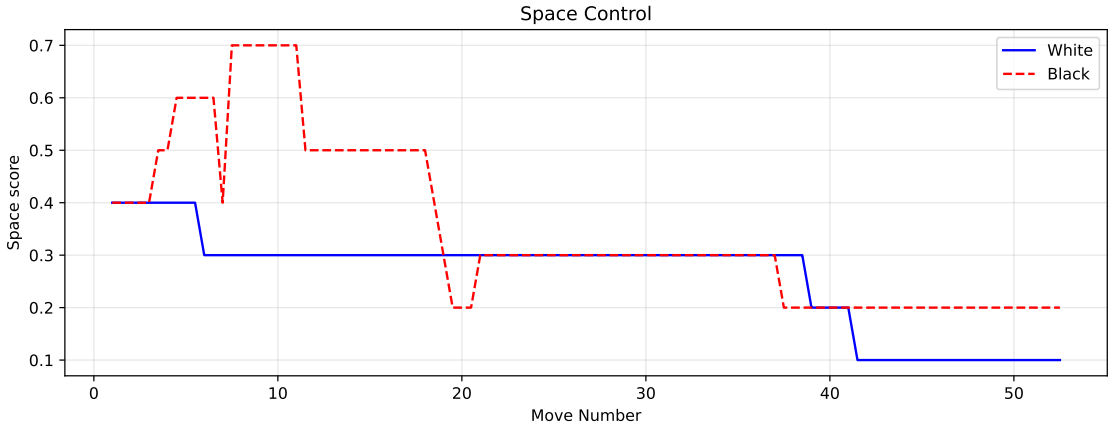
**Game Character Classification.** Evaluation range: minimum $m_1 = -7.83$, maximum $m_2 = +0.91$, spread $d = m_2 - m_1 = 8.74$.

This game is classified as **chaotic** (normal). A chaotic game with moderate swings.




### 6.3.2 Space Control

- White average: 0.26 (range: 0.10 to 0.40)
- Black average: 0.35 (range: 0.20 to 0.70)



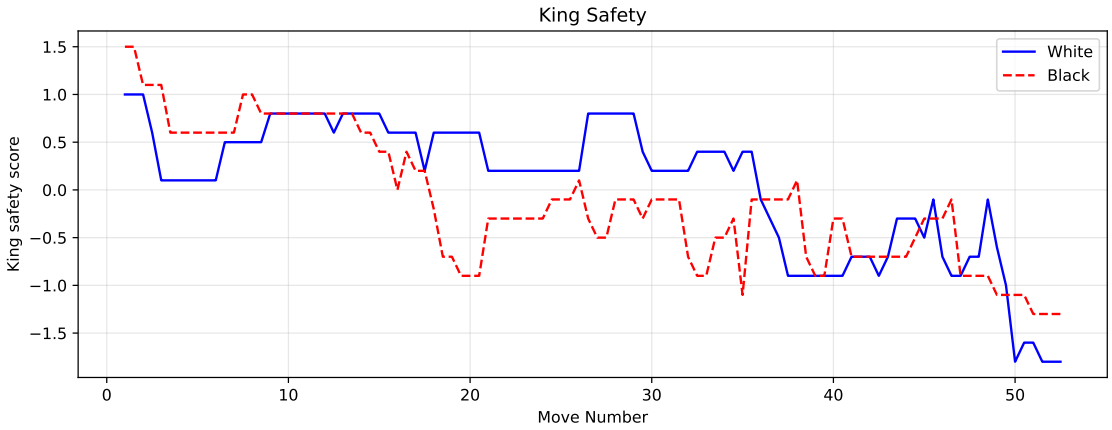
### 6.3.3 Mobility (Piece Activity)

- White average: 3.97 (range: 2.20 to 6.00)
- Black average: 4.17 (range: 2.40 to 5.80)



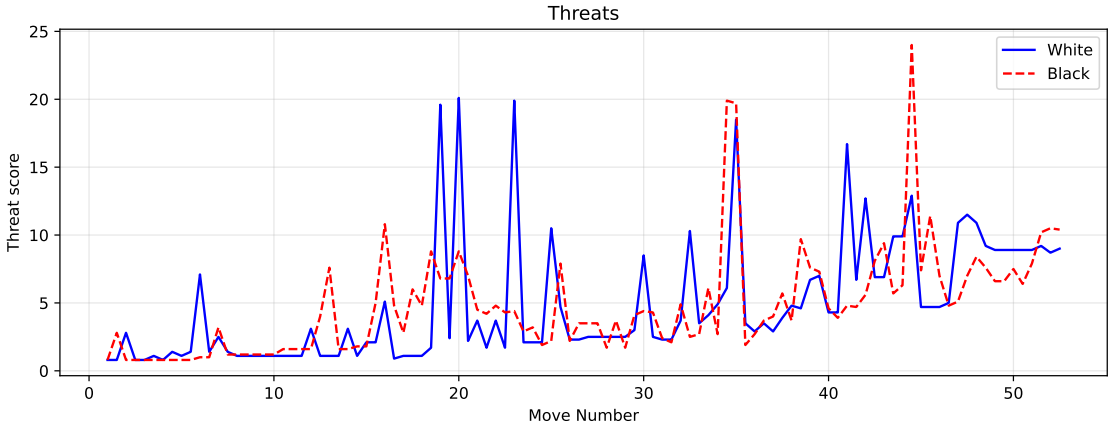

### 6.3.4 King Safety

- White average: 0.06
- Black average: -0.12




### 6.3.5 Threats

- White average: 4.87 (range: 0.80 to 20.10)
- Black average: 4.84 (range: 0.80 to 24.00)
- Average advantage: 0.02 (threats were balanced)




## 6.4 Annotated Game

1. e4 Nf6?! 2. e5 Nd5 3. d4 d6 4. Nf3 g6 5. Be2?! Bg7?! 6. c4 Nb6 7. exd6! cxd6! 8. h3! O-O 9. O-O Nc6 10. Nc3 Bf5 11. Be3?! d5! 12. c5?! Nc4! 13. Bxc4 dxc4! 14. d5?! Nb4! 15. Bd4! Bd3! 16. Bxg7! Kxg7 17. Re1 Qc7? *Black loses 87cp. Consider instead: Rc8, Nxd5, Nc2.*

18. Re5?! f6 19. d6? *White loses 72cp. Consider instead: a3.*

19...exd6! 20. cxd6?! Qxd6 21. Rb5! b6 22. a3 Na6 23. Rd5! Qe7! 24. Ne1 Bf5 25. Qa4? *White loses 112cp. Consider instead: Nc2, Nf3, Rd4.*

25...Nc5! 26. Qxc4 Be6! 27. Qe2 Rfe8 28. Rdd1 Nb3 29. Rab1! Bf5! 30. Qb5?? *White loses 129cp. Consider instead: Qxe7+.*

30...Qf7! 31. Nd3! Rad8! 32. Nf4? *White loses 77cp. Consider instead: a4, f4, Kh2.*

32...Bxb1 33. Rxb1! Re5! 34. Qc6 Qd7 35. Qxd7+! Rxd7 36. g3 Rd2 37. Kg2 g5?! 38. Nh5+ Kg6 39. g4! Nc5 40. Ng3! Nd3! 41. Nge4! Rxb2 42. Rd1 Nf4+! 43. Kf1 Ne2?! 44. a4?! Nxc3 45. Nxc3! Rb3 46. Ne2! Rxh3! 47. Rd7 Ra3 48. Ng3 Rxa4! 49. f3 Ra1+ 50. Kg2 Ra2+! 51. Kh3 Re3 52. Rc7 Rb3

0-1

## 6.5 Critical Positions

This section highlights critical moments where the evaluation shifted substantially—either due to errors or missed opportunities. Each diagram shows the position *after* the move was played.

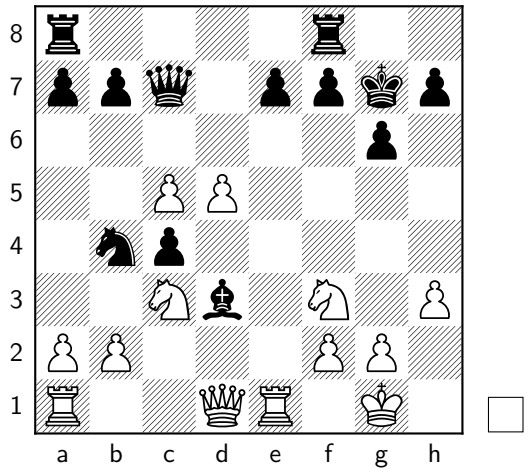
- **Instead of [move]**: The move(s) that should have been played instead.
- **Best continuation**: The optimal sequence of moves from the diagrammed position.

Positions marked with ★ represent the biggest evaluation swings in the game. The positional metrics table summarizes the spatial control, piece activity, king safety, and tactical threats for each side.

## Position 1: After 17...Qc7 ★

*Biggest evaluation swing(s) (82cp)*

Evaluation: +0.63



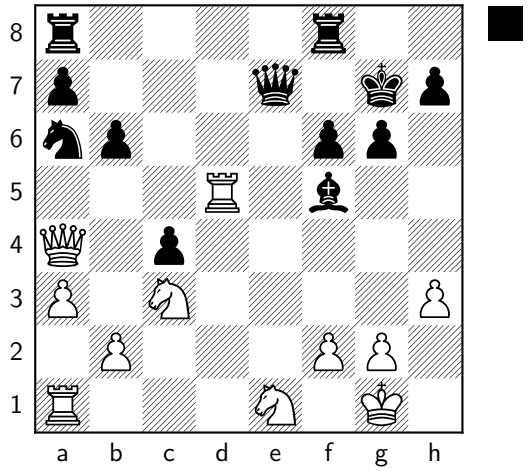
| Metric | White | Black |
|---|---|---|
| Space | 0.30 | 0.50 |
| Mobility | 4.40 | 4.50 |
| King Safety | 0.20 | 0.20 |
| Threats | 1.10 | 6.00 |

Instead of Qc7: Rc8, Nxd5, Nc2 Best continuation: Ne5 Qxc5 Nd7

## Position 2: After 25. Qa4

*Critical moment*

Evaluation: -1.72



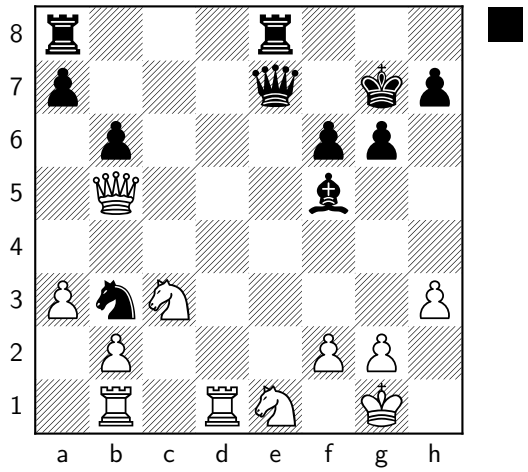
| Metric | White | Black |
|---|---|---|
| Space | 0.30 | 0.30 |
| Mobility | 4.20 | 5.30 |
| King Safety | 0.20 | -0.10 |
| Threats | 10.50 | 2.20 |

Instead of Qa4: Nc2, Nf3, Rd4 Best continuation: Nc5 Qb4 Be6

### Position 3: After 30. Qb5

*White blunders (-126cp)*

Evaluation: -3.24



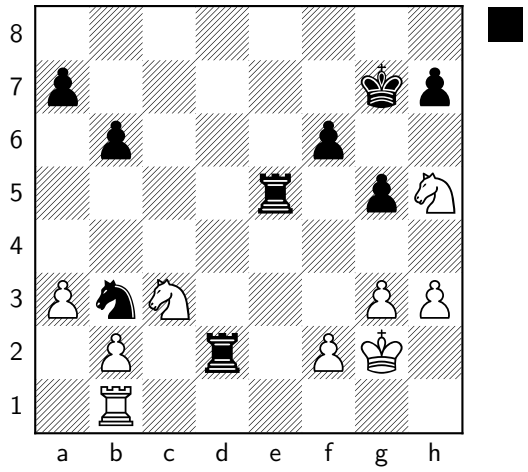
| Metric | White | Black |
|---|---|---|
| Space | 0.30 | 0.30 |
| Mobility | 4.60 | 5.20 |
| King Safety | 0.20 | -0.10 |
| Threats | 8.50 | 4.40 |

Instead of Qb5: Qxe7+ Best continuation: Qf7 Nd3 Rad8

### Position 4: After 38. Nh5+ ★

*Biggest evaluation swing(s) (45cp)*

Evaluation: -4.61



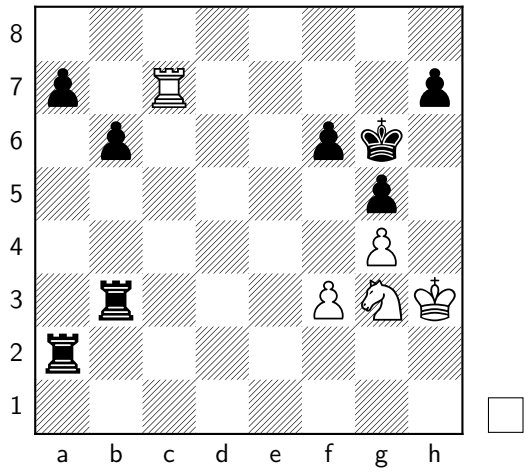
| Metric | White | Black |
|---|---|---|
| Space | 0.30 | 0.20 |
| Mobility | 2.50 | 3.50 |
| King Safety | -0.90 | 0.10 |
| Threats | 4.80 | 3.70 |

Best continuation: Kf7 g4 Nc5

### Position 5: After 52...Rb3

*Critical moment*

Evaluation: -7.83



| Metric | White | Black |
|---|---|---|
| Space | 0.10 | 0.20 |
| Mobility | 2.30 | 3.10 |
| King Safety | -1.80 | -1.30 |
| Threats | 9.00 | 10.40 |

Instead of Rb3: a5, Raa3, Rf2

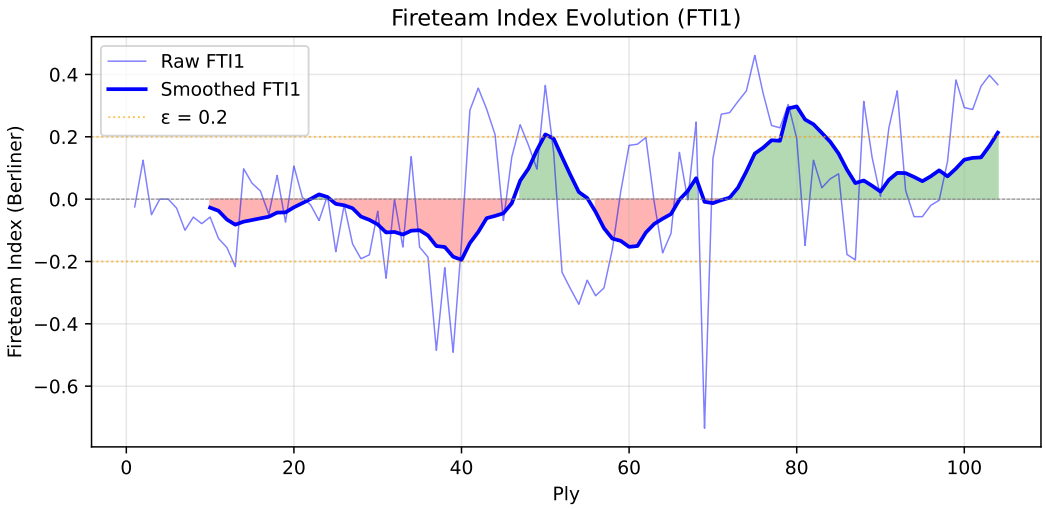
Best continuation: Rc1 Rxf3 Rc3

## 6.6 Fireteam Index Prediction

Tracking: **Berliner** (Black)

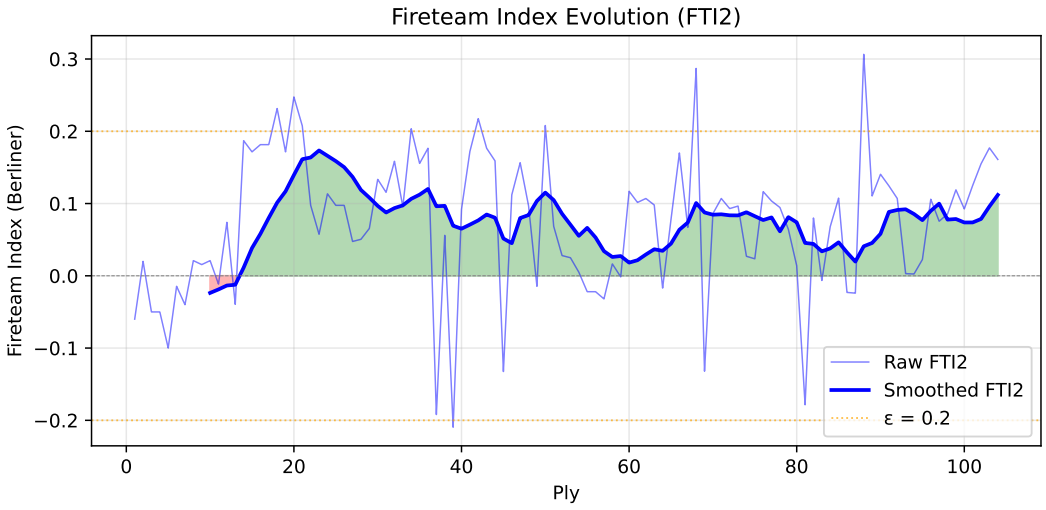
**FTI1: Harmonious** $(\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4})$

| Algorithm | Prediction | Max Streak / Confidence |
|---|---|---|
| Per-ply | DRAW | 6 plies |
| Windowed | DRAW | 3 plies |
| Balance | DRAW | $B_n$=+0.16, C=0.11 |



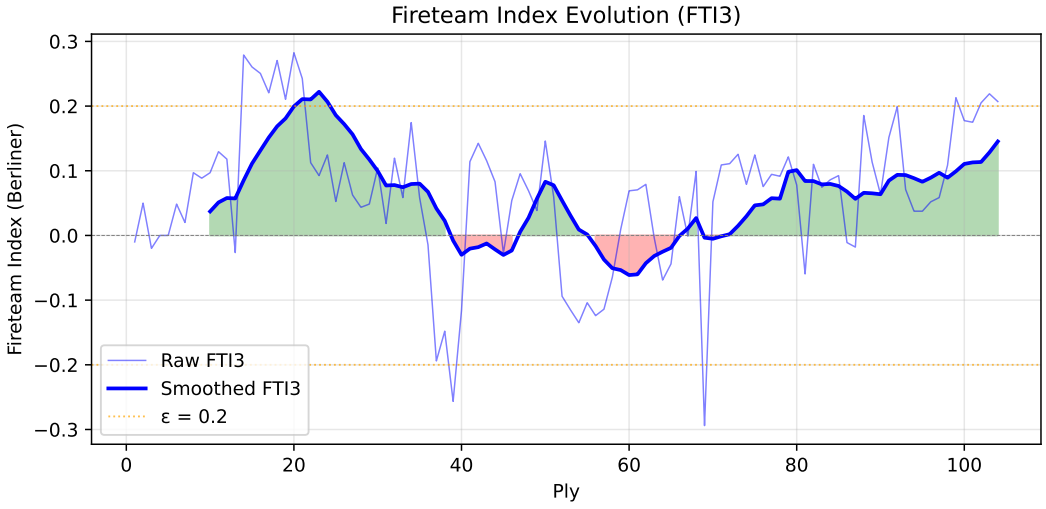

**FTI2: Tactical** $(0.6, 0.1, 0, 0.3)$

| Algorithm | Prediction | Max Streak / Confidence |
|---|---|---|
| Per-ply | DRAW | 1 plies |
| Windowed | DRAW | 0 plies |
| Balance | DRAW | $B_n$=+0.08, C=0.07 |



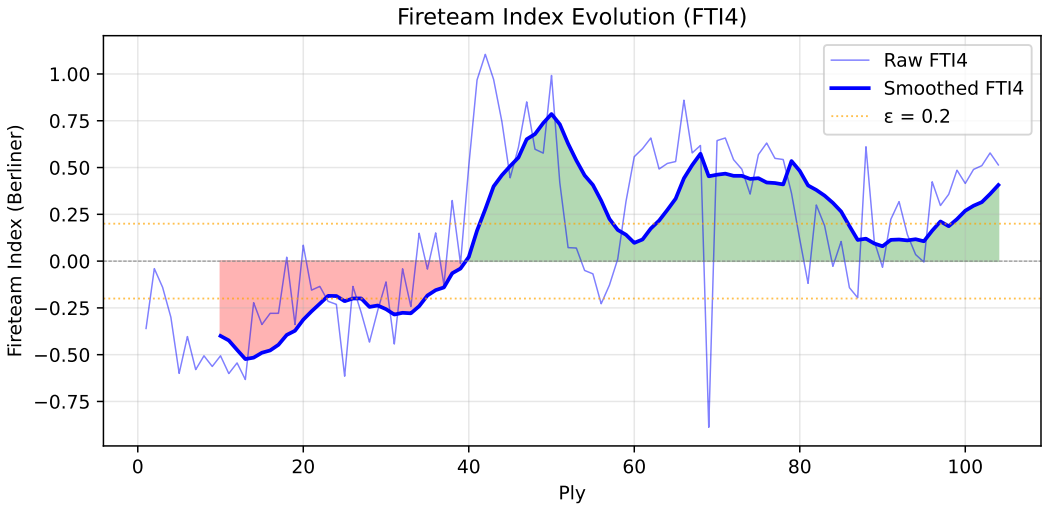

**FTI3: Strategic** $(0.7, 0.1, 0.1, 0.1)$

| Algorithm | Prediction | Max Streak / Confidence |
|---|---|---|
| Per-ply | DRAW | 1 plies |
| Windowed | DRAW | 0 plies |
| Balance | DRAW | $B_n$=+0.08, C=0.07 |



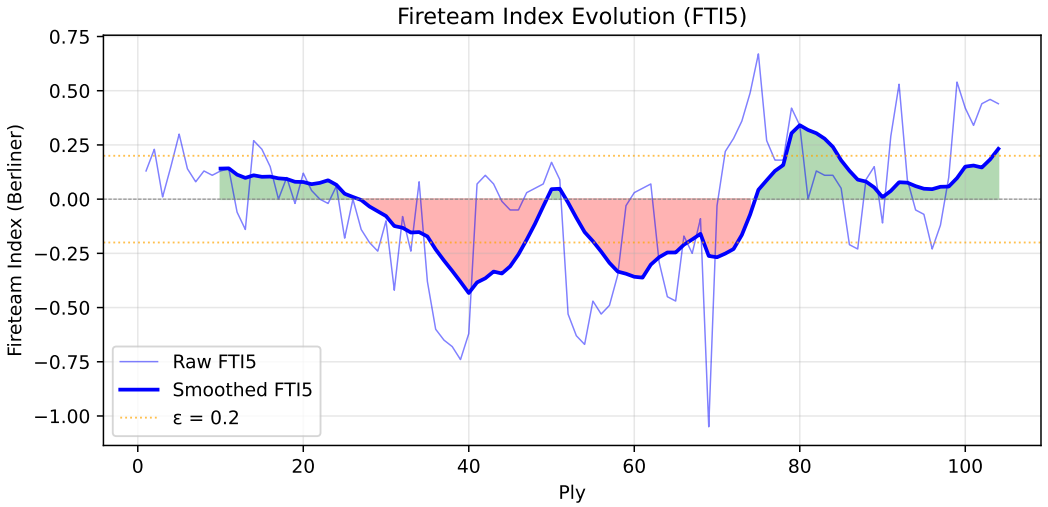

**FTI4: Dynamic** $(0.2, 0.6, 0, 0.2)$

| Algorithm | Prediction | Max Streak / Confidence |
|---|---|---|
| Per-ply | WIN | 12 plies |
| Windowed | WIN | 22 plies |
| Balance | WIN | $B_n$=+0.39, C=0.28 |




**FTI5: Solid** $(0.3, 0.2, 0.5, 0)$

| Algorithm | Prediction | Max Streak / Confidence |
|---|---|---|
| Per-ply | DRAW | 6 plies |
| Windowed | DRAW | 5 plies |
| Balance | DRAW | $B_n$=-0.09, C=0.07 |




Actual result: 0-1 (Berliner's outcome: WIN)

FTI1: Per-ply × Windowed × Balance ×
FTI2: Per-ply × Windowed × Balance ×
FTI3: Per-ply × Windowed × Balance ×
FTI4: Per-ply ✓ Windowed ✓ Balance ✓
FTI5: Per-ply × Windowed × Balance ×

## 6.7 Analysis Information

- Engine: Stockfish 16
- Depth: 20
- Analysis time: 249.5 seconds

# Chapter 7

# Yakov Estrin vs Hans Berliner

## 7.1 Game Information

| | |
|---|---|
| **Event** | 5th Correspondence World Championship |
| **Site** | ICCF |
| **Date** | 1965.07.01 |
| **Round** | 12 |
| **White** | Yakov Estrin |
| **Black** | Hans Berliner |
| **Result** | 0-1 |
| **Opening** | C57 – Unknown |

## 7.2 Player Statistics

### 7.2.1 Yakov Estrin (White)

- Total moves: 42
- Accuracy: 88.1%
- Average centipawn loss: 25.3
- Best/Excellent moves: 14 / 9
- Good moves: 12
- Inaccuracies: 1
- Mistakes: 4
- Blunders: 2

### 7.2.2 Hans Berliner (Black)

- Total moves: 42
- Accuracy: 92.1%
- Average centipawn loss: 16.4

- Best/Excellent moves: 16 / 12
- Good moves: 7
- Inaccuracies: 3
- Mistakes: 3
- Blunders: 1

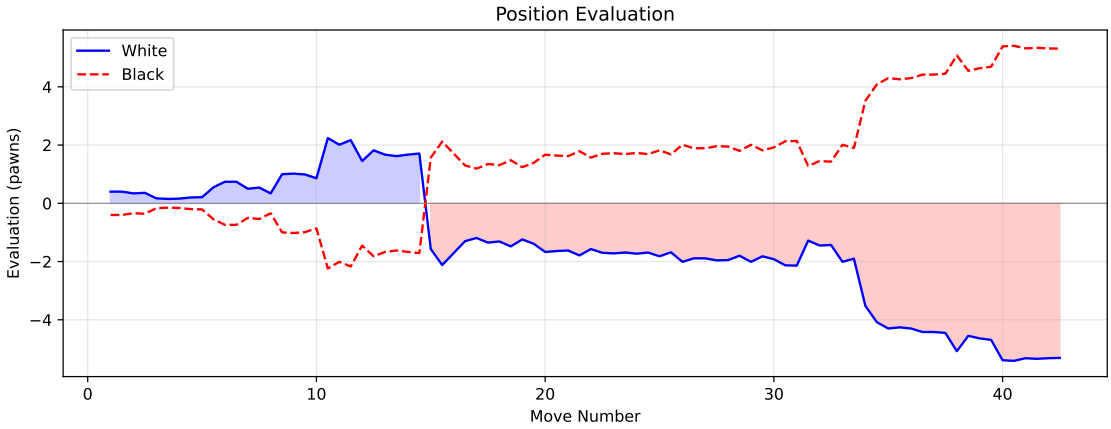
## 7.3 Positional Analysis

This section shows how key positional factors evolved throughout the game.

### 7.3.1 Position Evaluation Over Time

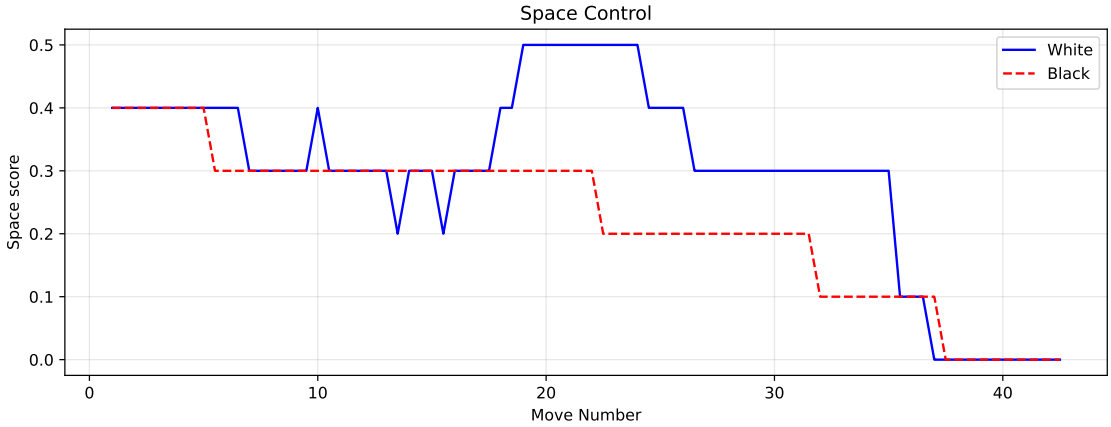
**Game Character Classification.** Evaluation range: minimum $m_1 = -5.41$, maximum $m_2 = +2.24$, spread $d = m_2 - m_1 = 7.65$.

This game is classified as **chaotic** (seesaw). A chaotic seesaw battle with the advantage changing hands.




### 7.3.2 Space Control

- White average: 0.30 (range: 0.00 to 0.50)
- Black average: 0.22 (range: 0.00 to 0.40)



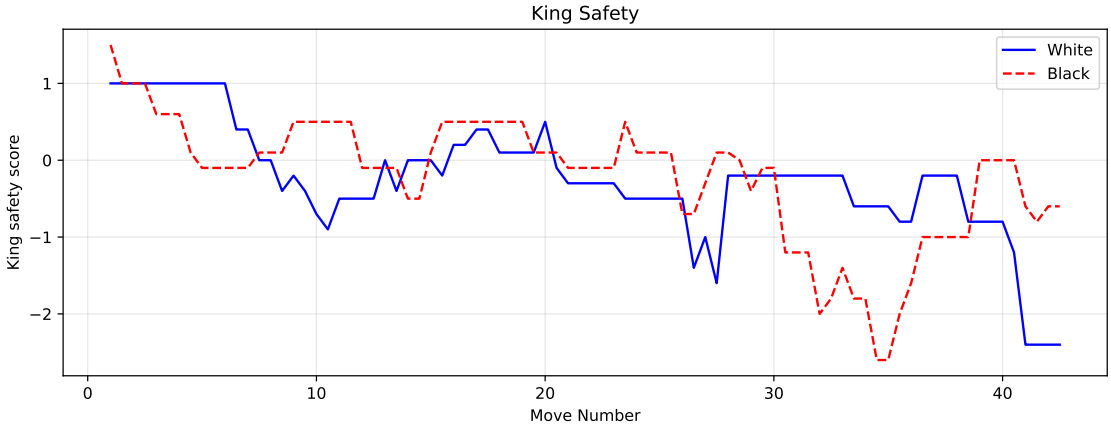

### 7.3.3 Mobility (Piece Activity)

- White average: 3.03 (range: 1.00 to 5.30)
- Black average: 3.34 (range: 1.30 to 5.40)



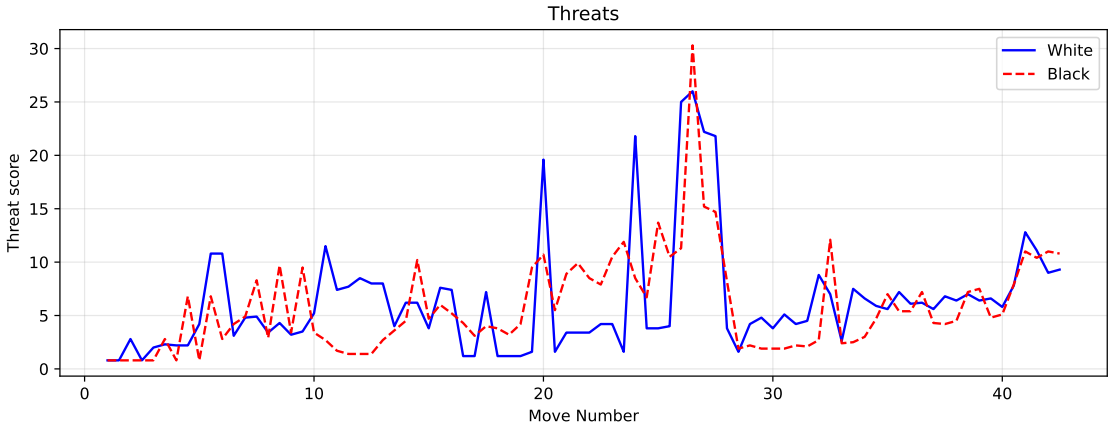

### 7.3.4 King Safety

- White average: -0.25
- Black average: -0.22




### 7.3.5 Threats

- White average: 6.35 (range: 0.80 to 26.00)
- Black average: 5.91 (range: 0.80 to 30.30)
- Average advantage: 0.43 (White maintained more threats)




## 7.4 Annotated Game

1. e4 e5 2. Nf3 Nc6 3. Bc4 Nf6 4. Ng5 d5 5. exd5 b5?! 6. Bf1 Nd4 7. c3! Nxd5! 8. Ne4 Qh4? *Black loses 69cp. Consider instead: Ne6, Nc6.*

9. Ng3! Bg4 10. f3 e4?? *Black loses 139cp. Consider instead: Bd7.*

11. cxd4 Bd6 12. Bxb5+? *White loses 74cp. Consider instead: Qe2.*

12...Kd8?! 13. O-O exf3! 14. Rxf3! Rb8! 15. Be2?? *White loses 330cp. Consider instead: Bf1.*

15...Bxf3! 16. Bxf3! Qxd4+? *Black loses 71cp. Consider instead: Re8.*

17. Kh1! Bxg3! 18. hxg3! Rb6 19. d3! Ne3 20. Bxe3 Qxe3! 21. Bg4 h5! 22. Bh3! g5 23. Nd2 g4 24. Nc4! Qxg3 25. Nxb6 gxh3 26. Qf3 hxg2+ 27. Qxg2! Qxg2+! 28.

Kxg2! cxb6 29. Rf1 Ke7 30. Re1+ Kd6 31. Rf1 Rc8? *Black loses 79cp. Consider instead: Ke6, Ke7.*

32. Rxf7 Rc7 33. Rf2? *White loses 101cp. Consider instead: Rf5.*

33...Ke5 34. a4?? *White loses 143cp. Consider instead: Rf8.*

34...Kd4! 35. a5 Kxd3 36. Rf3+! Kc2! 37. b4?! b5! 38. a6? *White loses 63cp. Consider instead: Rf5.*

38...Rc4?! 39. Rf7 Rxb4! 40. Rb7? *White loses 62cp. Consider instead: Rxa7.*

40...Rg4+! 41. Kf3 b4! 42. Rxa7 b3!

0-1

## 7.5 Critical Positions

This section highlights critical moments where the evaluation shifted substantially—either due to errors or missed opportunities. Each diagram shows the position *after* the move was played.

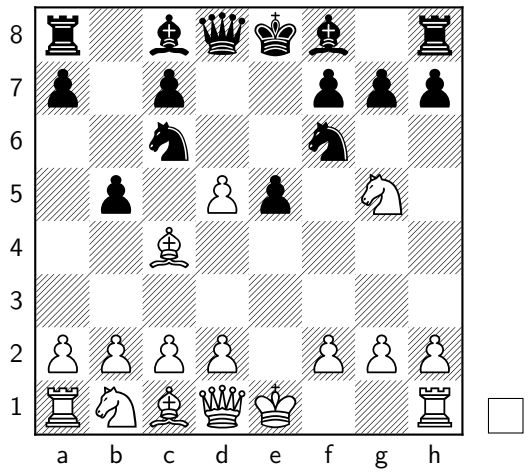
- **Instead of [move]**: The move(s) that should have been played instead.
- **Best continuation**: The optimal sequence of moves from the diagrammed position.

Positions marked with ★ represent the biggest evaluation swings in the game. The positional metrics table summarizes the spatial control, piece activity, king safety, and tactical threats for each side.

### Position 1: After 5...b5 ★

*Biggest evaluation swing(s) (34cp)*

Evaluation: +0.55



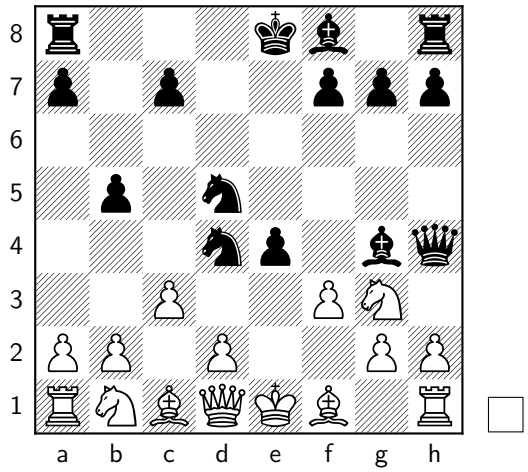
| Metric | White | Black |
|---|---|---|
| Space | 0.40 | 0.30 |
| Mobility | 4.40 | 5.10 |
| King Safety | 1.00 | -0.10 |
| Threats | 10.80 | 6.80 |

Instead of b5: Na5, Nd4 Best continuation: Bf1 Nd4 c3

### Position 2: After 10...e4

*Black blunders (+138cp)*

Evaluation: +2.24



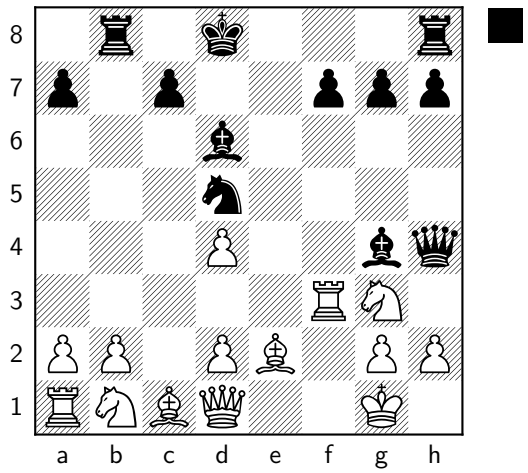
| Metric | White | Black |
|---|---|---|
| Space | 0.30 | 0.30 |
| Mobility | 3.60 | 4.70 |
| King Safety | -0.90 | 0.50 |
| Threats | 11.50 | 2.70 |

Instead of e4: Bd7 Best continuation: cxd4 Bd6 Qe2

### Position 3: After 15. Be2

*White blunders (-328cp)*

Evaluation: -1.57



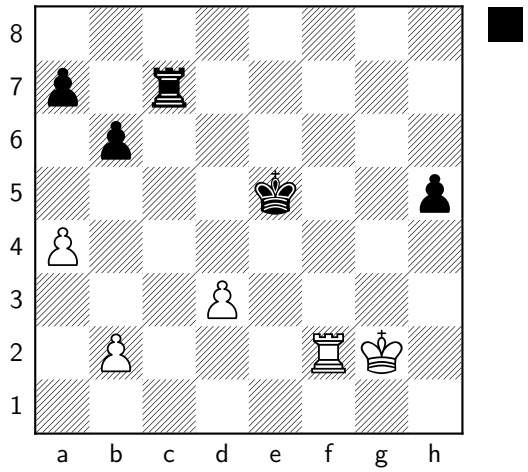
| Metric | White | Black |
|---|---|---|
| Space | 0.30 | 0.30 |
| Mobility | 5.00 | 4.60 |
| King Safety | 0.00 | 0.10 |
| Threats | 3.80 | 4.70 |

Instead of Be2: Bf1 Best continuation: Bxf3 Bxf3 Qxd4+

### Position 4: After 34. a4

*White blunders (-162cp)*

Evaluation: -3.52



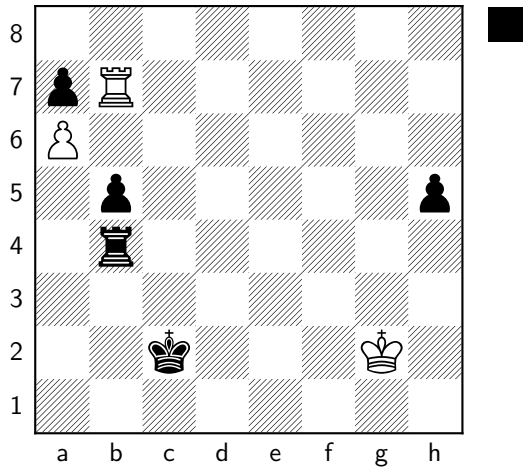
| Metric | White | Black |
|---|---|---|
| Space | 0.30 | 0.10 |
| Mobility | 1.70 | 1.60 |
| King Safety | -0.60 | -1.80 |
| Threats | 6.60 | 3.00 |

Instead of a4: Rf8 Best continuation: Kd4 Rf4+ Kxd3

### Position 5: After 40. Rb7 ★

*Biggest evaluation swing(s) (70cp)*

Evaluation: -5.39



| Metric | White | Black |
|---|---|---|
| Space | 0.00 | 0.00 |
| Mobility | 1.00 | 1.50 |
| King Safety | -0.80 | 0.00 |
| Threats | 5.80 | 5.10 |

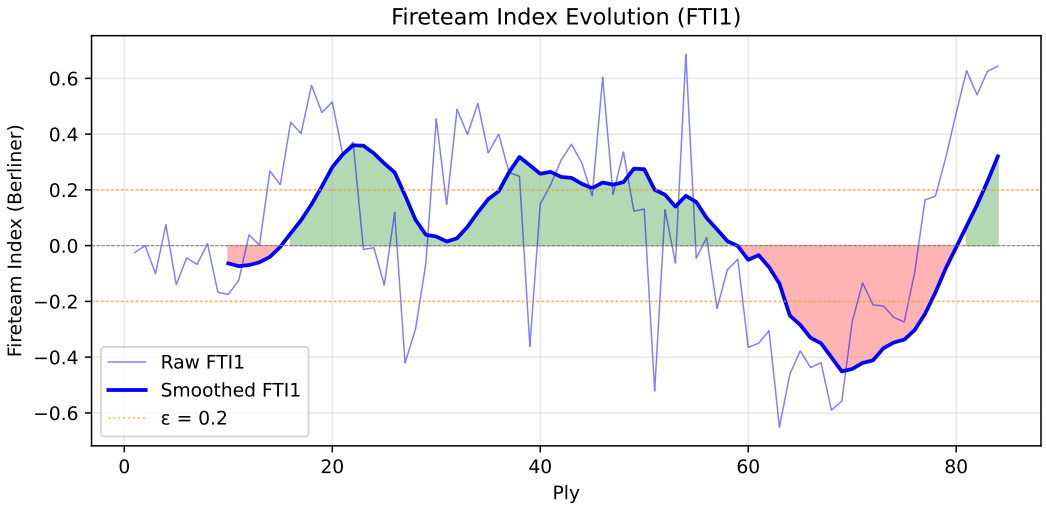
Instead of Rb7: Rxa7 Best continuation: Rg4+ Kf3 b4

## 7.6 Fireteam Index Prediction

Tracking: **Berliner** (Black)

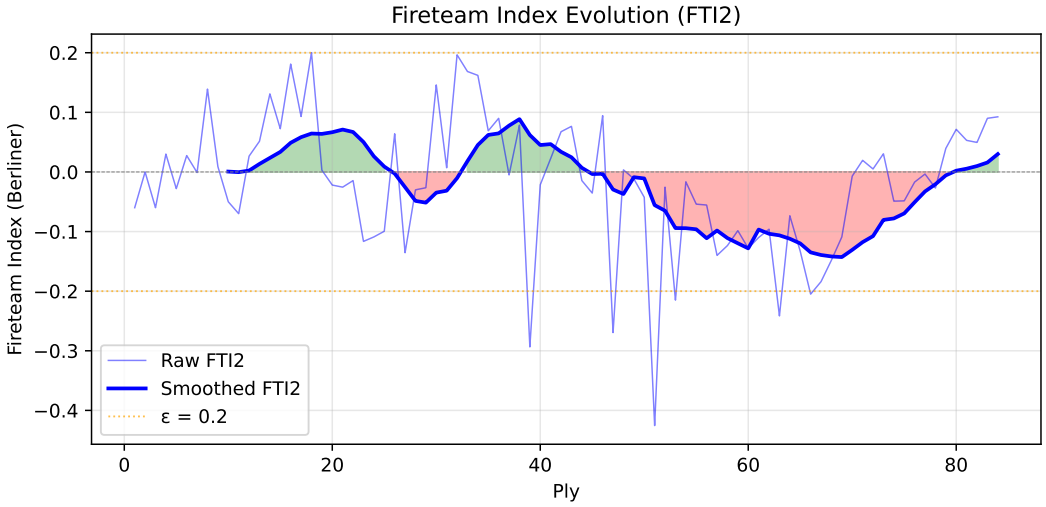
**FTI1: Harmonious** $(\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4})$

| **Algorithm** | **Prediction** | **Max Streak / Confidence** |
|---|---|---|
| Per-ply | DRAW | 6 plies |
| Windowed | DRAW | 7 plies |
| Balance | DRAW | $B_n$=+0.10, C=0.04 |



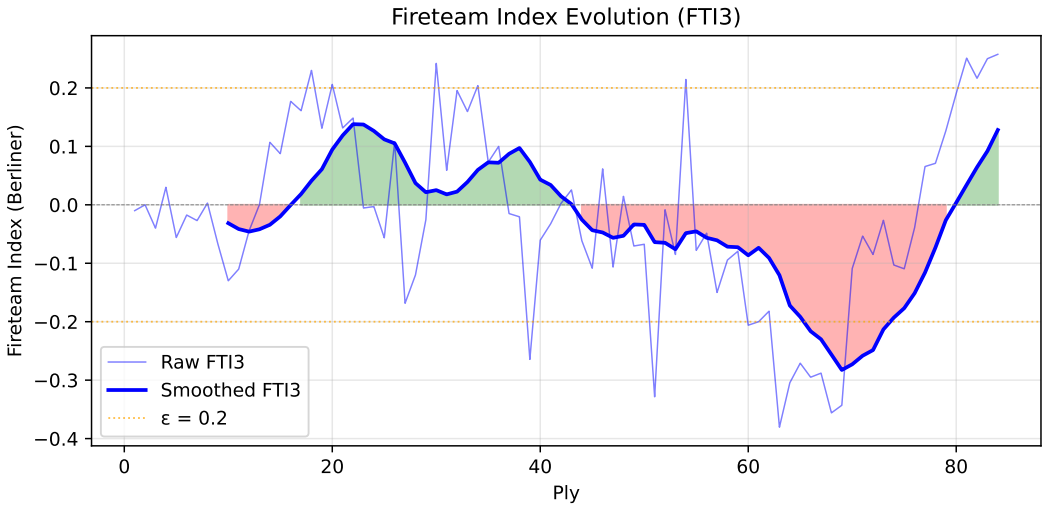

**FTI2: Tactical** $(0.6, 0.1, 0, 0.3)$

| Algorithm | Prediction | Max Streak / Confidence |
|---|---|---|
| Per-ply | DRAW | 0 plies |
| Windowed | DRAW | 0 plies |
| Balance | DRAW | $B_n$=-0.09, C=0.09 |



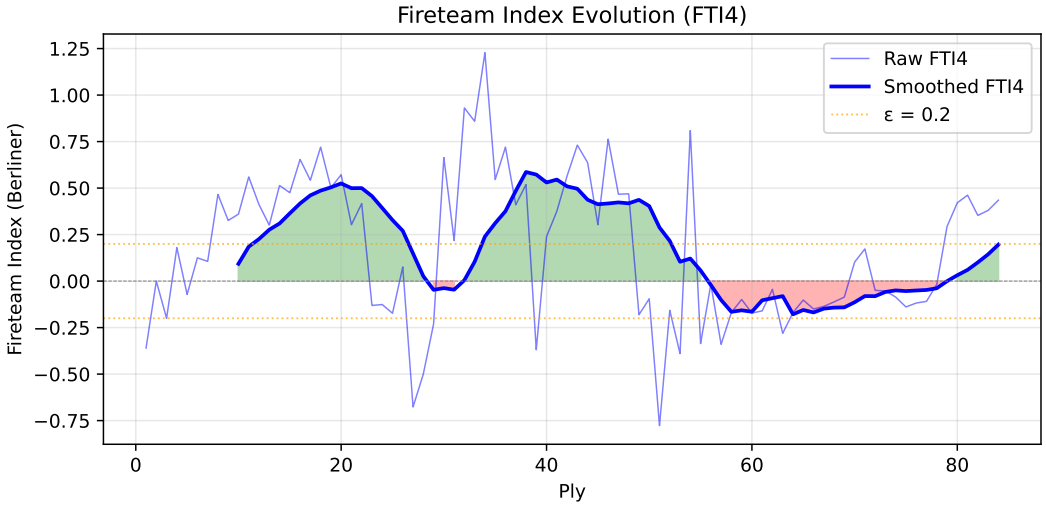

**FTI3: Strategic** $(0.7, 0.1, 0.1, 0.1)$

| Algorithm | Prediction | Max Streak / Confidence |
|---|---|---|
| Per-ply | DRAW | 2 plies |
| Windowed | DRAW | 0 plies |
| Balance | DRAW | $B_n$=-0.01, C=0.01 |



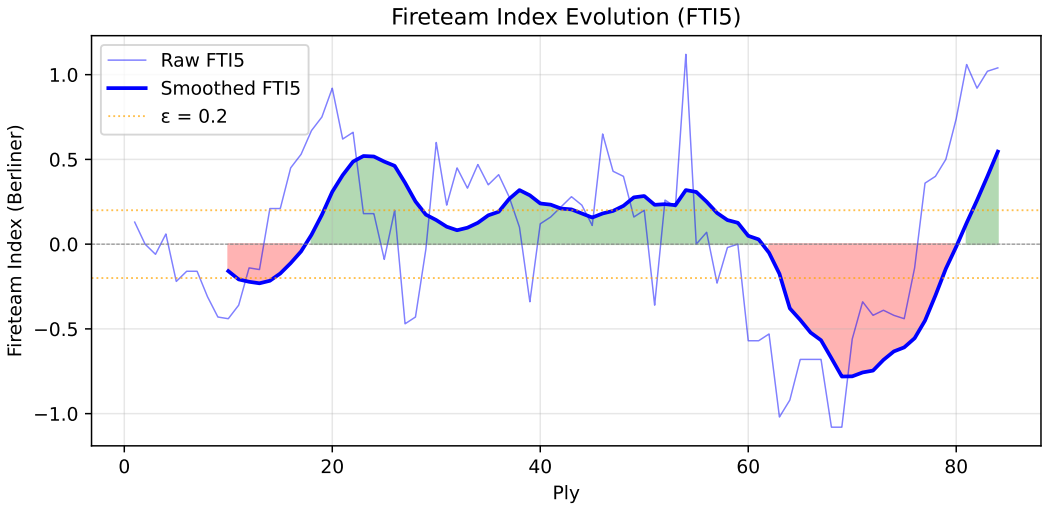

**FTI4: Dynamic** $(0.2, 0.6, 0, 0.2)$

| Algorithm | Prediction | Max Streak / Confidence |
|---|---|---|
| Per-ply | DRAW | 8 plies |
| Windowed | WIN | 17 plies |
| Balance | DRAW | $B_n$=+0.32, C=0.13 |




**FTI5: Solid** $(0.3, 0.2, 0.5, 0)$

| Algorithm | Prediction | Max Streak / Confidence |
|---|---|---|
| Per-ply | DRAW | 8 plies |
| Windowed | DRAW | 9 plies |
| Balance | DRAW | $B_n$=+0.15, C=0.06 |




Actual result: 0-1 (Berliner's outcome: WIN)
FTI1: Per-ply × Windowed × Balance ×
FTI2: Per-ply × Windowed × Balance ×
FTI3: Per-ply × Windowed × Balance ×
FTI4: Per-ply × Windowed ✓ Balance ×
FTI5: Per-ply × Windowed × Balance ×

## 7.7 Analysis Information

- Engine: Stockfish 16
- Depth: 20
- Analysis time: 228.4 seconds

# Chapter 8

# Berliner Hans J (USA) vs Husak Karel (CZE)

## 8.1 Game Information

| | |
|---|---|
| **Event** | Ch World (corr.) |
| **Site** | corr. |
| **Date** | 1965.??.?? |
| **Round** | 2 |
| **White** | Berliner Hans J (USA) |
| **Black** | Husak Karel (CZE) |
| **Result** | 1/2-1/2 |
| **Opening** | D87 – Unknown |

## 8.2 Player Statistics

### 8.2.1 Berliner Hans J (USA) (White)

- Total moves: 33
- Accuracy: 96.5%
- Average centipawn loss: 7.2
- Best/Excellent moves: 20 / 3
- Good moves: 8
- Inaccuracies: 1
- Mistakes: 1
- Blunders: 0

### 8.2.2 Husak Karel (CZE) (Black)

- Total moves: 33
- Accuracy: 97.1%

- Average centipawn loss: 5.8
- Best/Excellent moves: 19 / 5
- Good moves: 6
- Inaccuracies: 3
- Mistakes: 0
- Blunders: 0

## 8.3 Positional Analysis

This section shows how key positional factors evolved throughout the game.

### 8.3.1 Position Evaluation Over Time

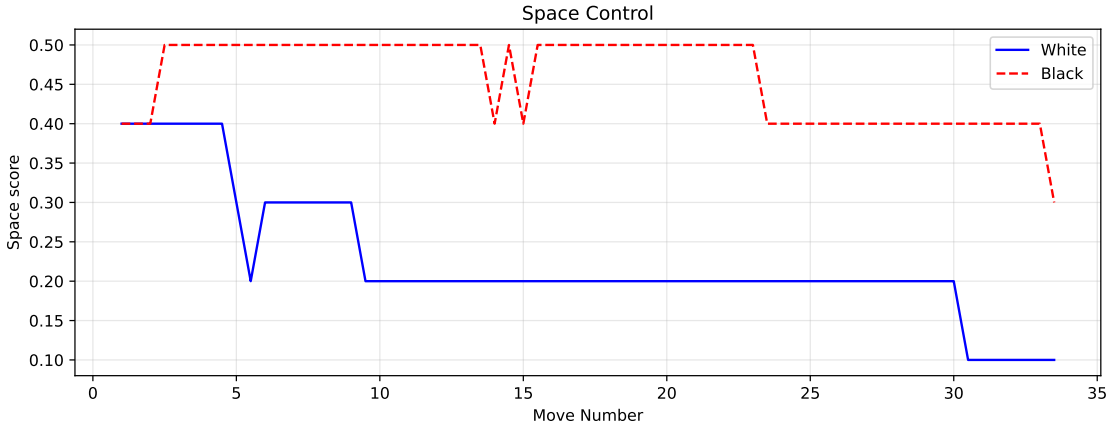
**Game Character Classification.** Evaluation range: minimum $m_1 = -0.30$, maximum $m_2 = +0.96$, spread $d = m_2 - m_1 = 1.26$.

This game is classified as **tense** (one-sided). A tense, one-sided game. White maintained the advantage throughout.




### 8.3.2 Space Control

- White average: 0.23 (range: 0.10 to 0.40)
- Black average: 0.46 (range: 0.30 to 0.50)



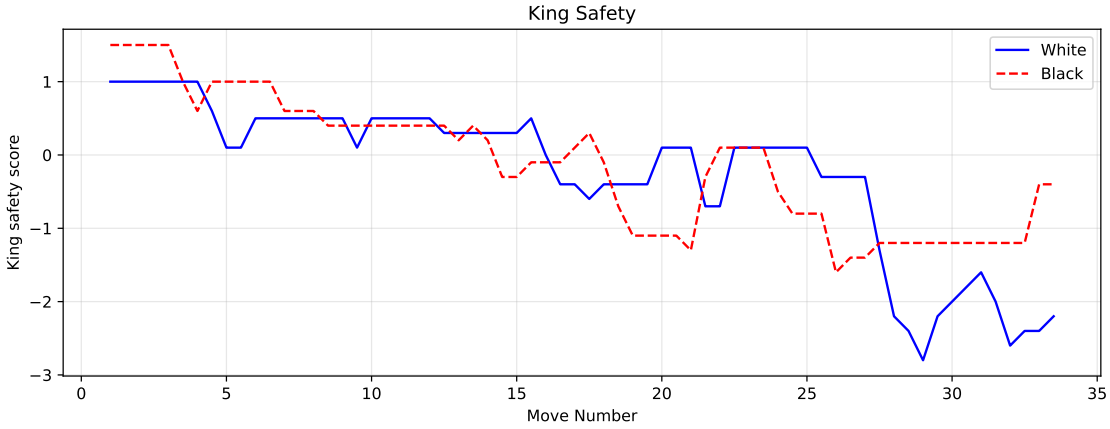

### 8.3.3 Mobility (Piece Activity)

- White average: 3.92 (range: 2.20 to 5.40)
- Black average: 3.69 (range: 2.30 to 4.60)



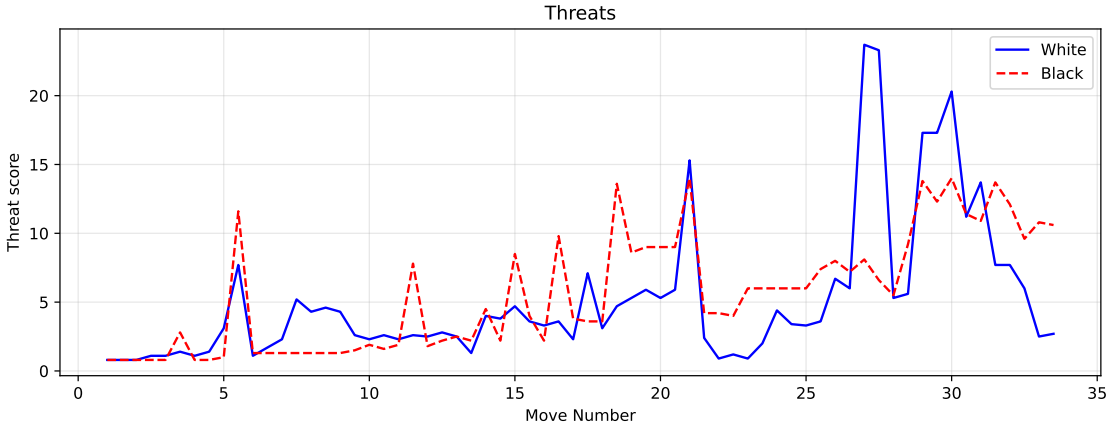

### 8.3.4 King Safety

- White average: -0.25
- Black average: -0.12




### 8.3.5 Threats

- White average: 5.17 (range: 0.80 to 23.70)
- Black average: 5.64 (range: 0.80 to 14.00)
- Average advantage: -0.47 (Black maintained more threats)




## 8.4 Annotated Game

1. d4 Nf6 2. c4 g6 3. Nc3 d5 4. cxd5 Nxd5 5. e4 Nxc3 6. bxc3 Bg7 7. Bc4 c5! 8. Ne2 O-O 9. h4 cxd4 10. cxd4! Nc6! 11. Be3! Na5! 12. Bd5 Qd6?! 13. h5! Be6?! 14. hxg6? *White loses 84cp. Consider instead: h6, Bxe6, Bf4.*

14...hxg6 15. Bxe6! Qxe6 16. f3?! Nc4?! 17. Bh6! Bxh6! 18. Rxh6! Kg7 19. Qc1! Qa6 20. Kf2! Rh8! 21. Rxh8 Rxh8! 22. Qc3 Rc8! 23. Rc1 b5 24. d5+ f6 25. Qd4! Qa3! 26. Rh1 Ne5 27. Qe3! Qxe3+! 28. Kxe3! Rc2! 29. Nd4! Nc4+ 30. Kd3! Rxg2! 31. Nxb5! Ne5+! 32. Ke3! Rxa2! 33. Rc1! f5

1/2-1/2

## 8.5 Critical Positions

This section highlights critical moments where the evaluation shifted substantially—either due to errors or missed opportunities. Each diagram shows the position *after* the move was played.

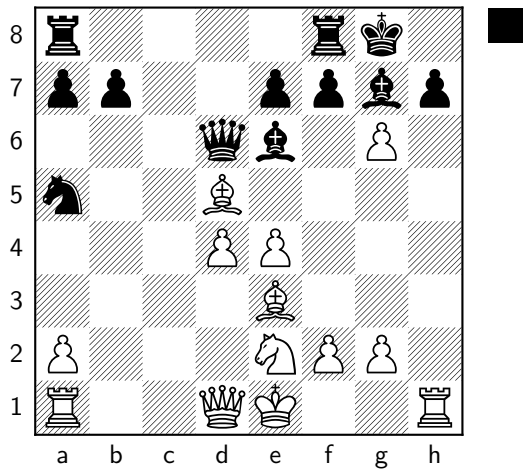
- **Instead of [move]**: The move(s) that should have been played instead.
- **Best continuation**: The optimal sequence of moves from the diagrammed position.

Positions marked with ★ represent the biggest evaluation swings in the game. The positional metrics table summarizes the spatial control, piece activity, king safety, and tactical threats for each side.

### Position 1: After 14. hxg6 ★

*Biggest evaluation swing(s) (89cp)*

Evaluation: +0.07



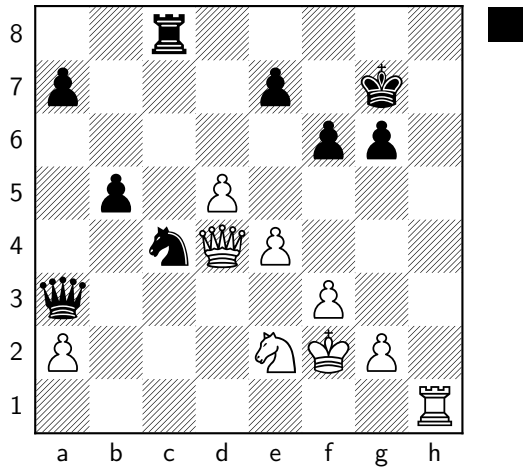
| Metric | White | Black |
|---|---|---|
| Space | 0.20 | 0.40 |
| Mobility | 5.30 | 4.40 |
| King Safety | 0.30 | 0.20 |
| Threats | 4.00 | 4.50 |

Instead of hxg6: h6, Bxe6, Bf4 Best continuation: hxg6 Bxe6 Qxe6

### Position 2: After 26. Rh1 ★

*Biggest evaluation swing(s) (30cp)*

Evaluation: -0.30



| Metric | White | Black |
|---|---|---|
| Space | 0.20 | 0.40 |
| Mobility | 3.90 | 4.00 |
| King Safety | -0.30 | -1.60 |
| Threats | 6.70 | 8.00 |

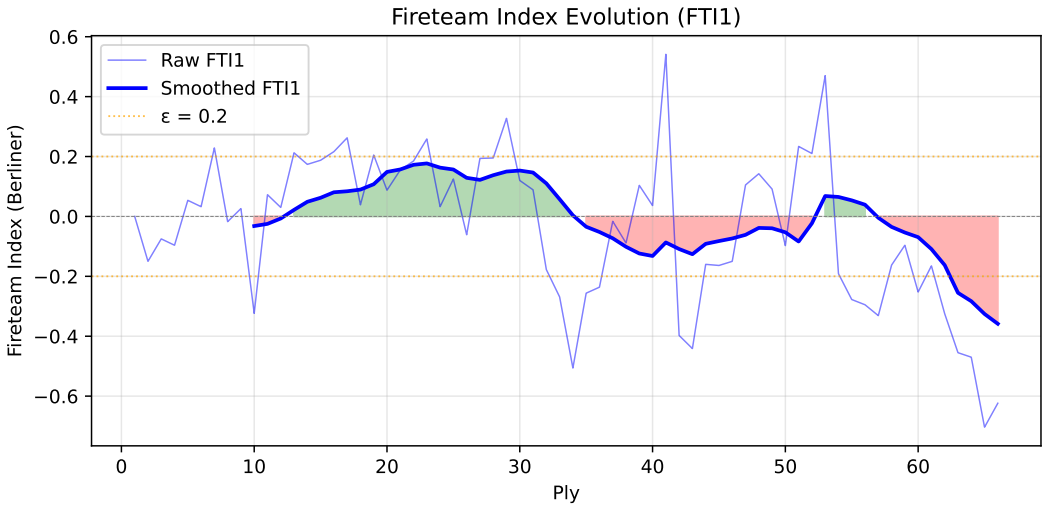
Instead of Rh1: f4, Rc2, Qc3 Best continuation: a6 Rc1 Rh8

## 8.6 Fireteam Index Prediction

Tracking: **Berliner** (White)

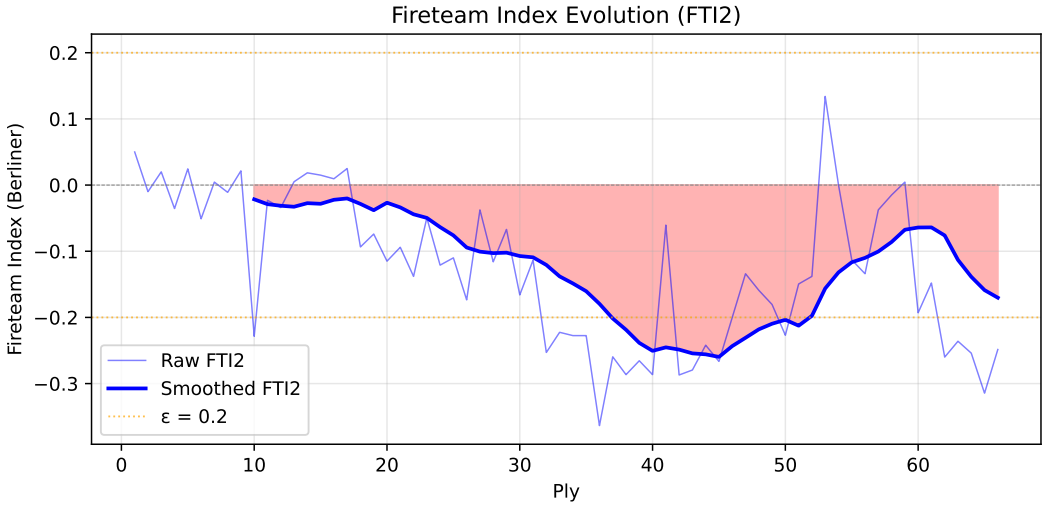
**FTI1: Harmonious** $(\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4})$

| Algorithm | Prediction | Max Streak / Confidence |
|---|---|---|
| Per-ply | DRAW | 1 plies |
| Windowed | DRAW | 0 plies |
| Balance | DRAW | $B_n$=-0.14, C=0.07 |



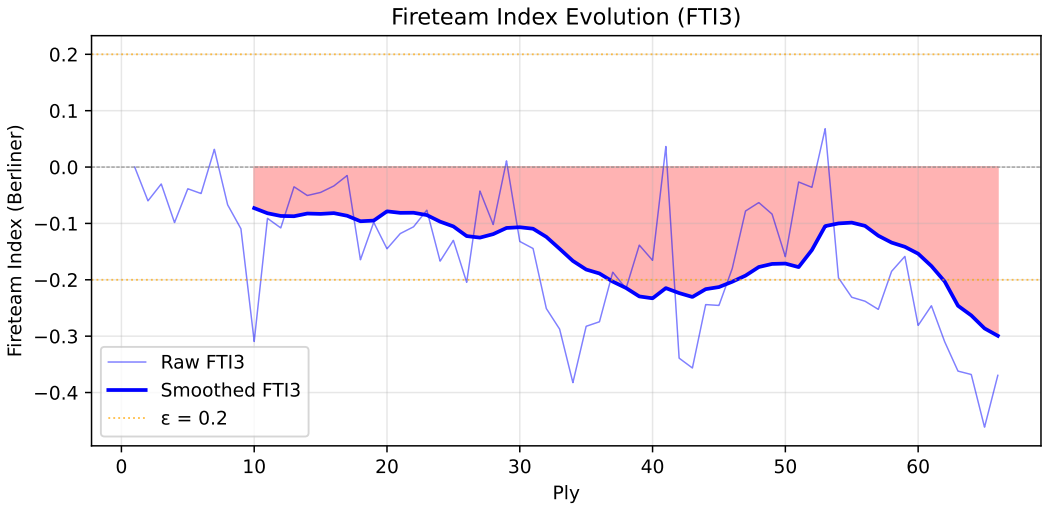

**FTI2: Tactical** $(0.6, 0.1, 0, 0.3)$

| Algorithm | Prediction | Max Streak / Confidence |
|---|---|---|
| Per-ply | DRAW | 0 plies |
| Windowed | DRAW | 0 plies |
| Balance | LOSS | $B_n$=-0.38, C=0.38 |



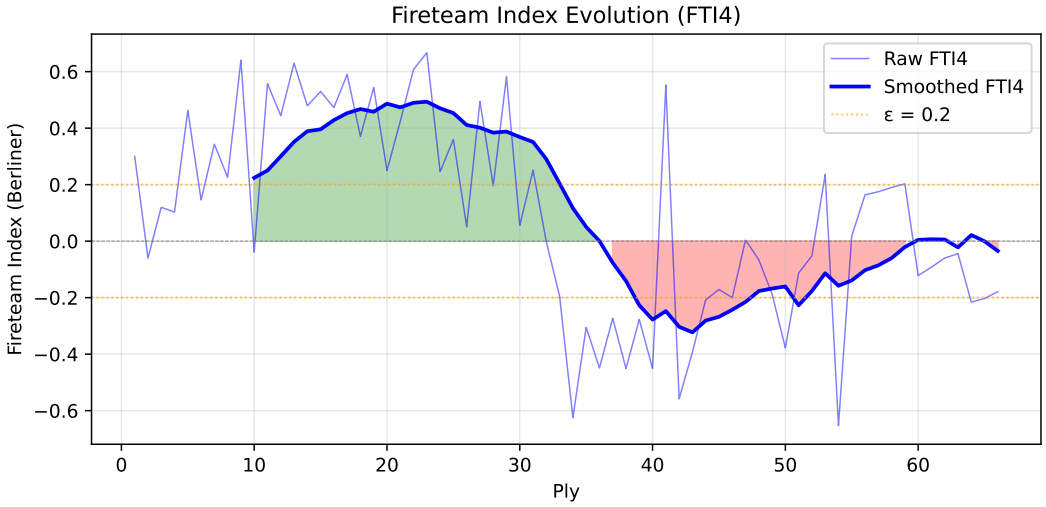

**FTI3: Strategic** $(0.7, 0.1, 0.1, 0.1)$

| Algorithm | Prediction | Max Streak / Confidence |
|---|---|---|
| Per-ply | DRAW | 0 plies |
| Windowed | DRAW | 0 plies |
| Balance | LOSS | $B_n$=-0.42, C=0.42 |



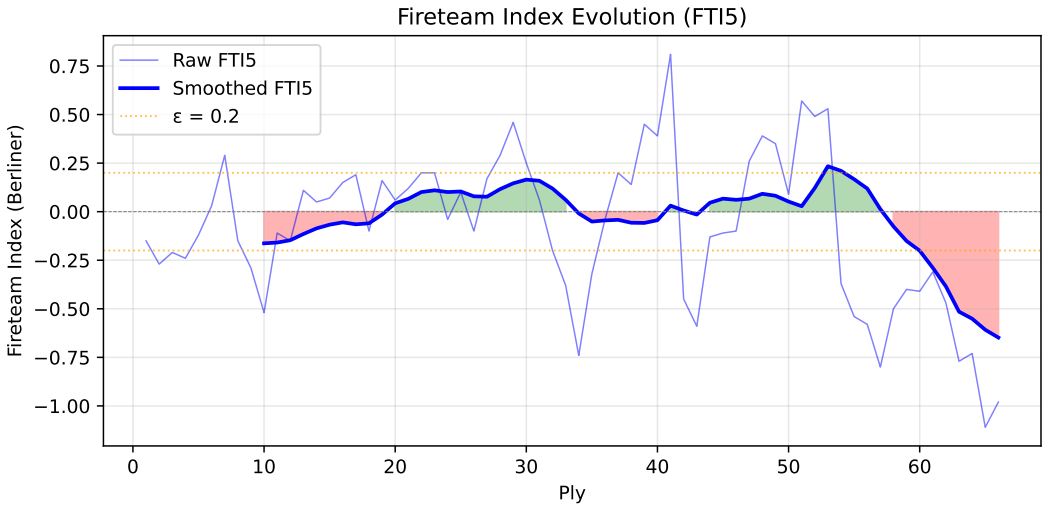

**FTI4: Dynamic** $(0.2, 0.6, 0, 0.2)$

| Algorithm | Prediction | Max Streak / Confidence |
|---|---|---|
| Per-ply | DRAW | 3 plies |
| Windowed | WIN | 16 plies |
| Balance | DRAW | $B_n$=+0.00, C=0.00 |




**FTI5: Solid** $(0.3, 0.2, 0.5, 0)$

| Algorithm | Prediction | Max Streak / Confidence |
|---|---|---|
| Per-ply | DRAW | 3 plies |
| Windowed | DRAW | 0 plies |
| Balance | DRAW | $B_n$=-0.14, C=0.07 |




Actual result: 1/2-1/2 (Berliner's outcome: DRAW)
FTI1: Per-ply ✓ Windowed ✓ Balance ✓
FTI2: Per-ply ✓ Windowed ✓ Balance ×
FTI3: Per-ply ✓ Windowed ✓ Balance ×
FTI4: Per-ply ✓ Windowed × Balance ✓
FTI5: Per-ply ✓ Windowed ✓ Balance ✓

## 8.7 Analysis Information

- Engine: Stockfish 16
- Depth: 20
- Analysis time: 170.9 seconds

# Chapter 9

# Berliner Hans J (USA) vs Hybl Jaroslav (CZE)

## 9.1 Game Information

| | |
|---|---|
| **Event** | Ch World (corr.) |
| **Site** | corr. |
| **Date** | 1965.??.?? |
| **Round** | 1 |
| **White** | Berliner Hans J (USA) |
| **Black** | Hybl Jaroslav (CZE) (2172) |
| **Result** | 1/2-1/2 |
| **Opening** | D86 – Unknown |

## 9.2 Player Statistics

### 9.2.1 Berliner Hans J (USA) (White)

- Total moves: 24
- Accuracy: 96.6%
- Average centipawn loss: 6.9
- Best/Excellent moves: 13 / 2
- Good moves: 7
- Inaccuracies: 2
- Mistakes: 0
- Blunders: 0

### 9.2.2 Hybl Jaroslav (CZE) (Black)

- Total moves: 24
- Accuracy: 97.9%

- Average centipawn loss: 4.2
- Best/Excellent moves: 15 / 2
- Good moves: 6
- Inaccuracies: 0
- Mistakes: 1
- Blunders: 0

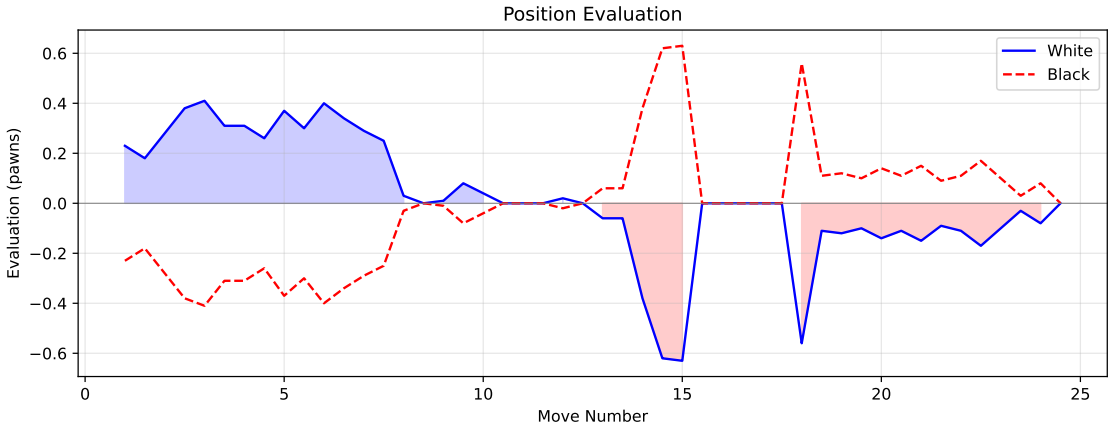
## 9.3 Positional Analysis

This section shows how key positional factors evolved throughout the game.

### 9.3.1 Position Evaluation Over Time

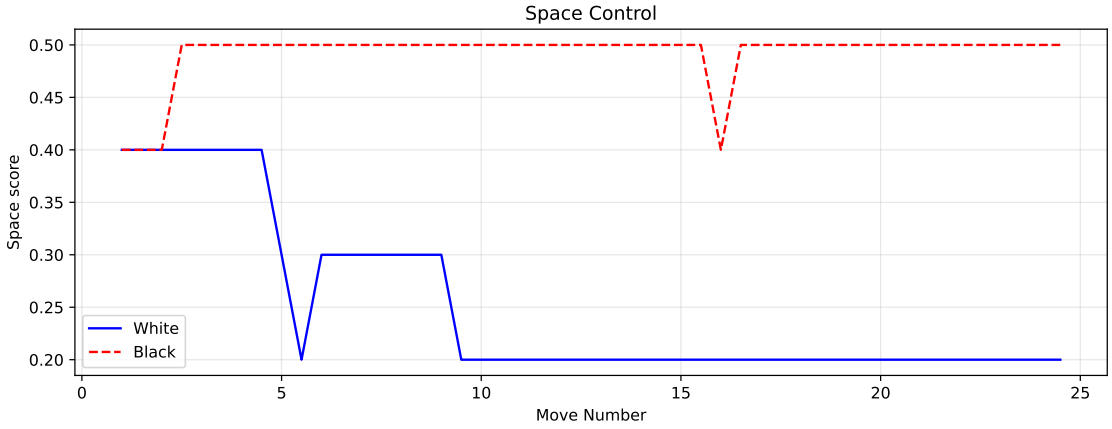
**Game Character Classification.** Evaluation range: minimum $m_1 = -0.63$, maximum $m_2 = +0.41$, spread $d = m_2 - m_1 = 1.04$.

This game is classified as **tense** (one-sided). A tense, one-sided game. Black maintained the advantage throughout.




### 9.3.2 Space Control

- White average: 0.25 (range: 0.20 to 0.40)
- Black average: 0.49 (range: 0.40 to 0.50)



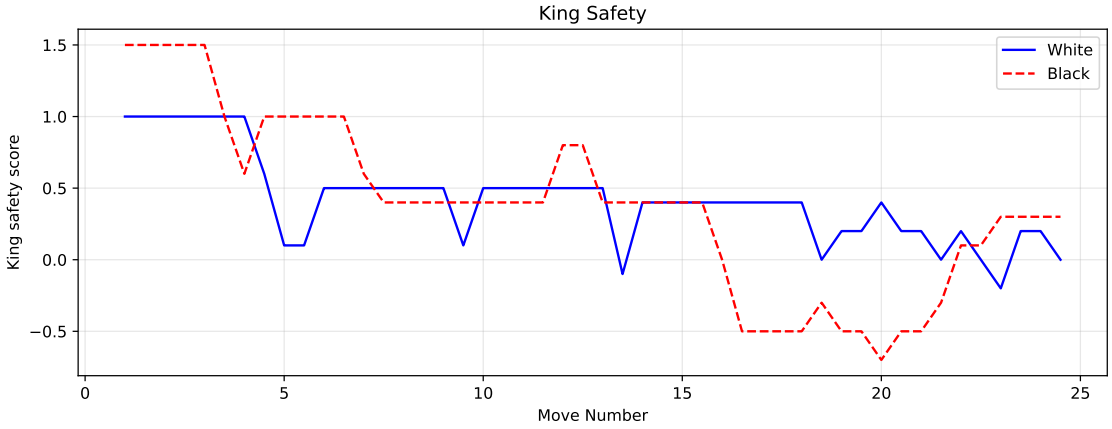

### 9.3.3 Mobility (Piece Activity)

- White average: 4.59 (range: 3.30 to 5.50)
- Black average: 4.31 (range: 3.20 to 6.10)



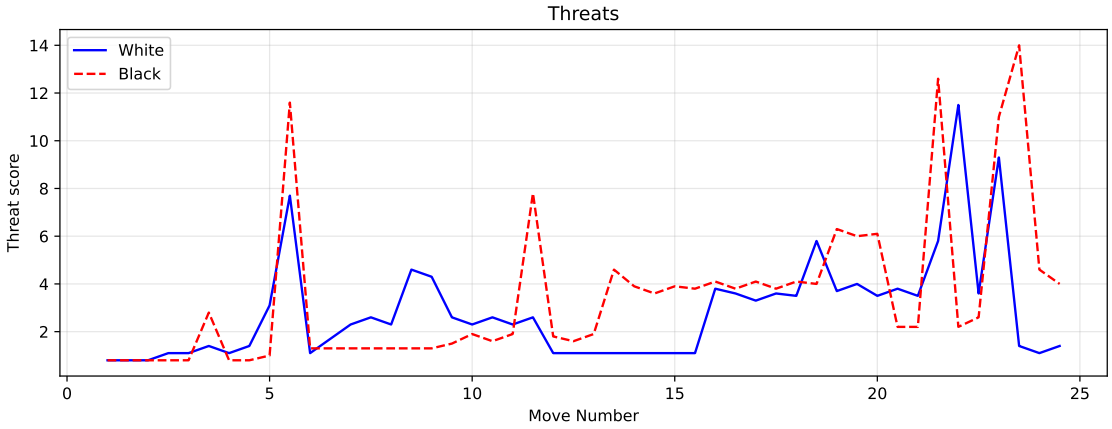

### 9.3.4 King Safety

- White average: 0.42
- Black average: 0.38




### 9.3.5 Threats

- White average: 2.82 (range: 0.80 to 11.50)
- Black average: 3.49 (range: 0.80 to 14.00)
- Average advantage: -0.67 (Black maintained more threats)




## 9.4 Annotated Game

1. d4 Nf6 2. c4 g6 3. Nc3 d5 4. cxd5 Nxd5 5. e4 Nxc3 6. bxc3 Bg7 7. Bc4! O-O 8. h4 c5! 9. Ne2! cxd4 10. cxd4! Nc6! 11. Be3! Na5! 12. Bd3! Nc6! 13. e5 Qa5+! 14. Kf1?! Rd8! 15. h5 Be6? *Black loses 65cp. Consider instead: Nxe5, Qa3.*

16. hxg6! hxg6! 17. Rh4! Rd7 18. Qb1?! Nxe5! 19. dxe5! Rad8! 20. Nd4! Qxe5! 21. Qc2! Rxd4 22. Rxd4! Rxd4! 23. Bxd4! Qxd4! 24. Rd1! Qh4

1/2-1/2

## 9.5 Critical Positions

This section highlights critical moments where the evaluation shifted substantially—either due to errors or missed opportunities. Each diagram shows the position *after* the move was played.

- **Instead of [move]**: The move(s) that should have been played instead.

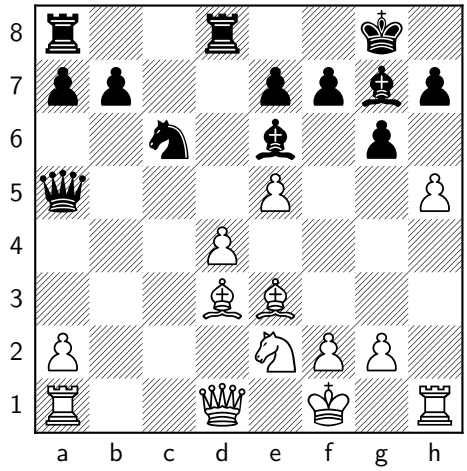
- **Best continuation**: The optimal sequence of moves from the diagrammed position.

Positions marked with ★ represent the biggest evaluation swings in the game. The positional metrics table summarizes the spatial control, piece activity, king safety, and tactical threats for each side.

## Position 1: After 15...Be6 ★

*Biggest evaluation swing(s) (63cp)*

Evaluation: +0.00



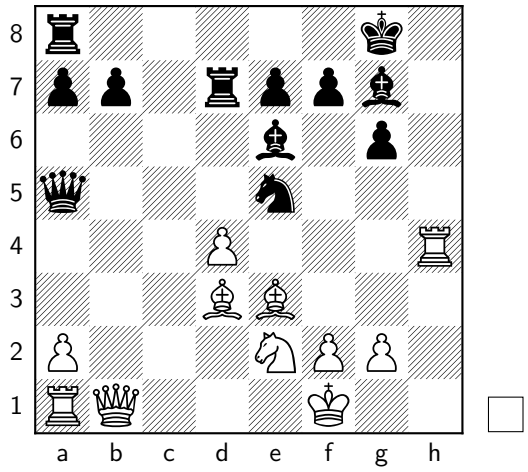
| Metric | White | Black |
|---|---|---|
| Space | 0.20 | 0.50 |
| Mobility | 4.60 | 5.10 |
| King Safety | 0.40 | 0.40 |
| Threats | 1.10 | 3.80 |

Instead of Be6: Nxe5, Qa3 Best continuation: hxg6 hxg6 Rh4

## Position 2: After 18...Nxe5 ★

*Biggest evaluation swing(s) (45cp)*

Evaluation: -0.11



| Metric | White | Black |
|---|---|---|
| Space | 0.20 | 0.50 |
| Mobility | 5.10 | 5.50 |
| King Safety | 0.00 | -0.30 |
| Threats | 5.80 | 4.00 |

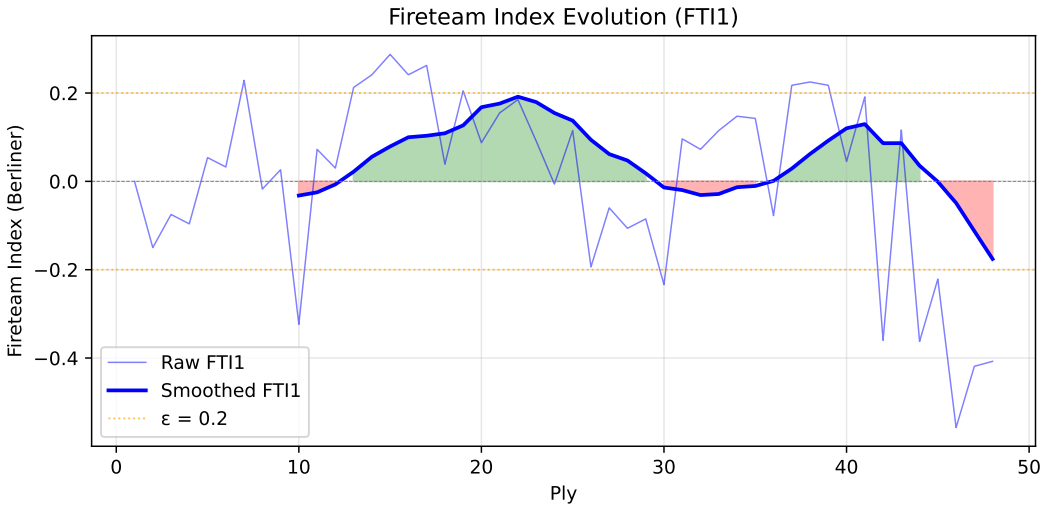
Instead of Nxe5: Qb4, Qb6 Best continuation: dxe5 Rad8 Nd4

## 9.6 Fireteam Index Prediction

Tracking: **Berliner** (White)

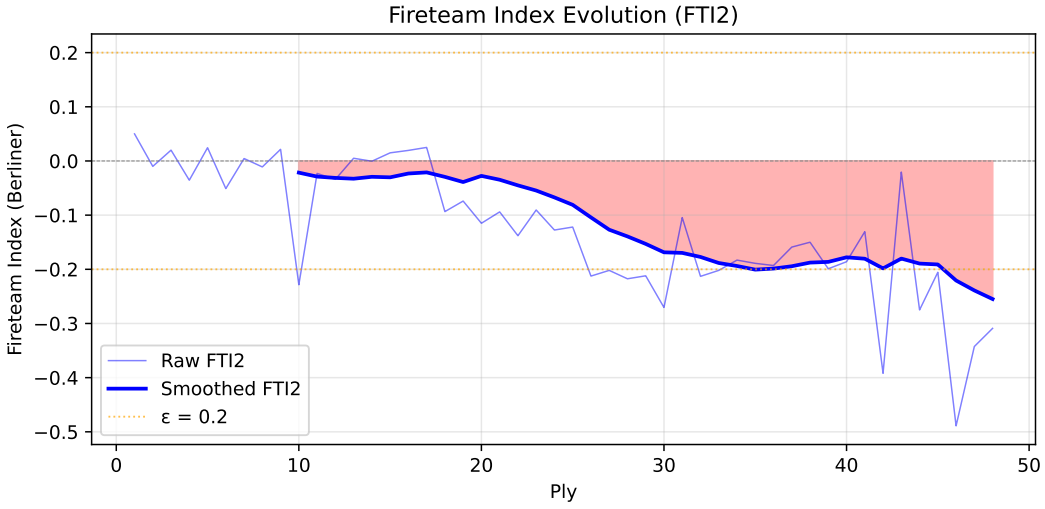
**FTI1: Harmonious** $(\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4})$

| **Algorithm** | **Prediction** | **Max Streak / Confidence** |
|---|---|---|
| Per-ply | DRAW | 1 plies |
| Windowed | DRAW | 0 plies |
| Balance | DRAW | $B_n$=-0.06, C=0.03 |



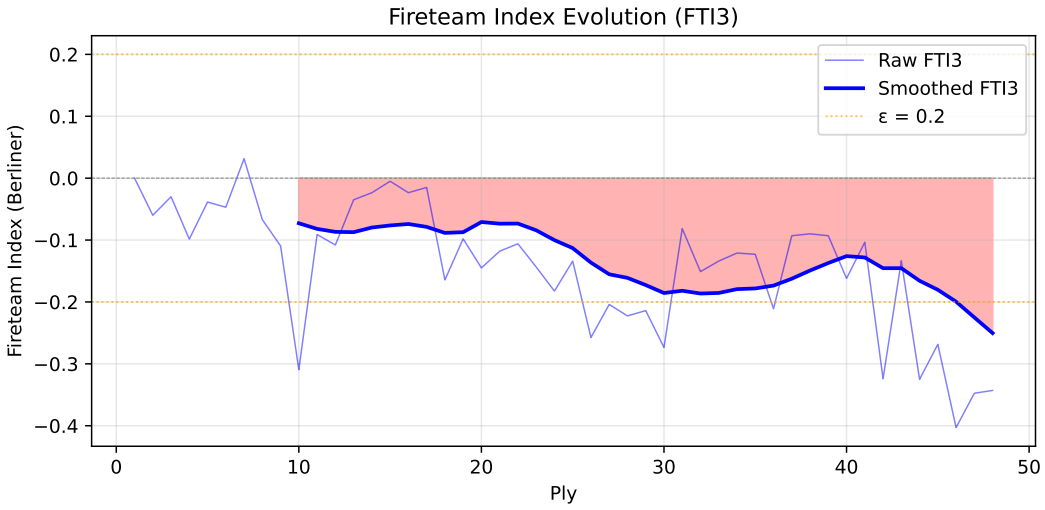

**FTI2: Tactical** $(0.6, 0.1, 0, 0.3)$

| Algorithm | Prediction | Max Streak / Confidence |
|---|---|---|
| Per-ply | DRAW | 0 plies |
| Windowed | DRAW | 0 plies |
| Balance | LOSS | $B_n$=-0.41, C=0.41 |



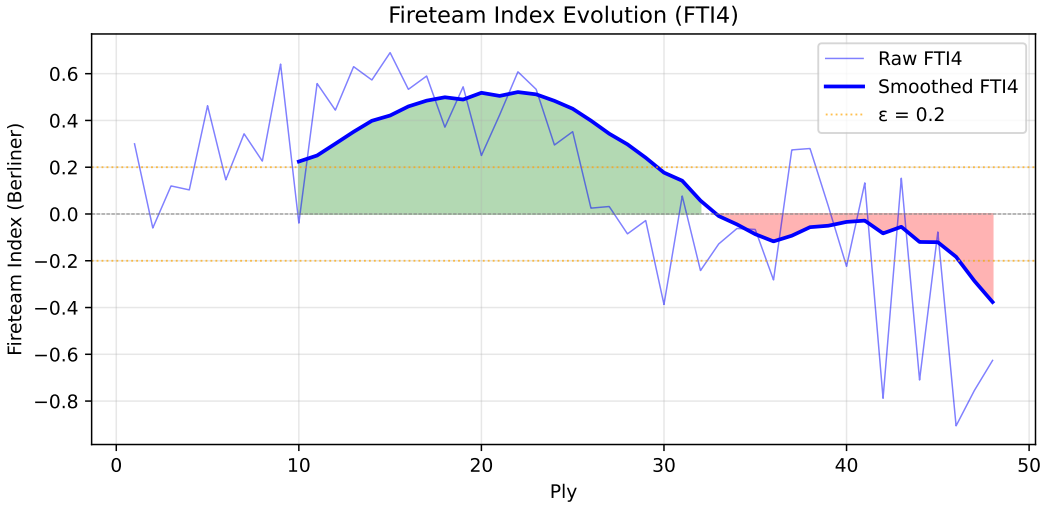

**FTI3: Strategic** $(0.7, 0.1, 0.1, 0.1)$

| Algorithm | Prediction | Max Streak / Confidence |
|---|---|---|
| Per-ply | DRAW | 0 plies |
| Windowed | DRAW | 0 plies |
| Balance | LOSS | $B_n$=-0.38, C=0.38 |



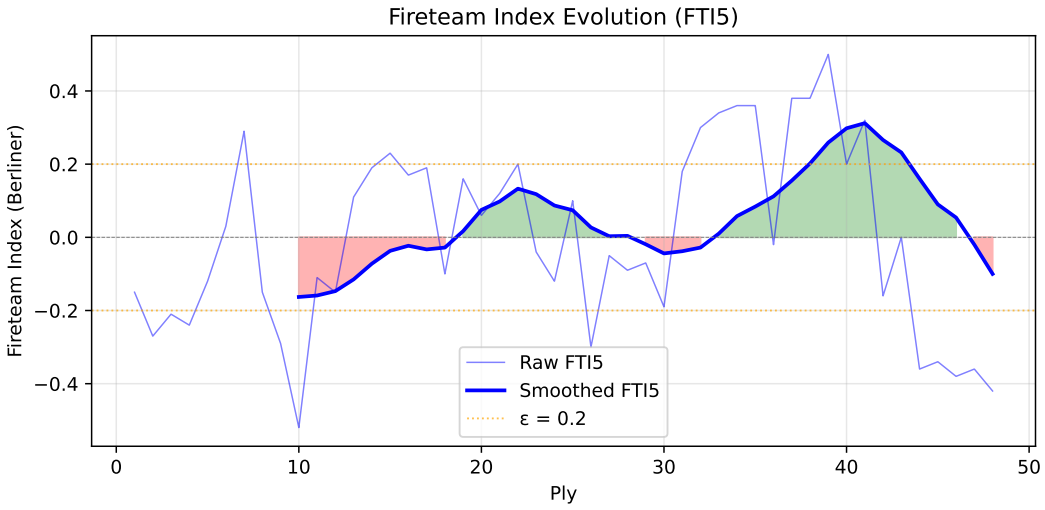

**FTI4: Dynamic** $(0.2, 0.6, 0, 0.2)$

| Algorithm | Prediction | Max Streak / Confidence |
|---|---|---|
| Per-ply | DRAW | 5 plies |
| Windowed | WIN | 12 plies |
| Balance | DRAW | $B_n$=+0.06, C=0.03 |




**FTI5: Solid** $(0.3, 0.2, 0.5, 0)$

| Algorithm | Prediction | Max Streak / Confidence |
|---|---|---|
| Per-ply | DRAW | 4 plies |
| Windowed | DRAW | 4 plies |
| Balance | DRAW | $B_n$=+0.09, C=0.06 |




Actual result: 1/2-1/2 (Berliner's outcome: DRAW)
FTI1: Per-ply ✓ Windowed ✓ Balance ✓
FTI2: Per-ply ✓ Windowed ✓ Balance ×
FTI3: Per-ply ✓ Windowed ✓ Balance ×
FTI4: Per-ply ✓ Windowed × Balance ✓
FTI5: Per-ply ✓ Windowed ✓ Balance ✓

## 9.7 Analysis Information

- Engine: Stockfish 16
- Depth: 20
- Analysis time: 137.2 seconds

# Chapter 10

# Hans Berliner vs Kenneth Charles Messere

## 10.1 Game Information

| | |
|---|---|
| **Event** | 5th Correspondence World Championship |
| **Site** | ICCF |
| **Date** | 1965.07.01 |
| **Round** | 13 |
| **White** | Hans Berliner |
| **Black** | Kenneth Charles Messere |
| **Result** | 1-0 |
| **Opening** | D86 – Unknown |

## 10.2 Player Statistics

### 10.2.1 Hans Berliner (White)

- Total moves: 22
- Accuracy: 95.1%
- Average centipawn loss: 10.1
- Best/Excellent moves: 9 / 5
- Good moves: 6
- Inaccuracies: 1
- Mistakes: 1
- Blunders: 0

### 10.2.2 Kenneth Charles Messere (Black)

- Total moves: 21
- Accuracy: 84.5%

- Average centipawn loss: 33.7
- Best/Excellent moves: 5 / 1
- Good moves: 9
- Inaccuracies: 1
- Mistakes: 4
- Blunders: 1

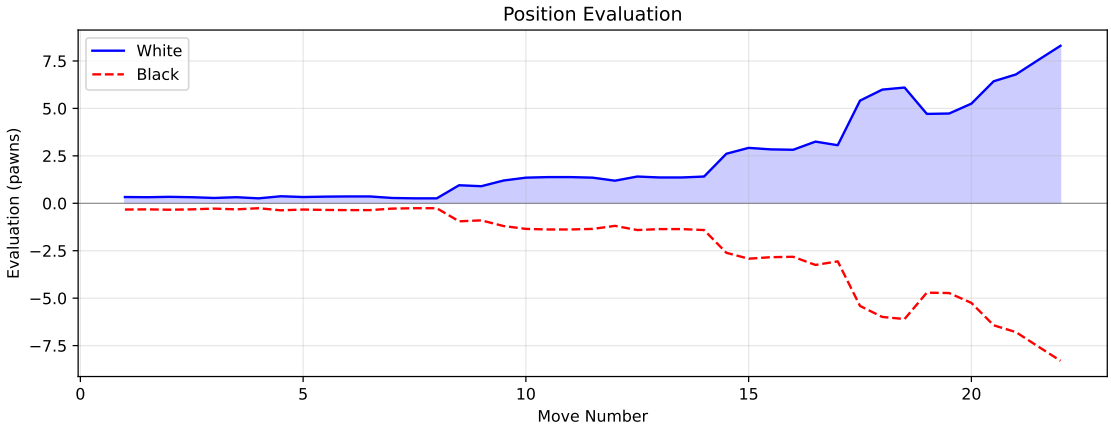
## 10.3 Positional Analysis

This section shows how key positional factors evolved throughout the game.

### 10.3.1 Position Evaluation Over Time

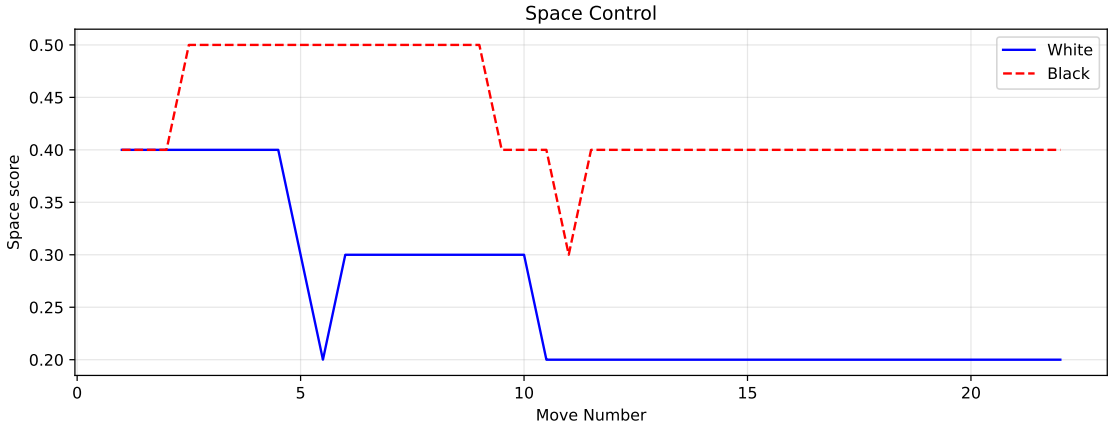
**Game Character Classification.** Evaluation range: minimum $m_1 = +0.26$, maximum $m_2 = +8.30$, spread $d = m_2 - m_1 = 8.04$.

This game is classified as **chaotic** (one-sided). A chaotic, one-sided game. White maintained the advantage throughout.



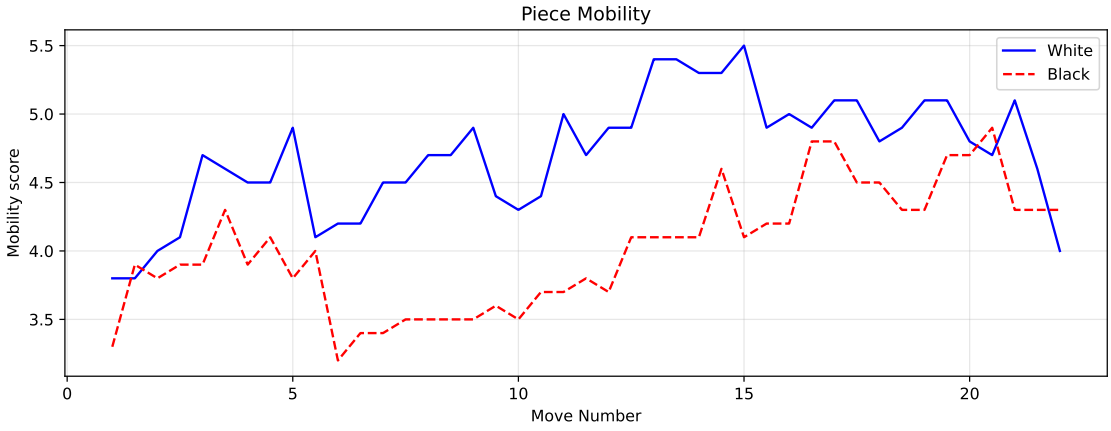

### 10.3.2 Space Control

- White average: 0.26 (range: 0.20 to 0.40)
- Black average: 0.43 (range: 0.30 to 0.50)



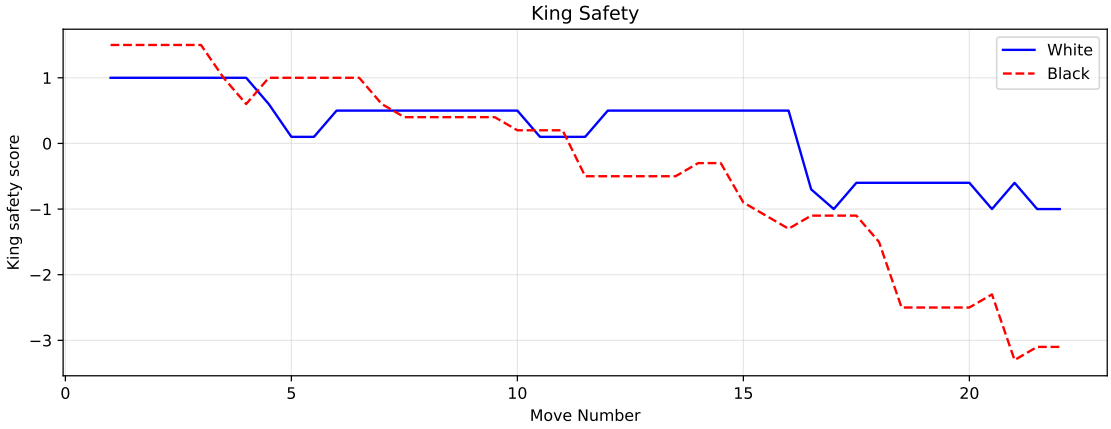

### 10.3.3 Mobility (Piece Activity)

- White average: 4.70 (range: 3.80 to 5.50)
- Black average: 4.02 (range: 3.20 to 4.90)



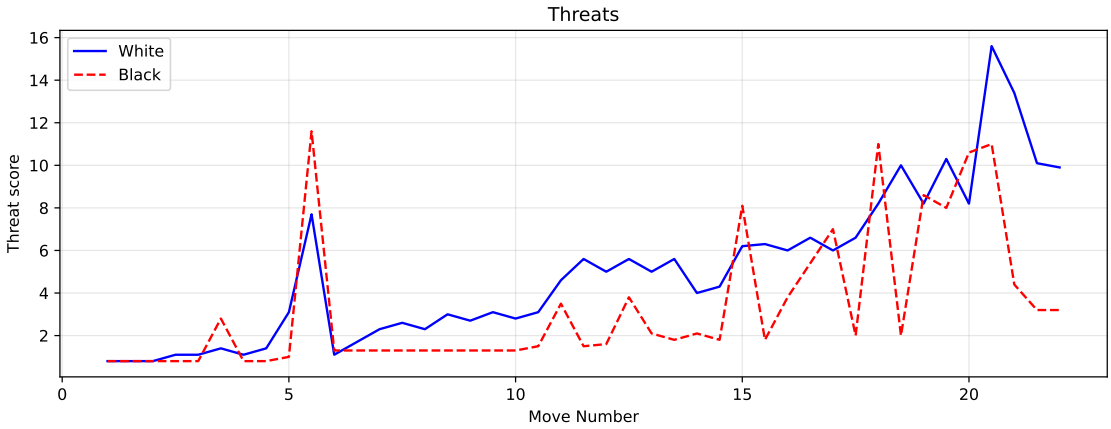

### 10.3.4 King Safety

- White average: 0.19
- Black average: -0.37




### 10.3.5 Threats

- White average: 5.01 (range: 0.80 to 15.60)
- Black average: 3.31 (range: 0.80 to 11.60)
- Average advantage: 1.69 (White maintained more threats)




## 10.4 Annotated Game

1. d4 Nf6 2. c4 g6 3. Nc3 d5 4. cxd5 Nxd5 5. e4 Nxc3 6. bxc3 Bg7 7. Bc4 O-O! 8. Ne2 b6? *Black loses 70cp. Consider instead: c5, Nc6, e5.*

9. h4! e5 10. h5! exd4 11. hxg6! hxg6! 12. cxd4?! Qe7 13. Qd3! Re8! 14. Bh6! Nc6? *Black loses 119cp. Consider instead: Bxh6, Ba6, Be6.*

15. Bxg7! Kxg7 16. Qd2 Qxe4?! 17. O-O-O Qe7?? *Black loses 214cp. Consider instead: Qf5, Nb4.*

18. Qh6+! Kf6! 19. Rd3? *White loses 116cp. Consider instead: Qh4+.*

19...Bg4! 20. Rhh3 Nxd4? *Black loses 101cp. Consider instead: Qe4, Qxe2, Bf5.*

21. Rxd4! Bxe2? *Black loses 79cp. Consider instead: Qe5, Bf5.*

22. Rf4+

1-0

## 10.5 Critical Positions

This section highlights critical moments where the evaluation shifted substantially—either due to errors or missed opportunities. Each diagram shows the position *after* the move was played.

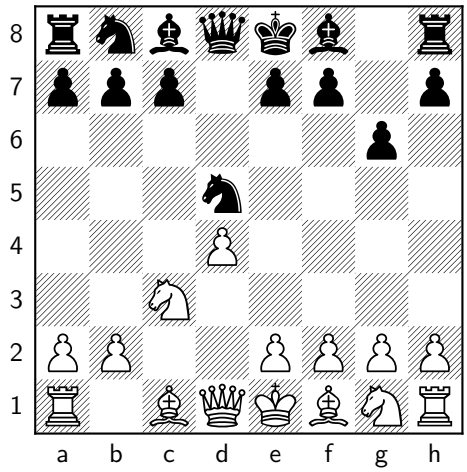
- **Instead of [move]**: The move(s) that should have been played instead.
- **Best continuation**: The optimal sequence of moves from the diagrammed position.

Positions marked with ★ represent the biggest evaluation swings in the game. The positional metrics table summarizes the spatial control, piece activity, king safety, and tactical threats for each side.

### Position 1: After 4...Nxd5 ★

*Biggest evaluation swing(s) (11cp)*

Evaluation: +0.37



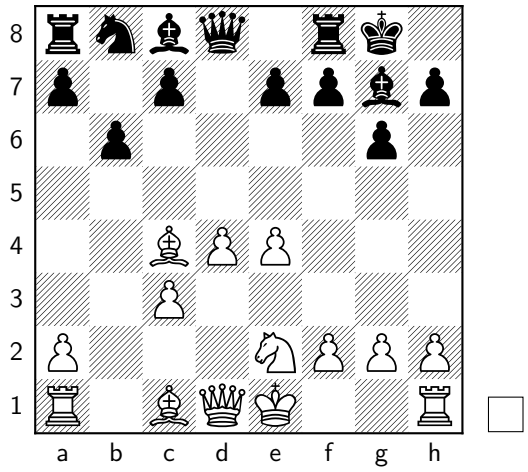
| Metric | White | Black |
|---|---|---|
| Space | 0.40 | 0.50 |
| Mobility | 4.50 | 4.10 |
| King Safety | 0.60 | 1.00 |
| Threats | 1.40 | 0.80 |

Best continuation: e4 Nxc3 bxc3

### Position 2: After 8...b6 ★

*Biggest evaluation swing(s) (69cp)*

Evaluation: +0.95



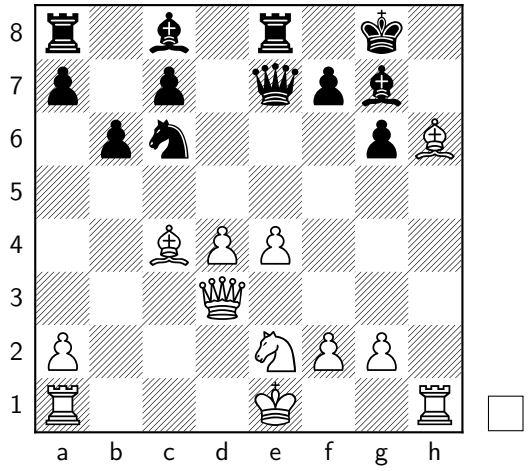
| Metric | White | Black |
|---|---|---|
| Space | 0.30 | 0.50 |
| Mobility | 4.70 | 3.50 |
| King Safety | 0.50 | 0.40 |
| Threats | 3.00 | 1.30 |

Instead of b6: c5, Nc6, e5 Best continuation: h4 Nc6 e5

### Position 3: After 14...Nc6

*Critical moment*

Evaluation: +2.61



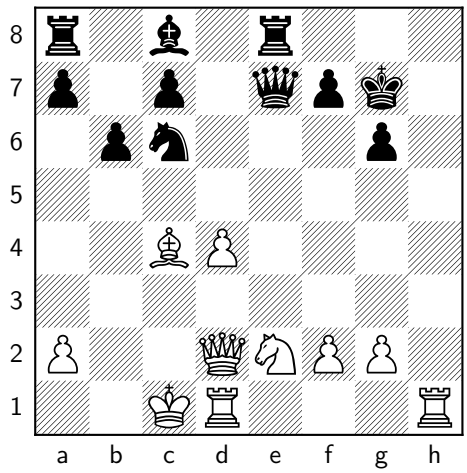
| Metric | White | Black |
|---|---|---|
| Space | 0.20 | 0.40 |
| Mobility | 5.30 | 4.60 |
| King Safety | 0.50 | -0.30 |
| Threats | 4.30 | 1.80 |

Instead of Nc6: Bxh6, Ba6, Be6 Best continuation: Bxg7 Kxg7 Qd2

### Position 4: After 17...Qe7

*Black blunders (+235cp)*

Evaluation: +5.41



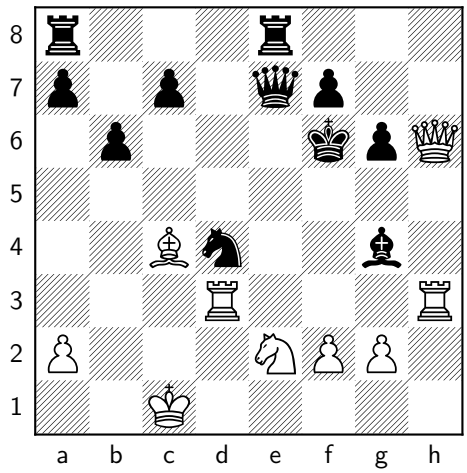
| Metric | White | Black |
|---|---|---|
| Space | 0.20 | 0.40 |
| Mobility | 5.10 | 4.50 |
| King Safety | -0.60 | -1.10 |
| Threats | 6.60 | 2.00 |

Instead of Qe7: Qf5, Nb4 Best continuation: Qh6+ Kf6 Qh4+

### Position 5: After 20...Nxd4

*Critical moment*

Evaluation: +6.43



| Metric | White | Black |
|---|---|---|
| Space | 0.20 | 0.40 |
| Mobility | 4.70 | 4.90 |
| King Safety | -1.00 | -2.30 |
| Threats | 15.60 | 11.00 |

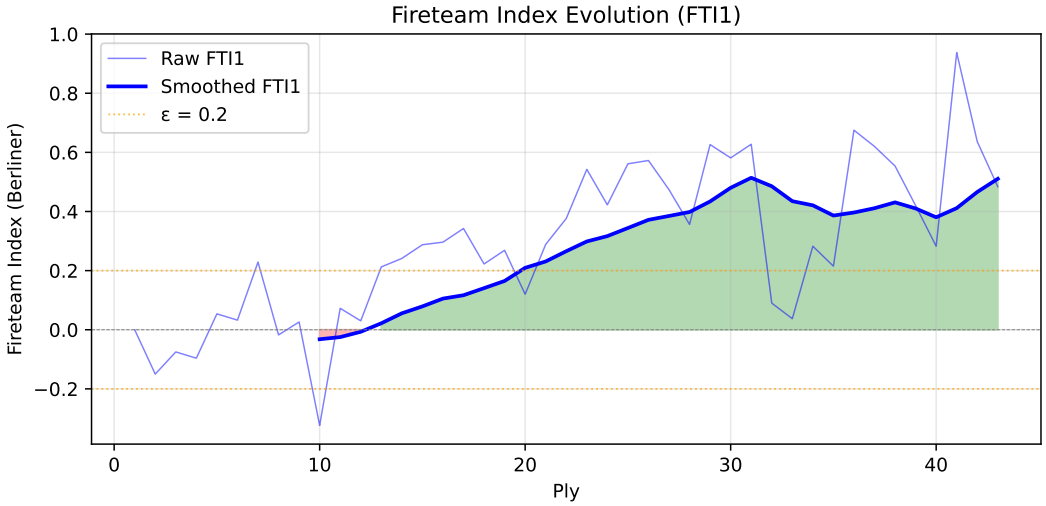
Instead of Nxd4: Qe4, Qxe2, Bf5 Best continuation: Qh4+ Kg7 Qh6+

## 10.6 Fireteam Index Prediction

Tracking: **Berliner** (White)

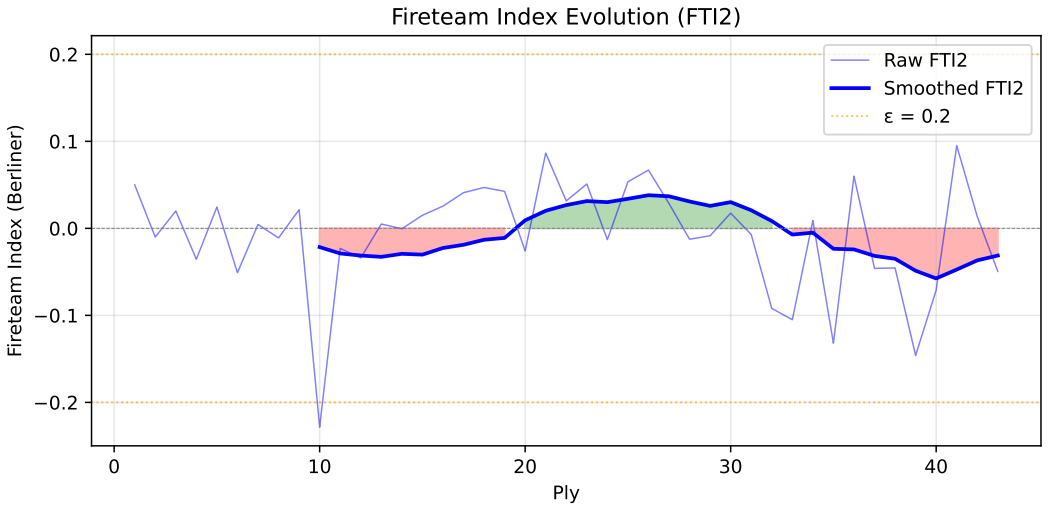
**FTI1: Harmonious** $(\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4})$

| Algorithm | Prediction | Max Streak / Confidence |
|---|---|---|
| Per-ply | WIN | 11 plies |
| Windowed | WIN | 22 plies |
| Balance | WIN | $B_n$=+0.89, C=0.89 |



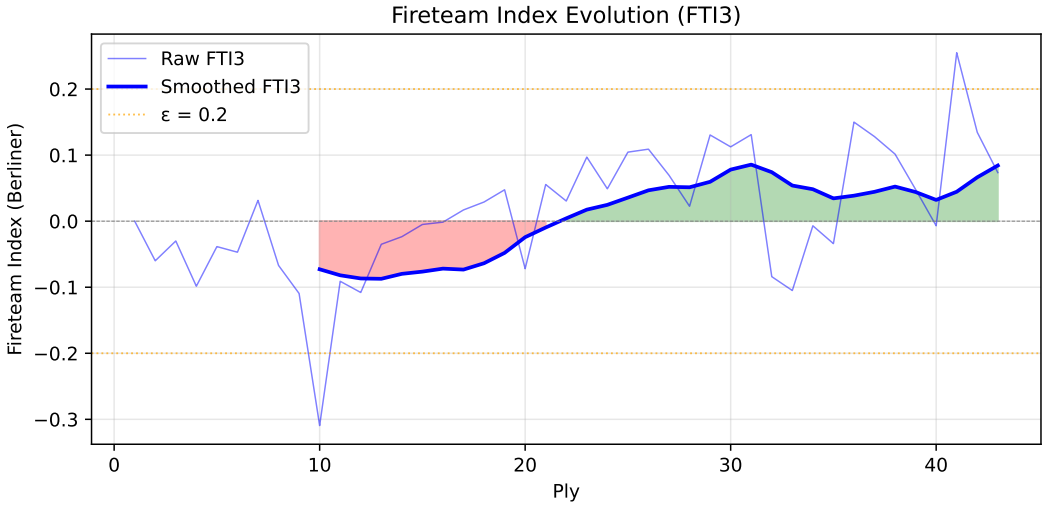

**FTI2: Tactical** $(0.6, 0.1, 0, 0.3)$

| Algorithm | Prediction | Max Streak / Confidence |
|---|---|---|
| Per-ply | DRAW | 0 plies |
| Windowed | DRAW | 0 plies |
| Balance | DRAW | $B_n$=+0.00, C=0.00 |



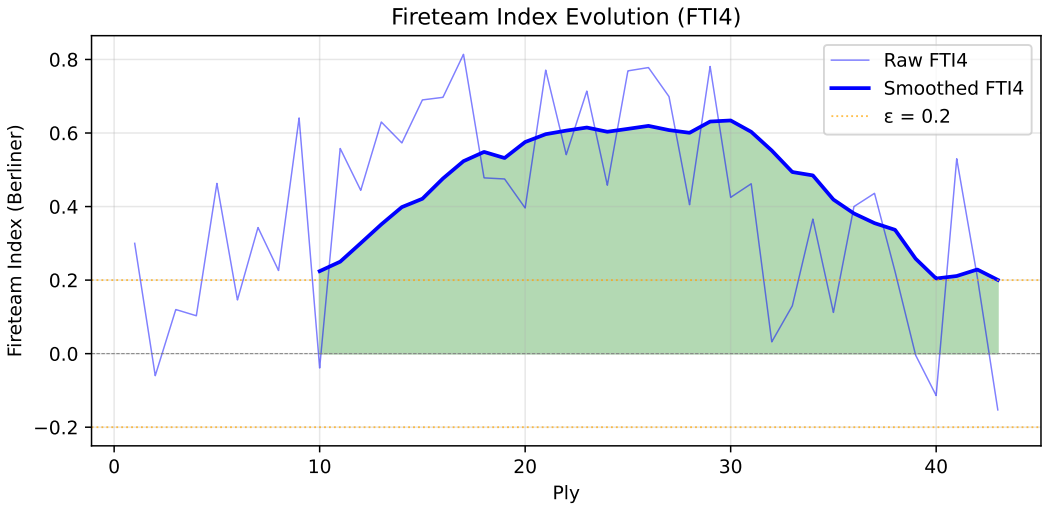

**FTI3: Strategic** $(0.7, 0.1, 0.1, 0.1)$

| Algorithm | Prediction | Max Streak / Confidence |
|---|---|---|
| Per-ply | DRAW | 1 plies |
| Windowed | DRAW | 0 plies |
| Balance | DRAW | $B_n$=+0.04, C=0.04 |



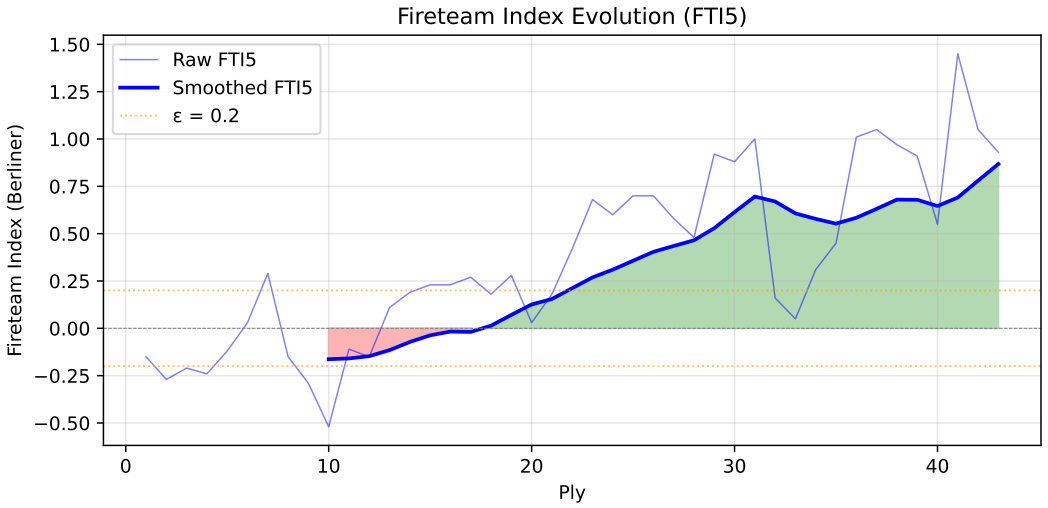

**FTI4: Dynamic** $(0.2, 0.6, 0, 0.2)$

| Algorithm | Prediction | Max Streak / Confidence |
|---|---|---|
| Per-ply | WIN | 15 plies |
| Windowed | WIN | 23 plies |
| Balance | WIN | $B_n$=+0.78, C=0.78 |




**FTI5: Solid** $(0.3, 0.2, 0.5, 0)$

| Algorithm | Prediction | Max Streak / Confidence |
|---|---|---|
| Per-ply | WIN | 10 plies |
| Windowed | WIN | 21 plies |
| Balance | WIN | $B_n$=+0.81, C=0.81 |




Actual result: 1-0 (Berliner's outcome: WIN)

FTI1: Per-ply ✓ Windowed ✓ Balance ✓
FTI2: Per-ply × Windowed × Balance ×
FTI3: Per-ply × Windowed × Balance ×
FTI4: Per-ply ✓ Windowed ✓ Balance ✓
FTI5: Per-ply ✓ Windowed ✓ Balance ✓

## 10.7 Analysis Information

- Engine: Stockfish 16
- Depth: 20
- Analysis time: 151.7 seconds

# Chapter 11

# Hans Berliner vs Julius Nielsen

## 11.1 Game Information

| | |
|---|---|
| **Event** | 5th Correspondence World Championship |
| **Site** | ICCF |
| **Date** | 1965.07.01 |
| **Round** | 7 |
| **White** | Hans Berliner |
| **Black** | Julius Nielsen |
| **Result** | 1-0 |
| **Opening** | D35 – Unknown |

## 11.2 Player Statistics

### 11.2.1 Hans Berliner (White)

- Total moves: 48
- Accuracy: 95.1%
- Average centipawn loss: 10.0
- Best/Excellent moves: 26 / 8
- Good moves: 7
- Inaccuracies: 5
- Mistakes: 2
- Blunders: 0

### 11.2.2 Julius Nielsen (Black)

- Total moves: 47
- Accuracy: 91.8%
- Average centipawn loss: 17.1

- Best/Excellent moves: 18 / 4
- Good moves: 17
- Inaccuracies: 6
- Mistakes: 2
- Blunders: 0

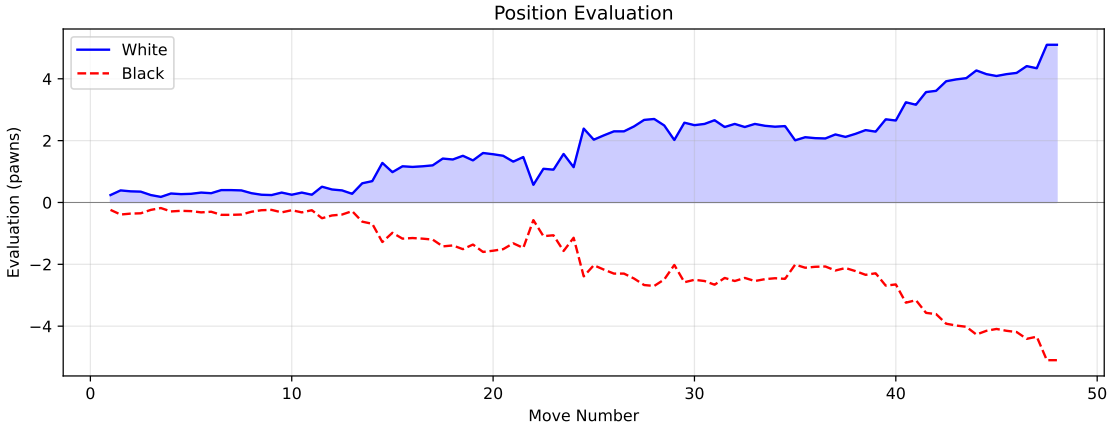
## 11.3 Positional Analysis

This section shows how key positional factors evolved throughout the game.

### 11.3.1 Position Evaluation Over Time

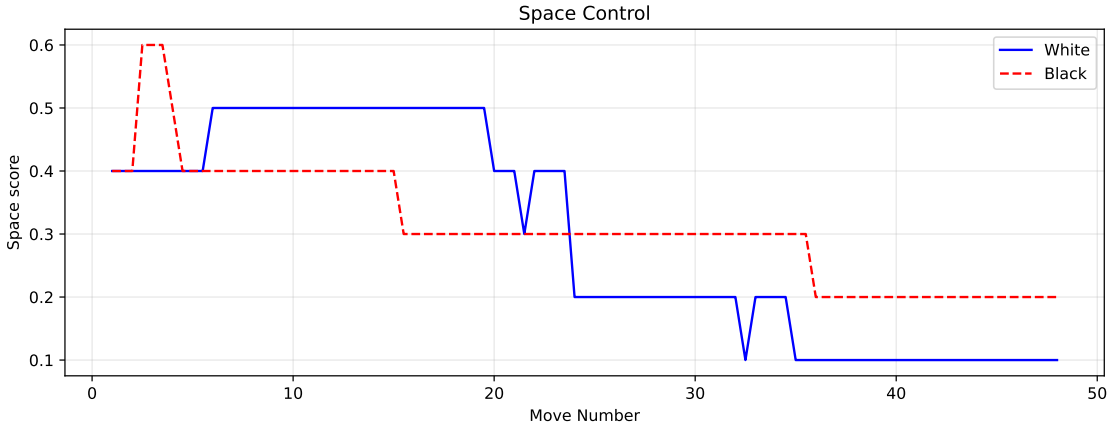
**Game Character Classification.** Evaluation range: minimum $m_1 = +0.18$, maximum $m_2 = +5.10$, spread $d = m_2 - m_1 = 4.92$.

This game is classified as **tactical** (one-sided). A tactical, one-sided game. White maintained the advantage throughout.




### 11.3.2 Space Control

- White average: 0.30 (range: 0.10 to 0.50)
- Black average: 0.31 (range: 0.20 to 0.60)



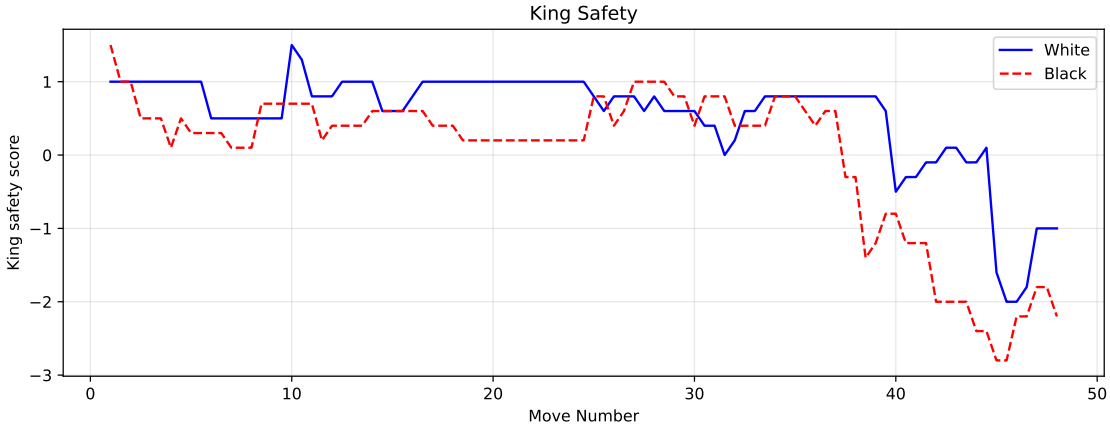

### 11.3.3 Mobility (Piece Activity)

- White average: 4.14 (range: 1.90 to 6.00)
- Black average: 3.69 (range: 1.80 to 5.50)



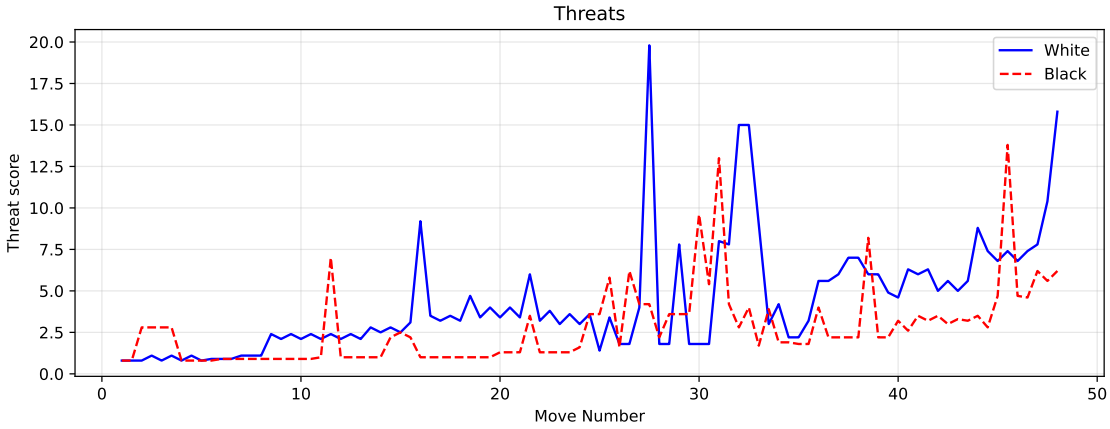

### 11.3.4 King Safety

- White average: 0.54
- Black average: 0.01




### 11.3.5 Threats

- White average: 4.33 (range: 0.80 to 19.80)
- Black average: 2.78 (range: 0.80 to 13.80)
- Average advantage: 1.55 (White maintained more threats)




## 11.4 Annotated Game

1. d4 d5 2. c4 e6 3. Nc3 Nf6 4. cxd5 exd5 5. Bg5 c6 6. e3 Nbd7 7. Bd3! Be7! 8. Nge2 O-O 9. Qc2 Re8 10. O-O Nf8! 11. f3 h6 12. Bh4! Ne6! 13. Rad1! b6?! 14. Bf2 Qc7?! 15. Kh1?! c5 16. Bg3! Qb7! 17. Be5! Rd8 18. Ng3! Ne8 19. Nf5 Bf6 20. f4 Nf8! 21. Be2?! cxd4 22. Bxd4? *White loses 83cp. Consider instead: exd4.*

22...Be6?! 23. Bf3! Qc8?! 24. e4?! Ng6? *Black loses 118cp. Consider instead: Kh8.*

25. Ne3?! dxe4 26. Qxe4! Rxd4! 27. Qxa8! Qxa8 28. Bxa8! Nxf4! 29. Bc6? *White loses 63cp. Consider instead: Ned5, Ncd5, Be4.*

29...Nc7 30. Ng4! Bxg4 31. Rxd4! Bxd4! 32. Rxf4! Bxc3! 33. bxc3! Be6! 34. Ra4! a5! 35. c4?! Kf8 36. c5! bxc5! 37. Rxa5! Ke7! 38. Kg1! Kd6 39. Bf3 Nd5?! 40. Kf2 Nb4?! 41. a4! Bb3 42. Ra8 c4 43. a5! c3 44. Rc8! c2! 45. Ke3 Be6 46. Rc3! Bf5 47. Kd2! h5? *Black loses 70cp. Consider instead: Bd3, Kd7.*

48. Rc4!

1-0

## 11.5 Critical Positions

This section highlights critical moments where the evaluation shifted substantially—either due to errors or missed opportunities. Each diagram shows the position *after* the move was played.

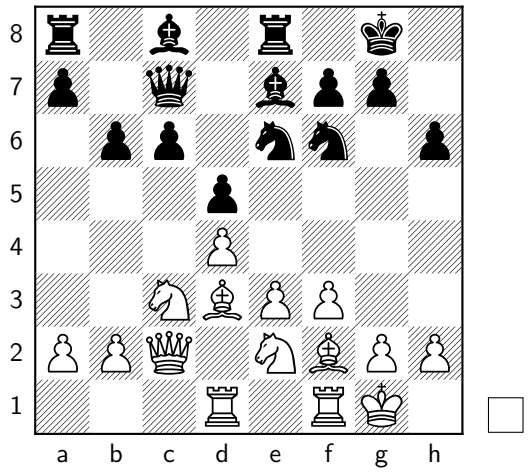
- **Instead of [move]**: The move(s) that should have been played instead.
- **Best continuation**: The optimal sequence of moves from the diagrammed position.

Positions marked with ★ represent the biggest evaluation swings in the game. The positional metrics table summarizes the spatial control, piece activity, king safety, and tactical threats for each side.

### Position 1: After 14...Qc7 ★

*Biggest evaluation swing(s) (59cp)*

Evaluation: +1.28



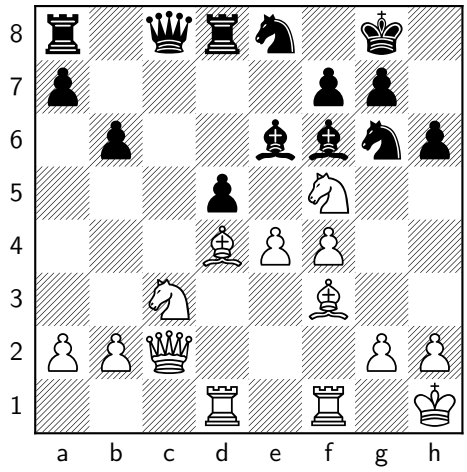
| Metric | White | Black |
|---|---|---|
| Space | 0.50 | 0.40 |
| Mobility | 5.20 | 4.90 |
| King Safety | 0.60 | 0.60 |
| Threats | 2.80 | 2.20 |

Instead of Qc7: b5, a5, Nh5 Best continuation: e4 dxe4 fxe4

### Position 2: After 24...Ng6

*Critical moment*

Evaluation: +2.39



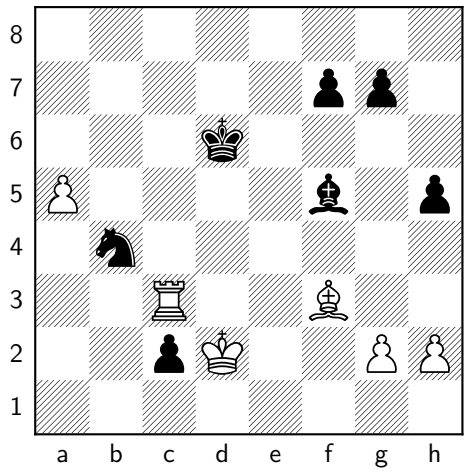
| Metric | White | Black |
|---|---|---|
| Space | 0.20 | 0.30 |
| Mobility | 5.90 | 4.40 |
| King Safety | 1.00 | 0.20 |
| Threats | 3.60 | 3.60 |

Instead of Ng6: Kh8 Best continuation: Nxh6+ gxh6 f5

## Position 3: After 47...h5 ★

*Biggest evaluation swing(s) (76cp)*

Evaluation: +5.10



| Metric | White | Black |
|---|---|---|
| Space | 0.10 | 0.20 |
| Mobility | 2.30 | 2.20 |
| King Safety | -1.00 | -1.80 |
| Threats | 10.40 | 5.60 |

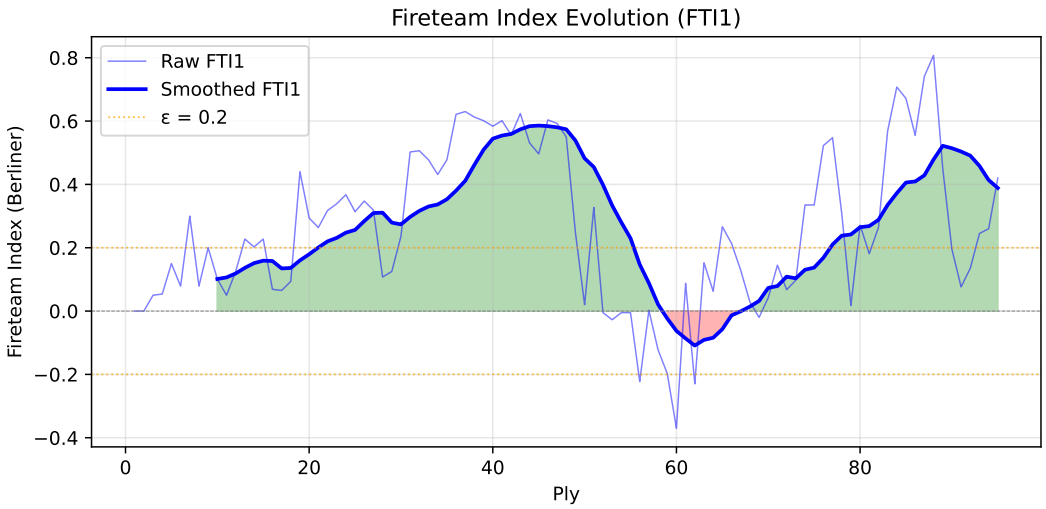
Instead of h5: Bd3, Kd7 Best continuation: Rc4 Na2 Rxc2

## 11.6 Fireteam Index Prediction

Tracking: **Berliner** (White)

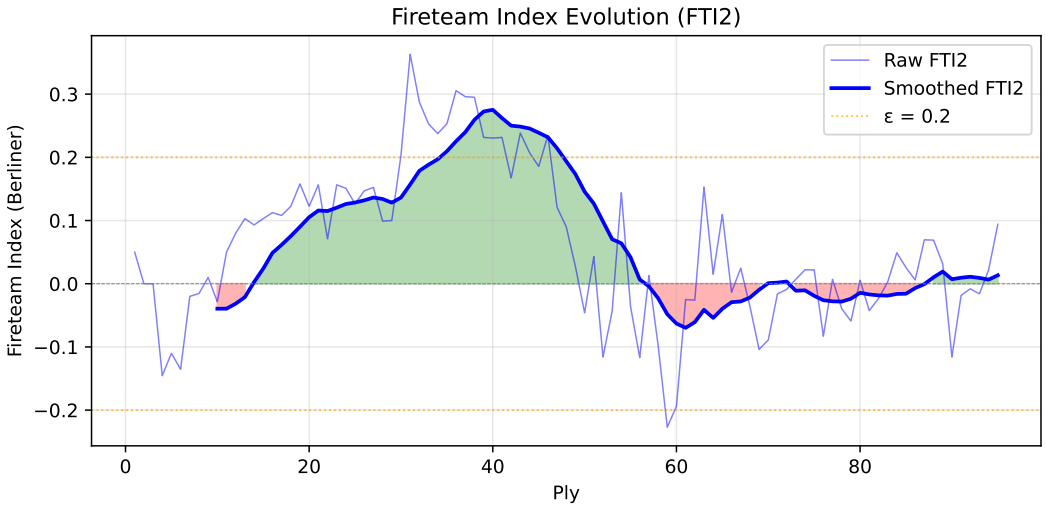
**FTI1: Harmonious** $(\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4})$

| Algorithm | Prediction | Max Streak / Confidence |
|---|---|---|
| Per-ply | WIN | 18 plies |
| Windowed | WIN | 30 plies |
| Balance | WIN | $B_n$=+0.58, C=0.51 |



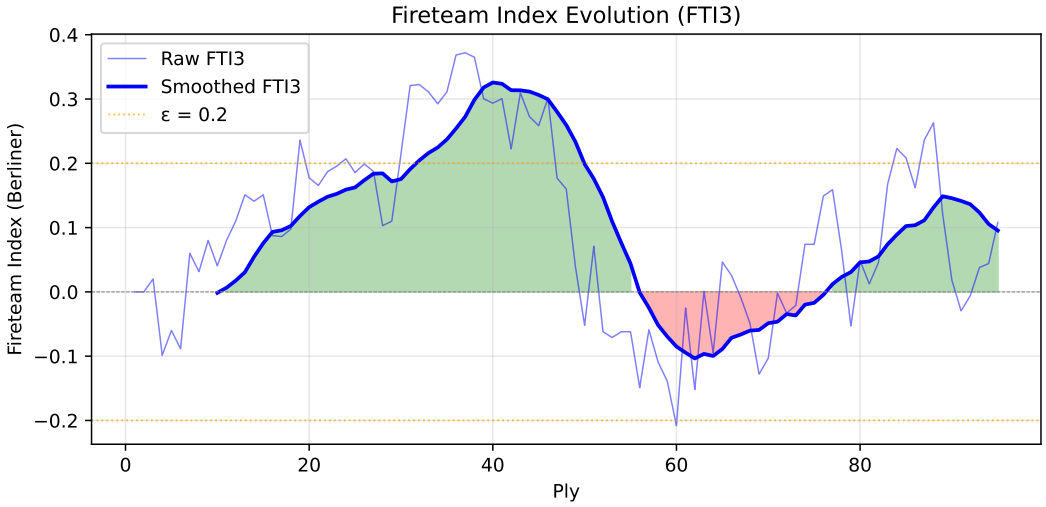

**FTI2: Tactical** $(0.6, 0.1, 0, 0.3)$

| Algorithm | Prediction | Max Streak / Confidence |
|---|---|---|
| Per-ply | DRAW | 4 plies |
| Windowed | DRAW | 5 plies |
| Balance | WIN | $B_n$=+0.18, C=0.17 |



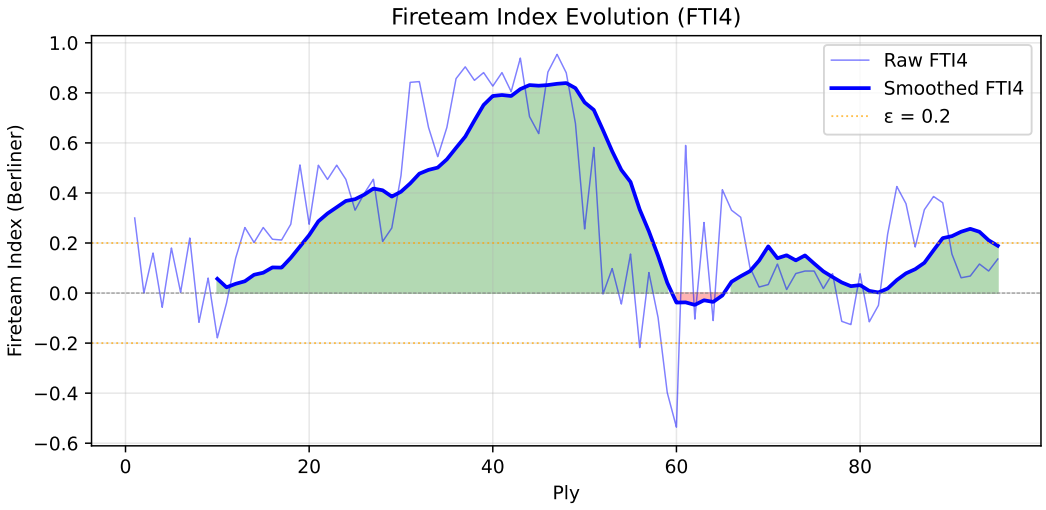

**FTI3: Strategic** $(0.7, 0.1, 0.1, 0.1)$

| Algorithm | Prediction | Max Streak / Confidence |
|---|---|---|
| Per-ply | WIN | 11 plies |
| Windowed | WIN | 13 plies |
| Balance | WIN | $B_n$=+0.28, C=0.24 |



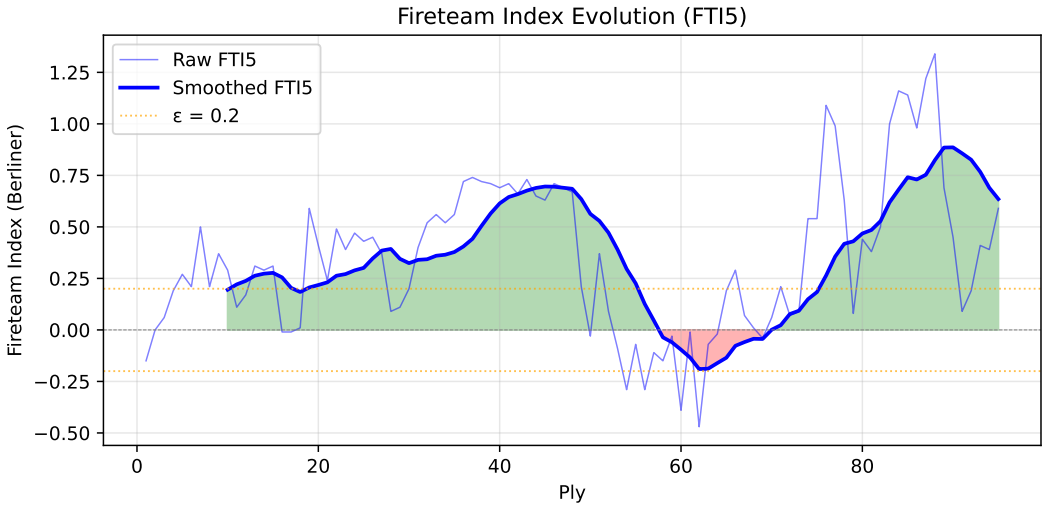

**FTI4: Dynamic** $(0.2, 0.6, 0, 0.2)$

| Algorithm | Prediction | Max Streak / Confidence |
|---|---|---|
| Per-ply | WIN | 23 plies |
| Windowed | WIN | 36 plies |
| Balance | WIN | $B_n$=+0.54, C=0.48 |




**FTI5: Solid** $(0.3, 0.2, 0.5, 0)$

| Algorithm | Prediction | Max Streak / Confidence |
|---|---|---|
| Per-ply | WIN | 18 plies |
| Windowed | WIN | 33 plies |
| Balance | WIN | $B_n$=+0.58, C=0.51 |




Actual result: 1-0 (Berliner's outcome: WIN)

FTI1: Per-ply ✓ Windowed ✓ Balance ✓
FTI2: Per-ply × Windowed × Balance ✓
FTI3: Per-ply ✓ Windowed ✓ Balance ✓
FTI4: Per-ply ✓ Windowed ✓ Balance ✓
FTI5: Per-ply ✓ Windowed ✓ Balance ✓

## 11.7 Analysis Information

- Engine: Stockfish 16
- Depth: 20
- Analysis time: 307.7 seconds

# Chapter 12

# Hans Berliner vs Sture Valentin Nyman

## 12.1 Game Information

| | |
|---|---|
| **Event** | 5th Correspondence World Championship |
| **Site** | ICCF |
| **Date** | 1965.07.01 |
| **Round** | 15 |
| **White** | Hans Berliner |
| **Black** | Sture Valentin Nyman |
| **Result** | 1-0 |
| **Opening** | A56 – Unknown |

## 12.2 Player Statistics

### 12.2.1 Hans Berliner (White)

- Total moves: 46
- Accuracy: 95.3%
- Average centipawn loss: 9.7
- Best/Excellent moves: 22 / 7
- Good moves: 11
- Inaccuracies: 6
- Mistakes: 0
- Blunders: 0

### 12.2.2 Sture Valentin Nyman (Black)

- Total moves: 45
- Accuracy: 92.8%

- Average centipawn loss: 14.9
- Best/Excellent moves: 19 / 5
- Good moves: 13
- Inaccuracies: 7
- Mistakes: 1
- Blunders: 0

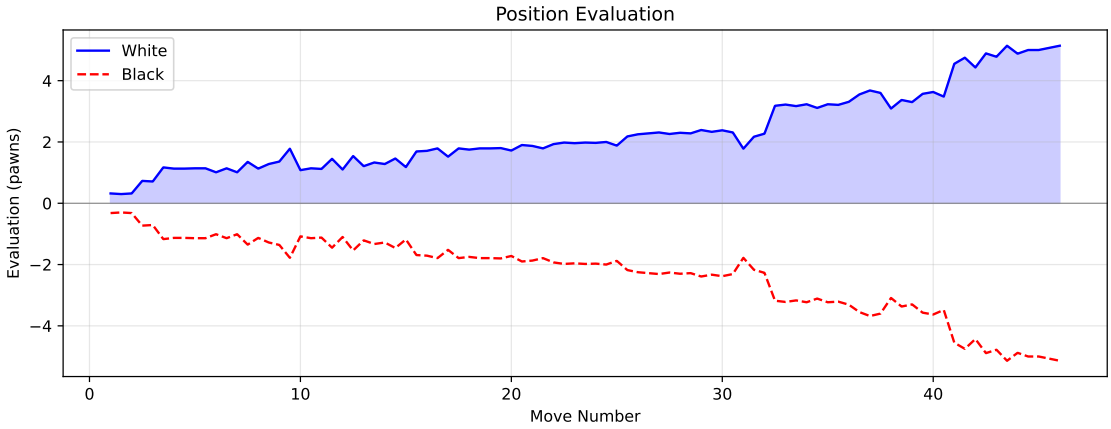
## 12.3 Positional Analysis

This section shows how key positional factors evolved throughout the game.

### 12.3.1 Position Evaluation Over Time

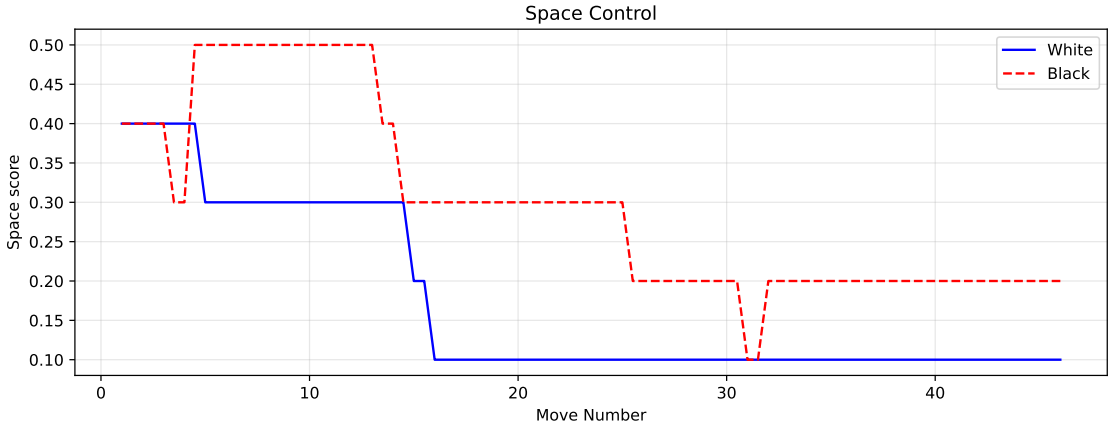
**Game Character Classification.** Evaluation range: minimum $m_1 = +0.30$, maximum $m_2 = +5.14$, spread $d = m_2 - m_1 = 4.84$.

This game is classified as **tactical** (one-sided). A tactical, one-sided game. White maintained the advantage throughout.



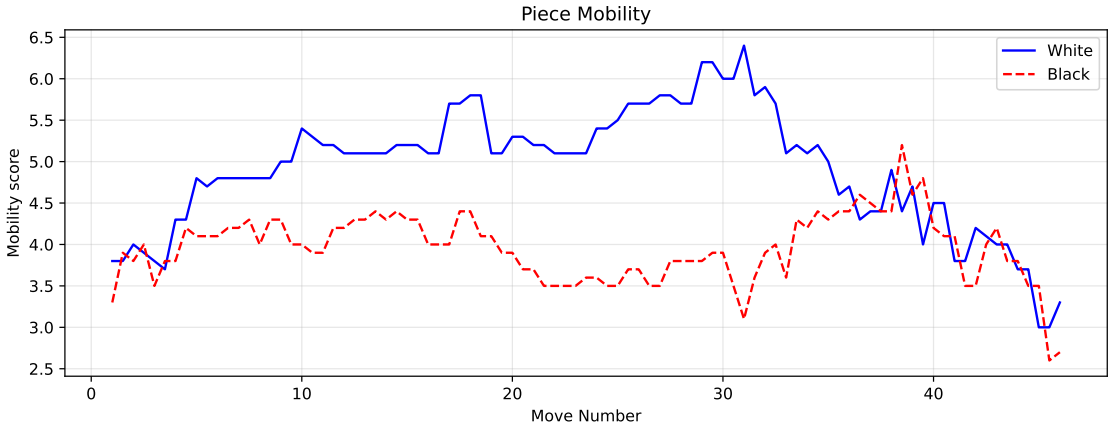

### 12.3.2 Space Control

- White average: 0.17 (range: 0.10 to 0.40)
- Black average: 0.30 (range: 0.10 to 0.50)



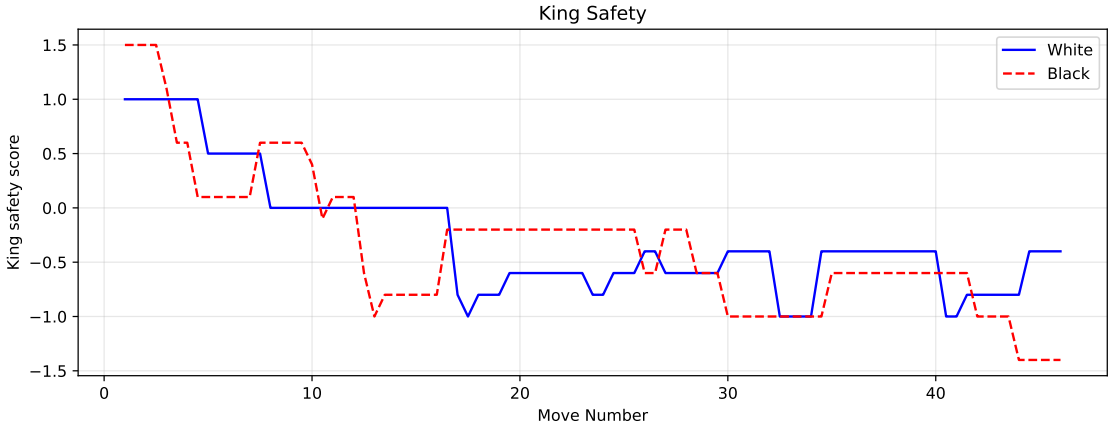

### 12.3.3 Mobility (Piece Activity)

- White average: 4.91 (range: 3.00 to 6.40)
- Black average: 3.95 (range: 2.60 to 5.20)



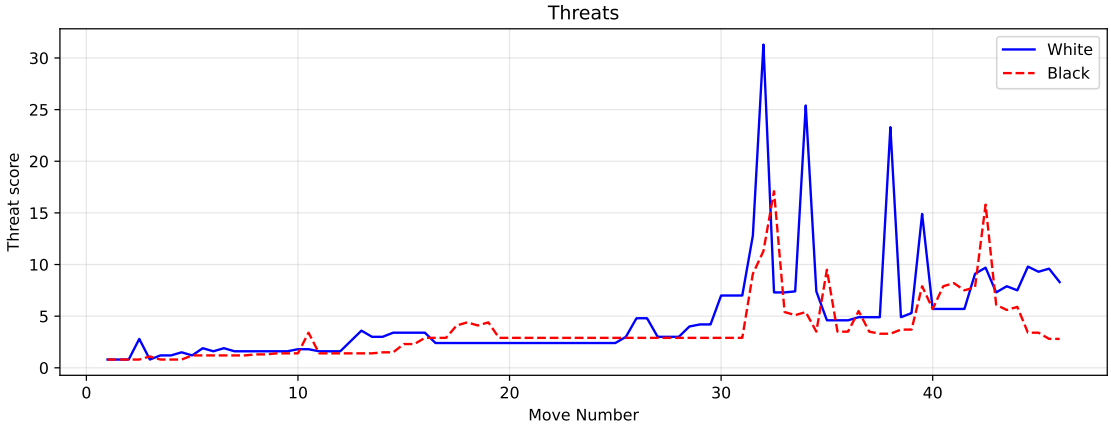

### 12.3.4 King Safety

- White average: -0.27
- Black average: -0.34




### 12.3.5 Threats

- White average: 4.79 (range: 0.80 to 31.30)
- Black average: 3.55 (range: 0.80 to 17.10)
- Average advantage: 1.24 (White maintained more threats)




## 12.4 Annotated Game

1. d4 Nf6 2. c4 c5?! 3. d5 e5?! 4. Nc3 d6 5. e4 g6 6. Bd3 Bg7 7. Nge2 O-O?! 8. f3 Na6 9. h4 Nc7?! 10. Bg5?! h6! 11. Be3 Bd7 12. Qd2?! Kh7?! 13. h5 g5! 14. a3 b6 15. g4?! Ng8?! 16. b4 Kh8 17. Kf2 Qf6 18. Kg2! Rfb8! 19. Rhb1 Qd8 20. Ra2 Qc8 21. Bc2 Bf8 22. Rab2! Ne8! 23. Qd3! Ngf6 24. Ng3 Ng8 25. Bf2 f6 26. Nf5 Ne7 27. Ne3! Qc7 28. Qd1 Kh7 29. Ba4 Nc8 30. Nf5 Ng7 31. bxc5?! Bxa4?! 32. cxb6! Qxc4? *Black loses 94cp. Consider instead: Nxb6, axb6.*

33. Nxa4! axb6! 34. Rb4! Qc7! 35. Nxg7 Bxg7! 36. Qb3! b5 37. Nc3! Na7! 38. Nxb5?! Nxb5 39. Rxb5! Rxb5 40. Qxb5! Qc2! 41. Rb4! Qa2 42. Qd7?! Qxa3! 43. Rb7! Rg8 44. Qe7 Qa8 45. Rd7! Qf8! 46. Qe6!

1-0

## 12.5 Critical Positions

This section highlights critical moments where the evaluation shifted substantially—either due to errors or missed opportunities. Each diagram shows the position *after* the move was played.

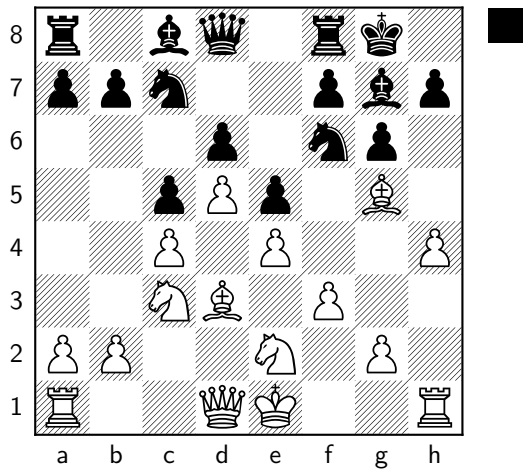
- **Instead of [move]**: The move(s) that should have been played instead.
- **Best continuation**: The optimal sequence of moves from the diagrammed position.

Positions marked with ★ represent the biggest evaluation swings in the game. The positional metrics table summarizes the spatial control, piece activity, king safety, and tactical threats for each side.

### Position 1: After 10. Bg5 ★

*Biggest evaluation swing(s) (70cp)*

Evaluation: +1.08



| Metric | White | Black |
|---|---|---|
| Space | 0.30 | 0.50 |
| Mobility | 5.40 | 4.00 |
| King Safety | 0.00 | 0.40 |
| Threats | 1.80 | 1.40 |

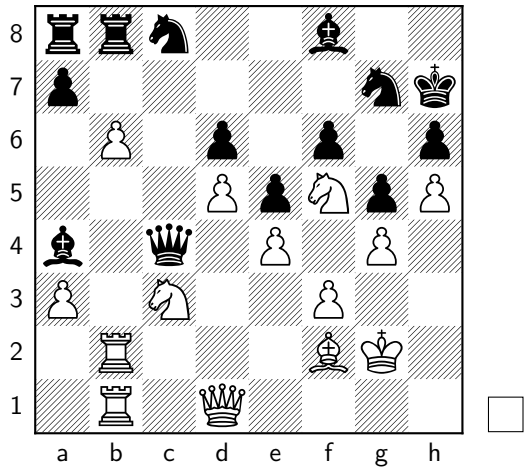
Instead of Bg5: h5, Qc2, Ng3

Best continuation: h6 Be3 h5

### Position 2: After 32...Qxc4 ★

*Biggest evaluation swing(s) (91cp)*

Evaluation: +3.18



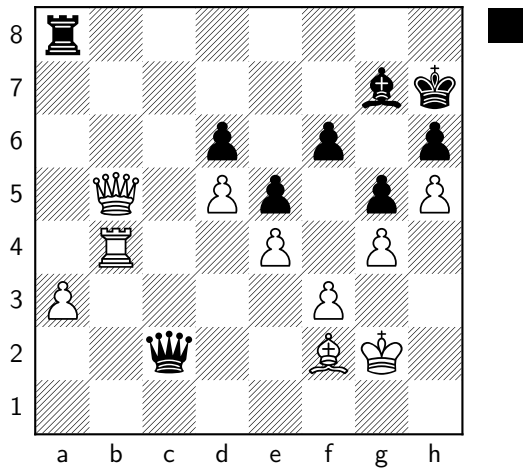
| Metric | White | Black |
|---|---|---|
| Space | 0.10 | 0.20 |
| Mobility | 5.70 | 4.00 |
| King Safety | -1.00 | -1.00 |
| Threats | 7.30 | 17.10 |

Instead of Qxc4: Nxb6, axb6 Best continuation: Nxa4 axb6 Rb4

### Position 3: After 41. Rb4

*Critical moment*

Evaluation: +4.55



| Metric | White | Black |
|---|---|---|
| Space | 0.10 | 0.20 |
| Mobility | 3.80 | 4.10 |
| King Safety | -1.00 | -0.60 |
| Threats | 5.70 | 8.20 |

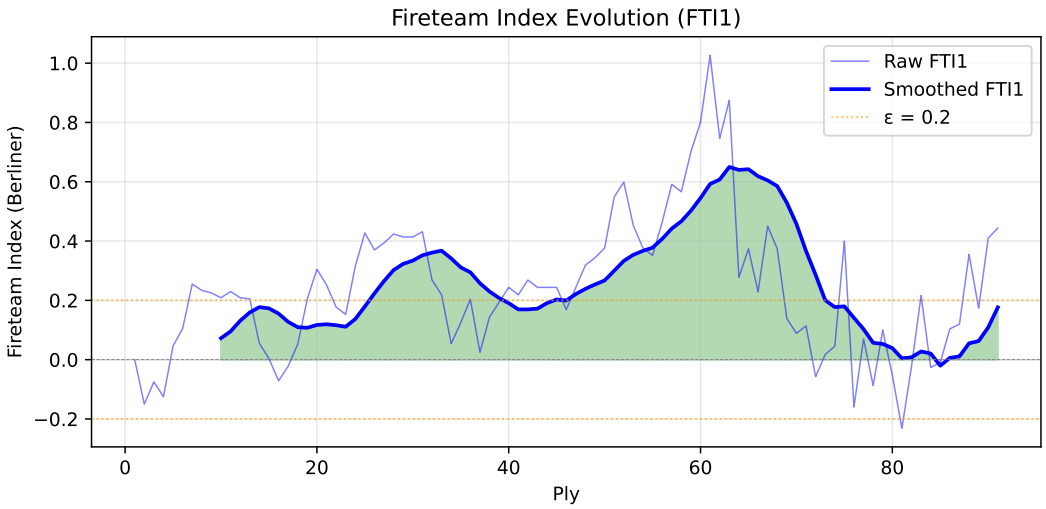
Instead of Rb4: Rb2 Best continuation: Qc8 a4 Qc3

## 12.6 Fireteam Index Prediction

Tracking: **Berliner** (White)

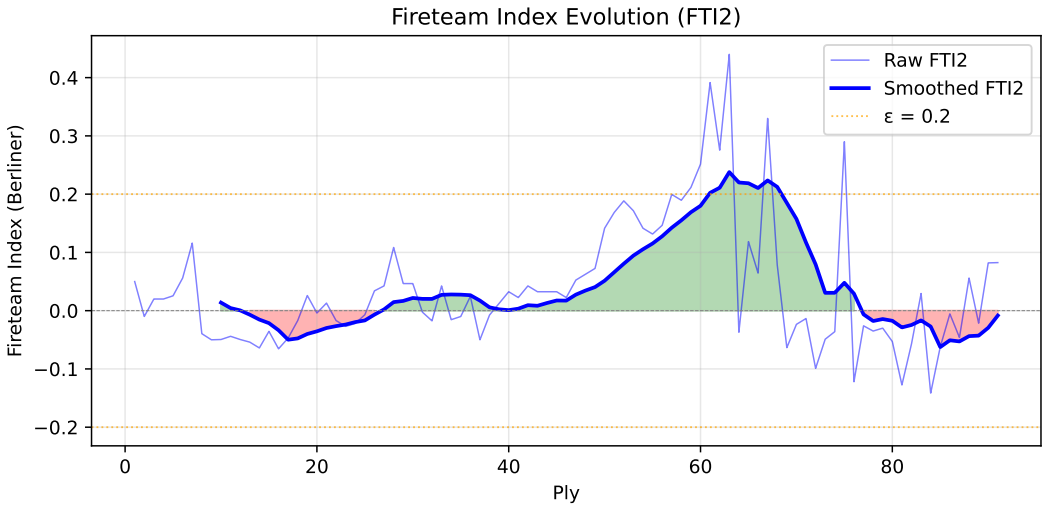
**FTI1: Harmonious** $(\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4})$

| Algorithm | Prediction | Max Streak / Confidence |
|---|---|---|
| Per-ply | WIN | 18 plies |
| Windowed | WIN | 24 plies |
| Balance | WIN | $B_n$=+0.61, C=0.53 |



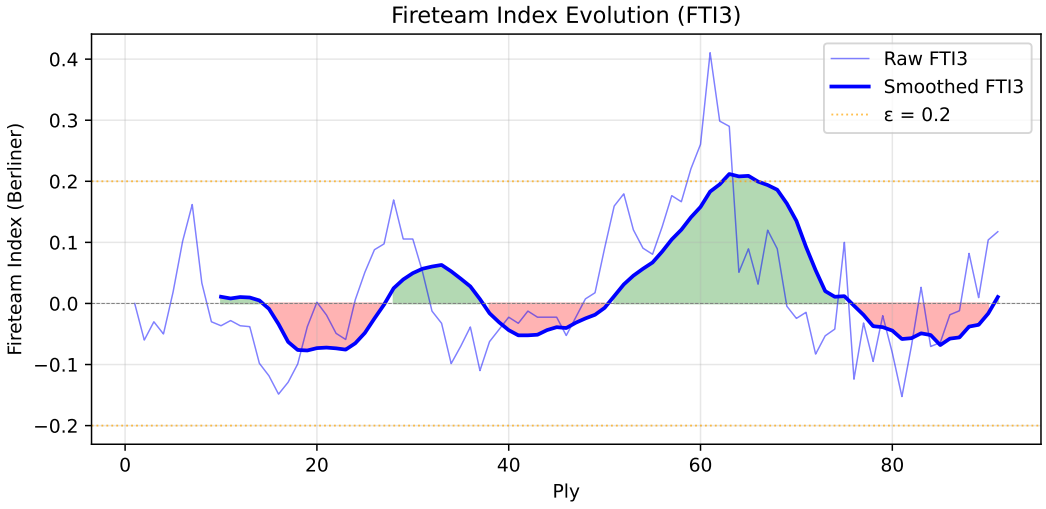

**FTI2: Tactical** $(0.6, 0.1, 0, 0.3)$

| Algorithm | Prediction | Max Streak / Confidence |
|---|---|---|
| Per-ply | DRAW | 4 plies |
| Windowed | DRAW | 0 plies |
| Balance | DRAW | $B_n$=+0.09, C=0.09 |



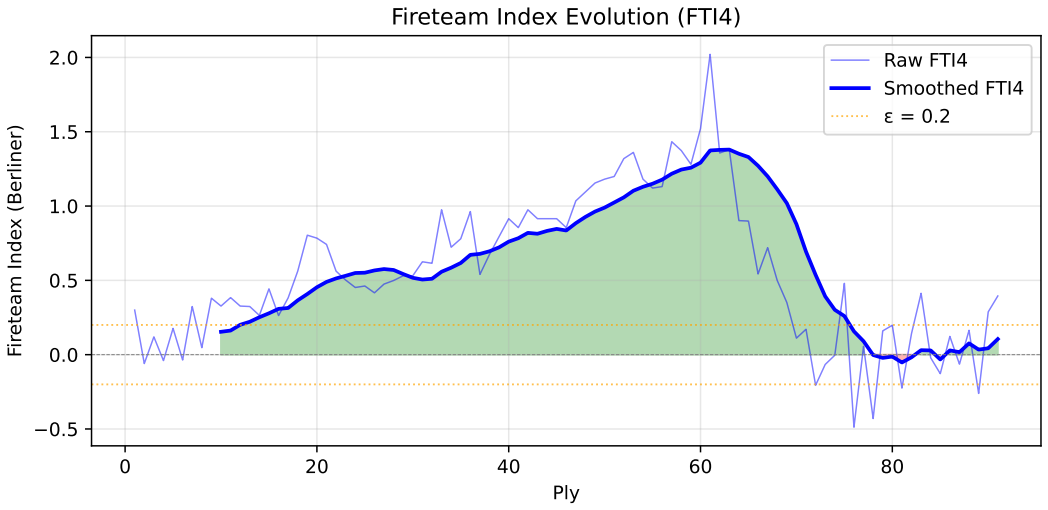

**FTI3: Strategic** $(0.7, 0.1, 0.1, 0.1)$

| Algorithm | Prediction | Max Streak / Confidence |
|---|---|---|
| Per-ply | DRAW | 4 plies |
| Windowed | DRAW | 0 plies |
| Balance | DRAW | $B_n$=+0.07, C=0.07 |



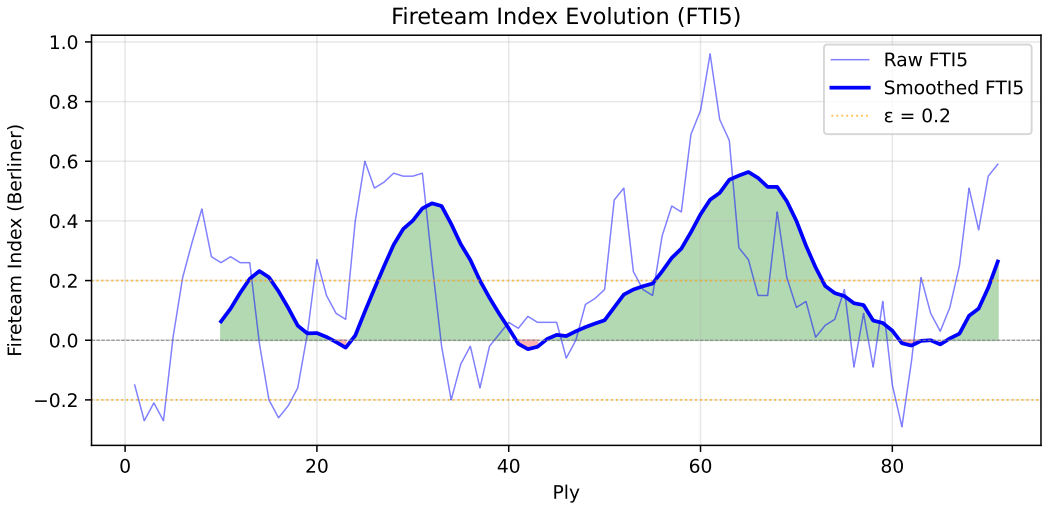

**FTI4: Dynamic** $(0.2, 0.6, 0, 0.2)$

| Algorithm | Prediction | Max Streak / Confidence |
|---|---|---|
| Per-ply | WIN | 53 plies |
| Windowed | WIN | 59 plies |
| Balance | WIN | $B_n$=+0.69, C=0.42 |




**FTI5: Solid** $(0.3, 0.2, 0.5, 0)$

| Algorithm | Prediction | Max Streak / Confidence |
|---|---|---|
| Per-ply | WIN | 10 plies |
| Windowed | WIN | 15 plies |
| Balance | WIN | $B_n$=+0.37, C=0.25 |




Actual result: 1-0 (Berliner's outcome: WIN)

FTI1: Per-ply ✓ Windowed ✓ Balance ✓
FTI2: Per-ply × Windowed × Balance ×
FTI3: Per-ply × Windowed × Balance ×
FTI4: Per-ply ✓ Windowed ✓ Balance ✓
FTI5: Per-ply ✓ Windowed ✓ Balance ✓

## 12.7 Analysis Information

- Engine: Stockfish 16
- Depth: 20
- Analysis time: 202.5 seconds

# Chapter 13

# Josef Richter vs Hans Berliner

## 13.1 Game Information

| | |
|---|---|
| **Event** | 5th Correspondence World Championship |
| **Site** | ICCF |
| **Date** | 1965.07.01 |
| **Round** | 8 |
| **White** | Josef Richter |
| **Black** | Hans Berliner |
| **Result** | 0-1 |
| **Opening** | B03 – Unknown |

## 13.2 Player Statistics

### 13.2.1 Josef Richter (White)

- Total moves: 39
- Accuracy: 90.2%
- Average centipawn loss: 20.6
- Best/Excellent moves: 14 / 5
- Good moves: 13
- Inaccuracies: 3
- Mistakes: 3
- Blunders: 1

### 13.2.2 Hans Berliner (Black)

- Total moves: 39
- Accuracy: 96.5%
- Average centipawn loss: 7.2

- Best/Excellent moves: 23 / 5
- Good moves: 8
- Inaccuracies: 2
- Mistakes: 1
- Blunders: 0

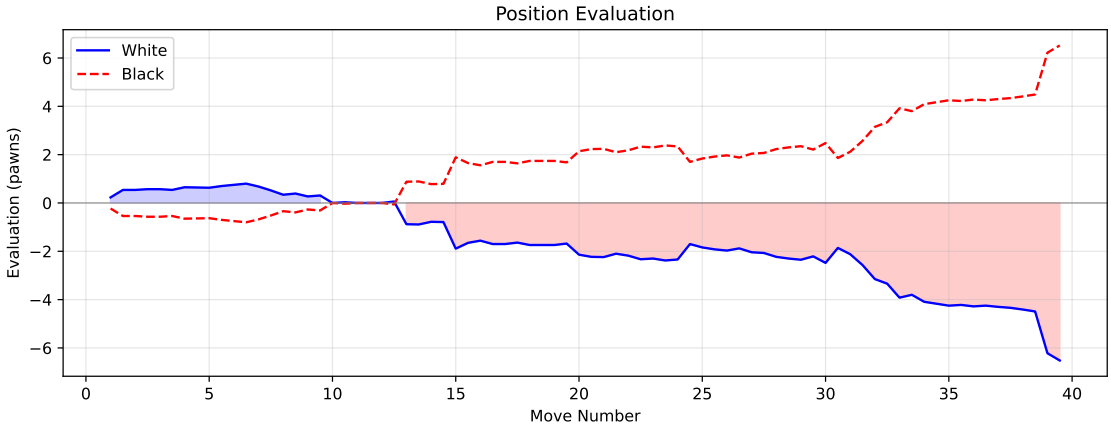
## 13.3 Positional Analysis

This section shows how key positional factors evolved throughout the game.

### 13.3.1 Position Evaluation Over Time

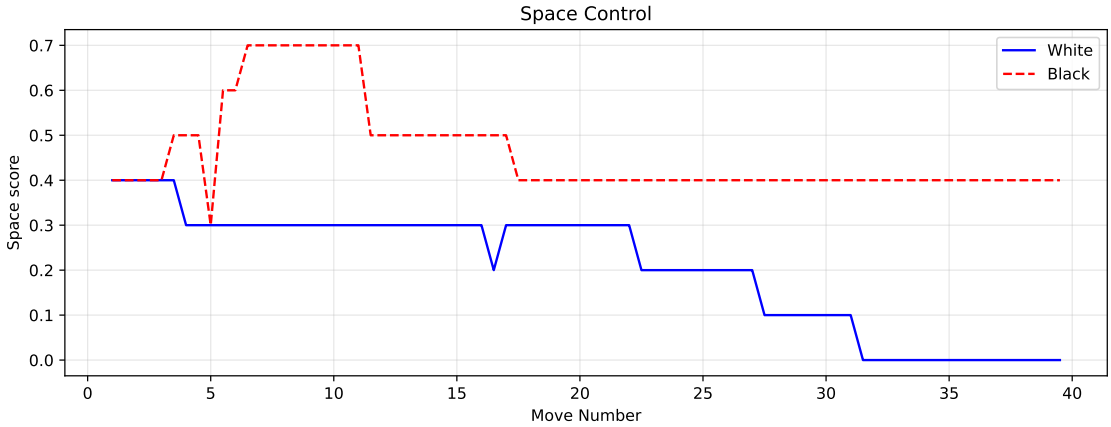
**Game Character Classification.** Evaluation range: minimum $m_1 = -6.52$, maximum $m_2 = +0.80$, spread $d = m_2 - m_1 = 7.32$.

This game is classified as **chaotic** (normal). A chaotic game with moderate swings.




### 13.3.2 Space Control

- White average: 0.21 (range: 0.00 to 0.40)
- Black average: 0.46 (range: 0.30 to 0.70)



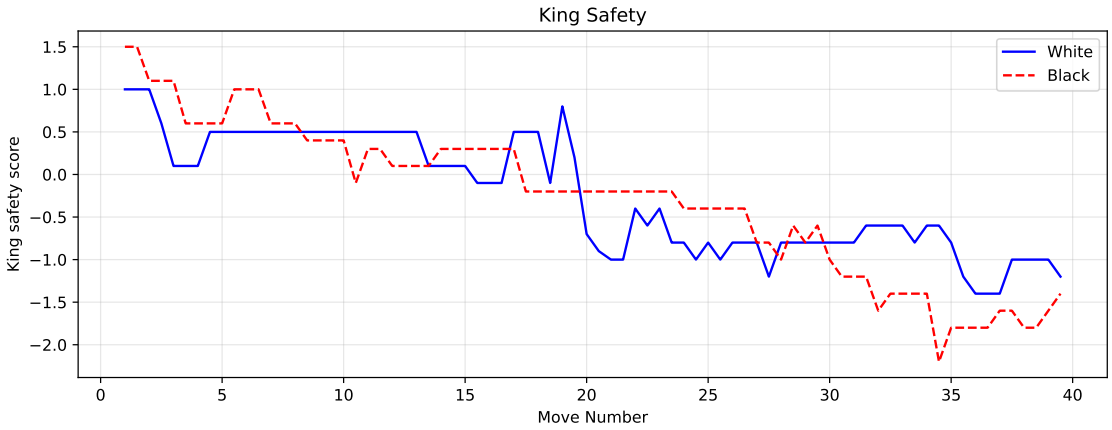
### 13.3.3 Mobility (Piece Activity)

- White average: 3.64 (range: 1.60 to 5.50)
- Black average: 3.68 (range: 2.30 to 5.90)



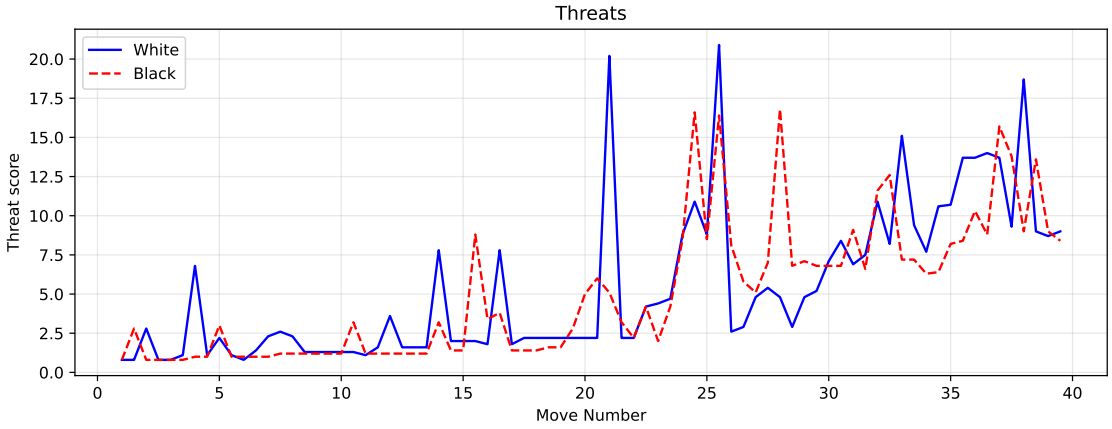

### 13.3.4 King Safety

- White average: -0.24
- Black average: -0.29




### 13.3.5 Threats

- White average: 5.37 (range: 0.80 to 20.90)
- Black average: 5.09 (range: 0.80 to 16.80)
- Average advantage: 0.28 (White maintained more threats)




## 13.4 Annotated Game

1. e4 Nf6?! 2. e5 Nd5 3. d4 d6 4. c4 Nb6 5. exd6 cxd6 6. Nc3 g6 7. Bg5 Bg7 8. Rc1 O-O! 9. h3 Nc6 10. Nf3 h6! 11. Be3! e5 12. d5! Ne7 13. Qb3? *White loses 78cp. Consider instead: Bd3, g4, Be2.*

13...Nf5! 14. Bxb6! axb6! 15. Be2? *White loses 114cp. Consider instead: Bd3, Ne4.*

15...Nd4 16. Qd1! Nxf3+ 17. Bxf3! f5 18. Be2 e4 19. O-O Qh4 20. Kh2 Be5+! 21. g3! Qg5 22. Kh1 f4! 23. Bg4! fxg3 24. Nxe4 Bxg4? *Black loses 66cp. Consider instead: g2+.*

25. Qxg4! Qxg4! 26. hxg4 Rxa2 27. fxg3 Rxb2! 28. Rxf8+ Kxf8! 29. g5! hxg5 30. Rf1+?! Kg7?! 31. Nxg5! Bxg3! 32. Rf7+? *White loses 75cp. Consider instead: Ne6+, Kg1, Rf3.*

32...Kh6! 33. Ne4?! Be5! 34. Nf2 Kg5! 35. Nd3 Rh2+! 36. Kg1! Bg3! 37. Ne1?! Rb2! 38. Nd3 Rd2 39. Nc1?? *White loses 182cp. Consider instead: Rf3.*

39...Bf4!

0-1

## 13.5 Critical Positions

This section highlights critical moments where the evaluation shifted substantially—either due to errors or missed opportunities. Each diagram shows the position *after* the move was played.

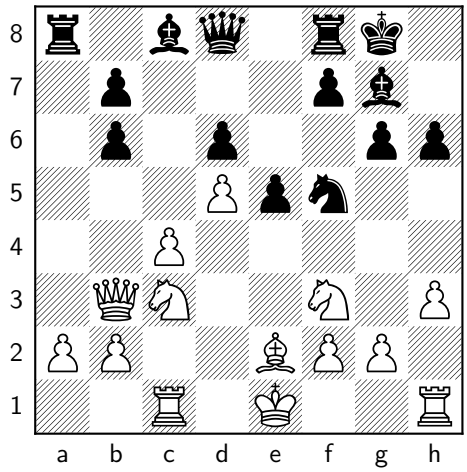
- **Instead of [move]**: The move(s) that should have been played instead.
- **Best continuation**: The optimal sequence of moves from the diagrammed position.

Positions marked with ★ represent the biggest evaluation swings in the game. The positional metrics table summarizes the spatial control, piece activity, king safety, and tactical threats for each side.

### Position 1: After 15. Be2

*Critical moment*

Evaluation: -1.89



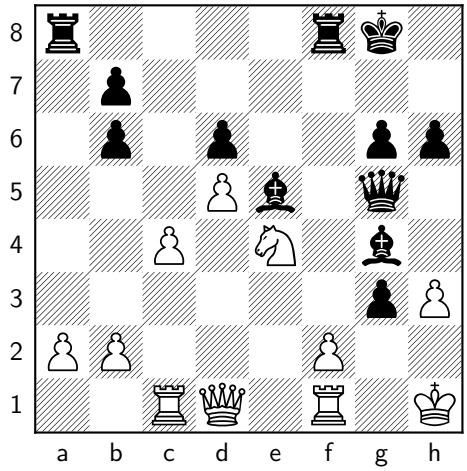
| Metric | White | Black |
|---|---|---|
| Space | 0.30 | 0.50 |
| Mobility | 5.00 | 4.70 |
| King Safety | 0.10 | 0.30 |
| Threats | 2.00 | 1.40 |

Instead of Be2: Bd3, Ne4 Best continuation: Re8 Qc2 Nd4

### Position 2: After 24...Bxg4 ★

*Biggest evaluation swing(s) (64cp)*

Evaluation: -1.70



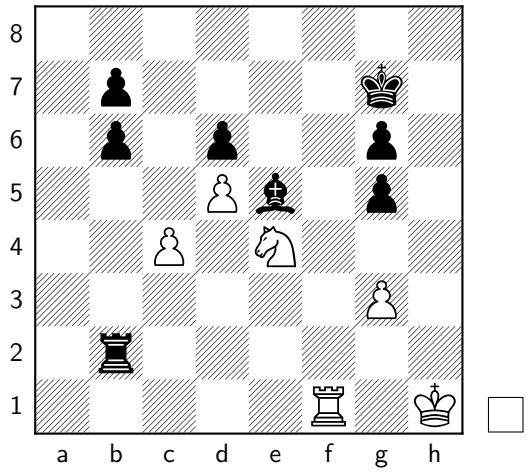
| Metric | White | Black |
|---|---|---|
| Space | 0.20 | 0.40 |
| Mobility | 3.70 | 5.90 |
| King Safety | -1.00 | -0.40 |
| Threats | 10.90 | 16.60 |

Instead of Bxg4: g2+ Best continuation: Qxg4 Qxg4 hxg4

## Position 3: After 30...Kg7 ★

*Biggest evaluation swing(s) (62cp)*

Evaluation: -1.86



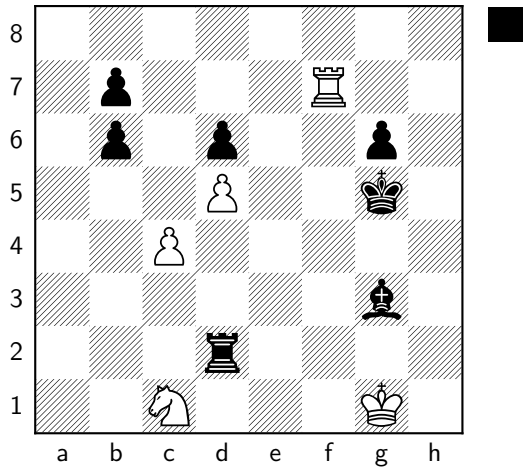
| Metric | White | Black |
|---|---|---|
| Space | 0.10 | 0.40 |
| Mobility | 2.20 | 2.50 |
| King Safety | -0.80 | -1.20 |
| Threats | 8.40 | 6.80 |

Instead of Kg7: Ke8 Best continuation: Nxg5 Rc2 Kg1

## Position 4: After 39. Nc1

*White blunders (-173cp)*

Evaluation: -6.22



| Metric | White | Black |
|---|---|---|
| Space | 0.00 | 0.40 |
| Mobility | 1.90 | 2.40 |
| King Safety | -1.00 | -1.60 |
| Threats | 8.70 | 9.00 |

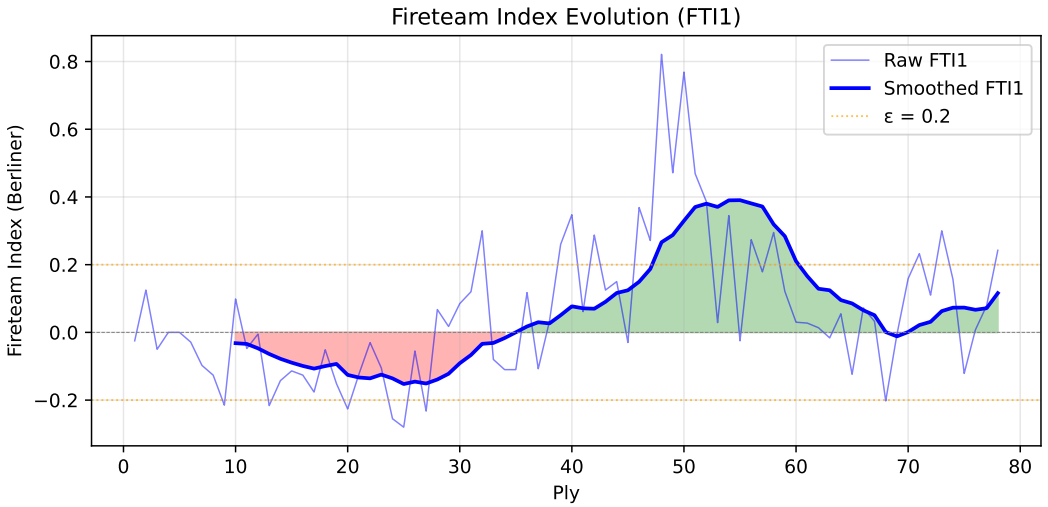
Instead of Nc1: Rf3 Best continuation: Bf4 c5 bxc5

## 13.6 Fireteam Index Prediction

Tracking: **Berliner** (Black)

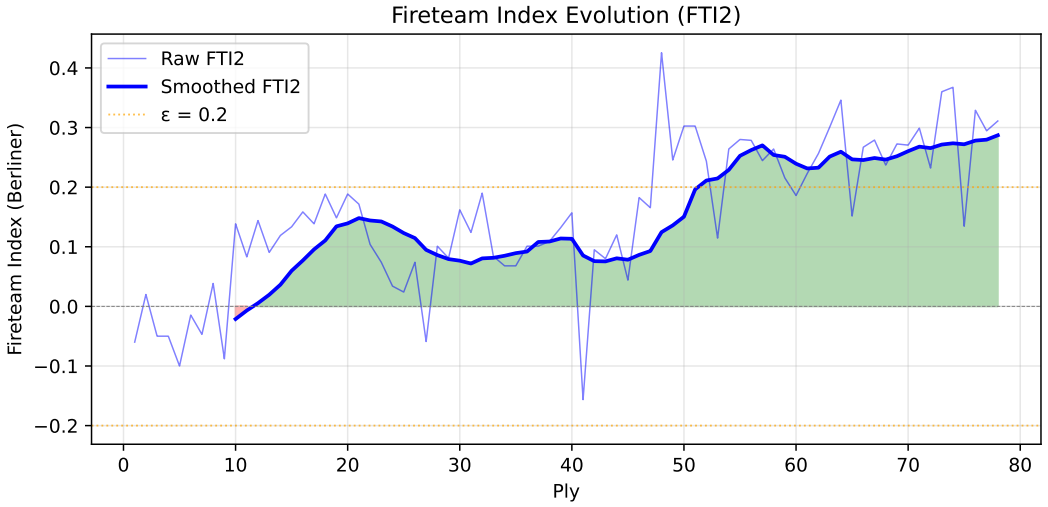
**FTI1: Harmonious** $(\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4})$

| **Algorithm** | **Prediction** | **Max Streak / Confidence** |
|---|---|---|
| Per-ply | DRAW | 7 plies |
| Windowed | WIN | 12 plies |
| Balance | DRAW | $B_n$=+0.19, C=0.15 |



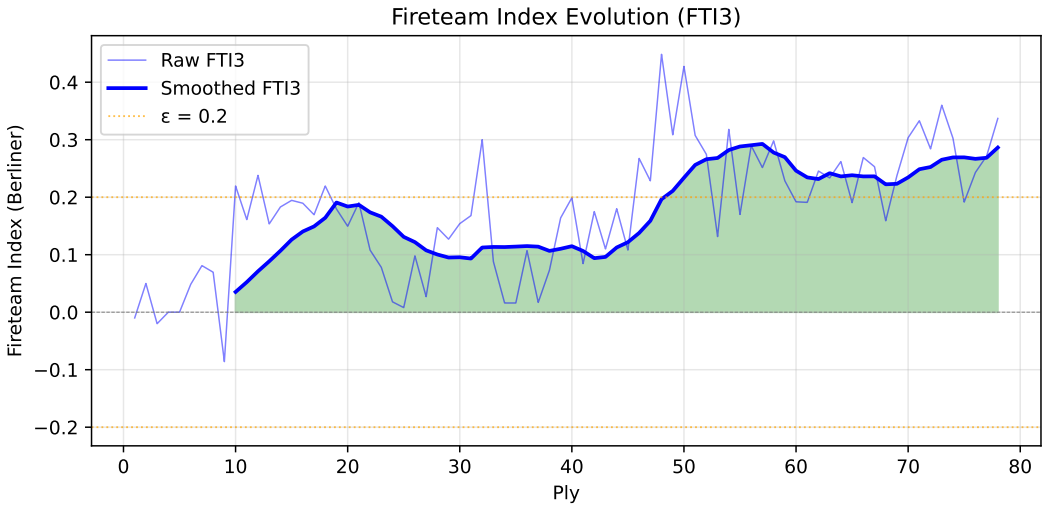

**FTI2: Tactical** $(0.6, 0.1, 0, 0.3)$

| Algorithm | Prediction | Max Streak / Confidence |
|---|---|---|
| Per-ply | DRAW | 3 plies |
| Windowed | WIN | 10 plies |
| Balance | WIN | $B_n$=+0.44, C=0.44 |



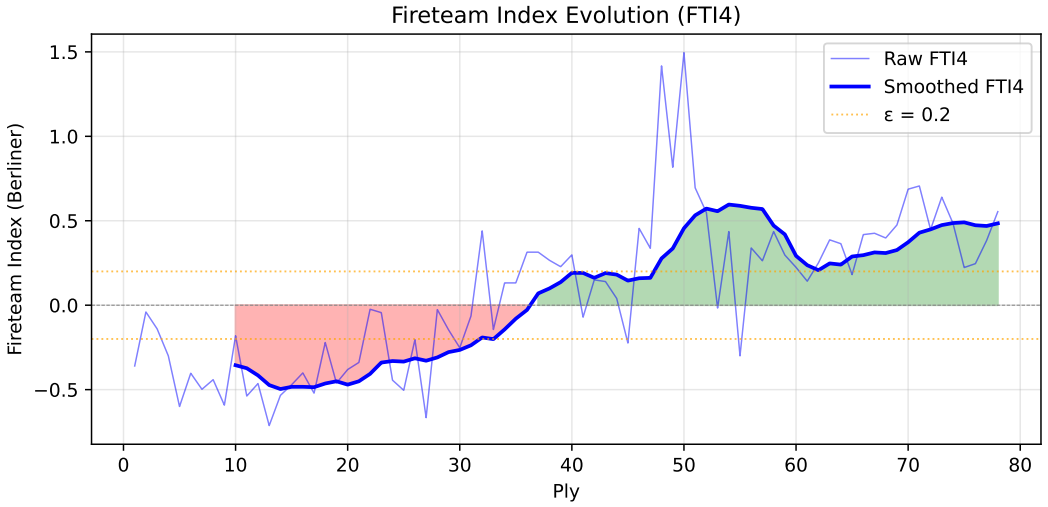

**FTI3: Strategic** $(0.7, 0.1, 0.1, 0.1)$

| Algorithm | Prediction | Max Streak / Confidence |
|---|---|---|
| Per-ply | DRAW | 5 plies |
| Windowed | DRAW | 9 plies |
| Balance | WIN | $B_n$=+0.45, C=0.45 |



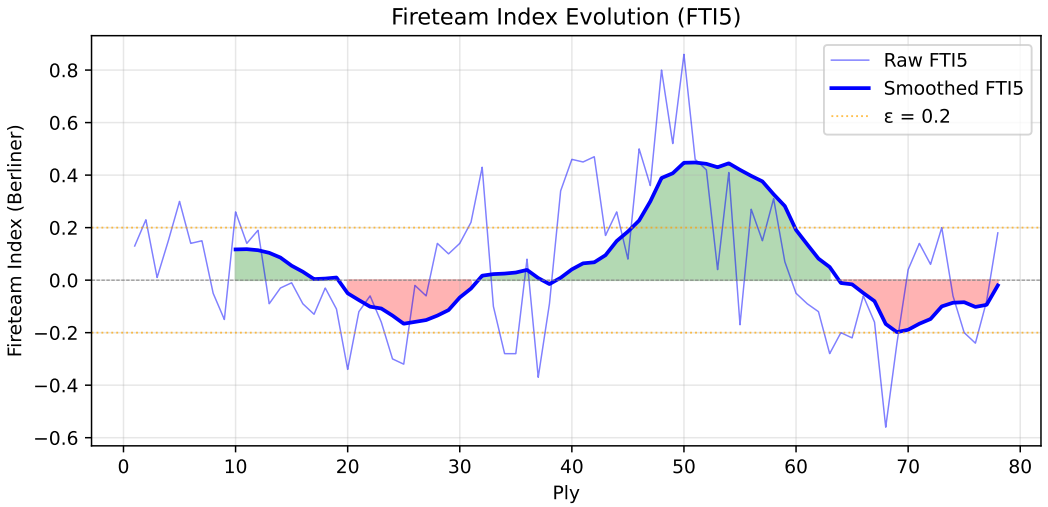

**FTI4: Dynamic** $(0.2, 0.6, 0, 0.2)$

| Algorithm | Prediction | Max Streak / Confidence |
|---|---|---|
| Per-ply | DRAW | 9 plies |
| Windowed | WIN | 14 plies |
| Balance | WIN | $B_n$=+0.37, C=0.22 |




**FTI5: Solid** $(0.3, 0.2, 0.5, 0)$

| Algorithm | Prediction | Max Streak / Confidence |
|---|---|---|
| Per-ply | DRAW | 7 plies |
| Windowed | WIN | 13 plies |
| Balance | DRAW | $B_n$=+0.11, C=0.06 |




Actual result: 0-1 (Berliner's outcome: WIN)
FTI1: Per-ply × Windowed ✓ Balance ×
FTI2: Per-ply × Windowed ✓ Balance ✓
FTI3: Per-ply × Windowed × Balance ✓
FTI4: Per-ply × Windowed ✓ Balance ✓
FTI5: Per-ply × Windowed ✓ Balance ×

## 13.7 Analysis Information

- Engine: Stockfish 16
- Depth: 20
- Analysis time: 184.7 seconds

# Chapter 14

# Berliner Hans J (USA) vs Rokhlin Yakov (RUS)

## 14.1 Game Information

| | |
|---|---|
| **Event** | Ch World (corr.) |
| **Site** | corr. |
| **Date** | 1965.??.?? |
| **Round** | 9 |
| **White** | Berliner Hans J (USA) |
| **Black** | Rokhlin Yakov (RUS) |
| **Result** | 1-0 |
| **Opening** | E31 – Unknown |

## 14.2 Player Statistics

### 14.2.1 Berliner Hans J (USA) (White)

- Total moves: 46
- Accuracy: 92.0%
- Average centipawn loss: 16.7
- Best/Excellent moves: 17 / 9
- Good moves: 9
- Inaccuracies: 8
- Mistakes: 3
- Blunders: 0

### 14.2.2 Rokhlin Yakov (RUS) (Black)

- Total moves: 46
- Accuracy: 89.8%

- Average centipawn loss: 21.6
- Best/Excellent moves: 15 / 9
- Good moves: 11
- Inaccuracies: 6
- Mistakes: 3
- Blunders: 2

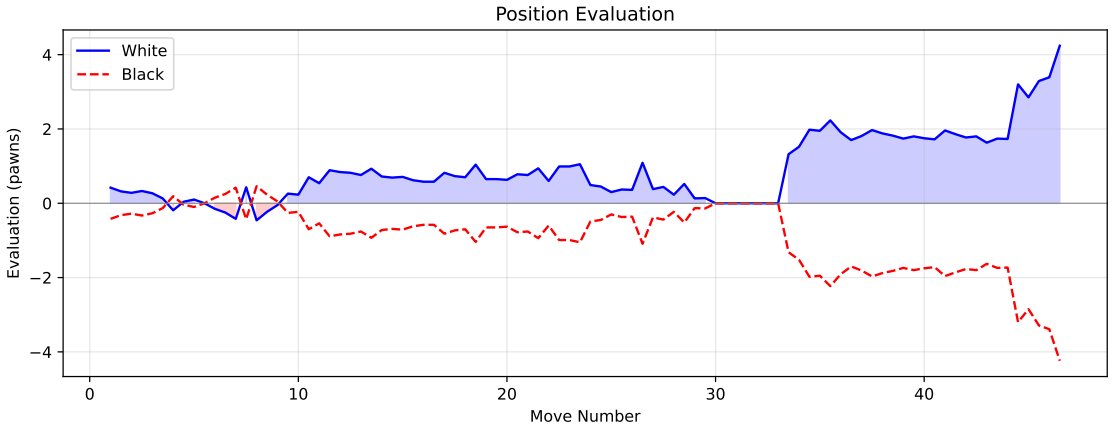
## 14.3 Positional Analysis

This section shows how key positional factors evolved throughout the game.

### 14.3.1 Position Evaluation Over Time

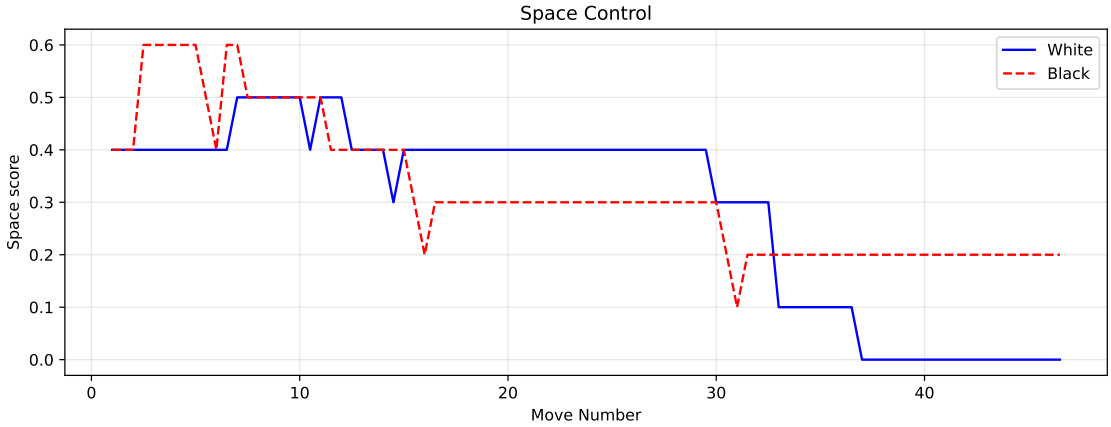
**Game Character Classification.** Evaluation range: minimum $m_1 = -0.46$, maximum $m_2 = +4.24$, spread $d = m_2 - m_1 = 4.70$.

This game is classified as **tactical** (one-sided). A tactical, one-sided game. White maintained the advantage throughout.




### 14.3.2 Space Control

- White average: 0.29 (range: 0.00 to 0.50)
- Black average: 0.32 (range: 0.10 to 0.60)



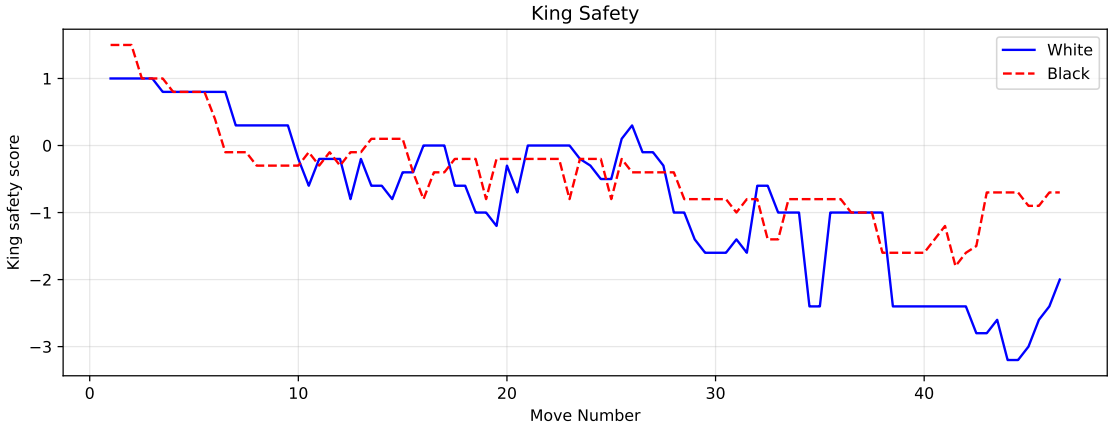

### 14.3.3 Mobility (Piece Activity)

- White average: 3.17 (range: 1.50 to 5.10)
- Black average: 2.83 (range: 1.20 to 4.20)



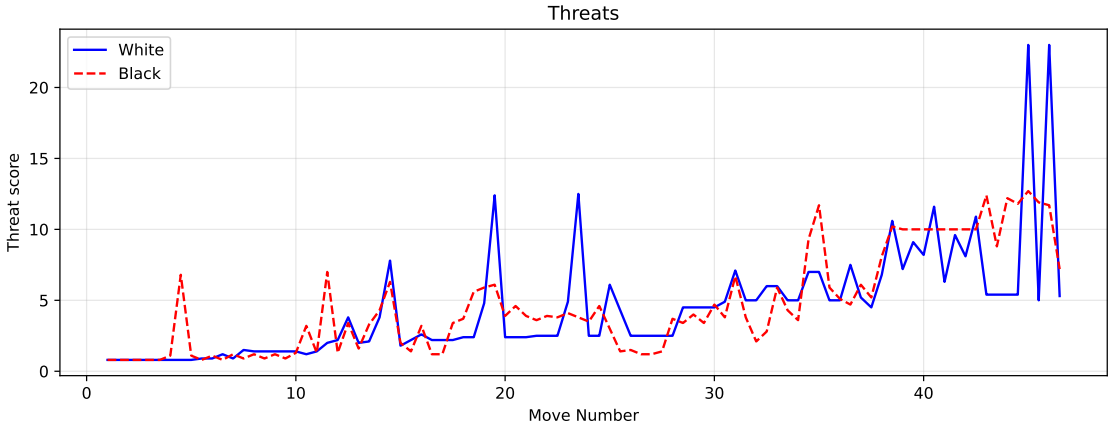

### 14.3.4 King Safety

- White average: -0.76
- Black average: -0.43




### 14.3.5 Threats

- White average: 4.46 (range: 0.80 to 23.00)
- Black average: 4.63 (range: 0.80 to 12.70)
- Average advantage: -0.17 (Black maintained more threats)




## 14.4 Annotated Game

1. d4 Nf6 2. c4 e6 3. Nc3 Bb4 4. Bg5?! h6 5. Bh4 c5 6. d5 d6 7. e3 e5? *Black loses 89cp. Consider instead: Bxc3+, g5, Qe7.*

8. Qc2? *White loses 81cp. Consider instead: Ne2.*

8...Nbd7 9. Bd3 Qe7?! 10. f3! e4?! 11. Bxe4 g5 12. Bg3 Nxe4 13. fxe4 Ne5 14. a3 Bxc3+! 15. Qxc3 f6! 16. Bxe5! fxe5! 17. Ne2! O-O 18. Ng3 Bg4?! 19. Rf1 Rxf1+! 20. Kxf1! Rf8+ 21. Kg1! h5 22. Qc2 Qh7?! 23. Rf1! Rxf1+ 24. Kxf1?! h4! 25. Nf5 Bxf5! 26. exf5 g4? *Black loses 73cp. Consider instead: Qh5.*

27. Qf2? *White loses 72cp. Consider instead: b4, Ke2, Ke1.*

27...Qh6 28. Ke2?! Kf7 29. Kd3?! Qg5! 30. b4 b6! 31. bxc5 bxc5! 32. Kc2 Kf6! 33. e4! Kf7?? *Black loses 132cp. Consider instead: g3.*

34. Kb3! Qc1?! 35. Qe2! Qf4 36. h3?! gxh3! 37. gxh3? *White loses 99cp. Consider instead: Qh5+.*

37...Ke7! 38. Qh5 Qc1! 39. Qxh4+ Kd7 40. Qh7+ Kd8 41. Qg8+?! Kc7 42. Qg7+! Kb6 43. Qg2?! Qb1+?! 44. Kc3! Qc1+?? *Black loses 127cp. Consider instead: Qe1+.*

45. Qc2?! Qe3+ 46. Qd3! Qf4? *Black loses 90cp. Consider instead: Qe1+, Qf2, Qg1.*

1-0

## 14.5 Critical Positions

This section highlights critical moments where the evaluation shifted substantially—either due to errors or missed opportunities. Each diagram shows the position *after* the move was played.

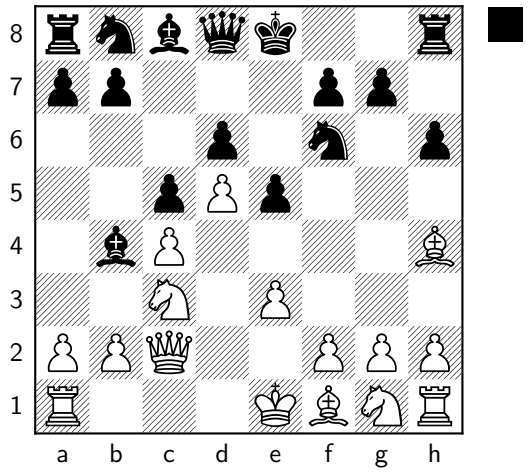
- **Instead of [move]**: The move(s) that should have been played instead.
- **Best continuation**: The optimal sequence of moves from the diagrammed position.

Positions marked with ★ represent the biggest evaluation swings in the game. The positional metrics table summarizes the spatial control, piece activity, king safety, and tactical threats for each side.

### Position 1: After 8. Qc2 ★

*Biggest evaluation swing(s) (89cp)*

Evaluation: -0.46



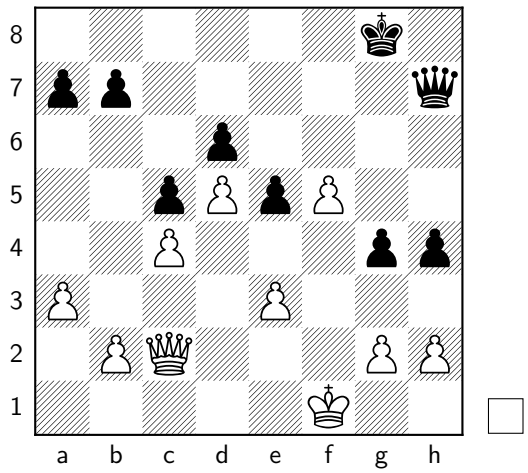
| Metric | White | Black |
|---|---|---|
| Space | 0.50 | 0.50 |
| Mobility | 4.80 | 4.00 |
| King Safety | 0.30 | -0.30 |
| Threats | 1.40 | 1.20 |

Instead of Qc2: Ne2 Best continuation: Bxc3+ bxc3 g5

## Position 2: After 26...g4 ★

*Biggest evaluation swing(s) (73cp)*

Evaluation: +1.09



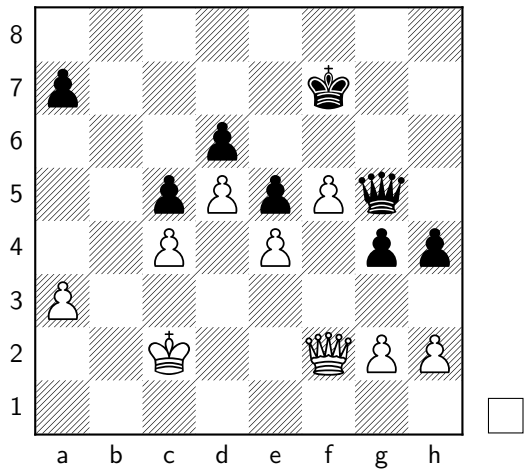
| Metric | White | Black |
|---|---|---|
| Space | 0.40 | 0.30 |
| Mobility | 2.20 | 1.60 |
| King Safety | -0.10 | -0.40 |
| Threats | 2.50 | 1.20 |

Instead of g4: Qh5 Best continuation: b4 g3 hxg3

## Position 3: After 33...Kf7

*Black blunders (+132cp)*

Evaluation: +1.32



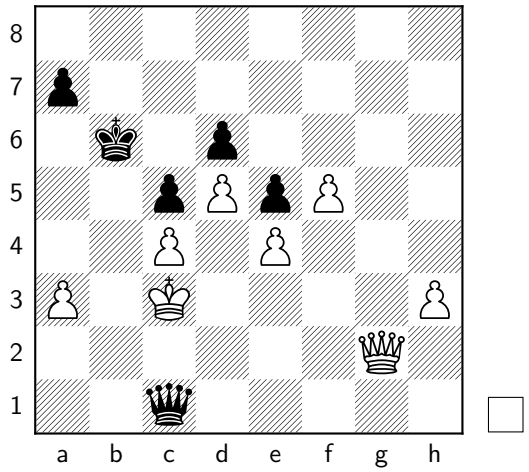
| Metric | White | Black |
|---|---|---|
| Space | 0.10 | 0.20 |
| Mobility | 1.80 | 1.90 |
| King Safety | -1.00 | -0.80 |
| Threats | 5.00 | 4.30 |

Instead of Kf7: g3 Best continuation: Kb3 Qd8 Qe3

### Position 4: After 44...Qc1+

*Black blunders (+147cp)*

Evaluation: +3.20



| Metric | White | Black |
|---|---|---|
| Space | 0.00 | 0.20 |
| Mobility | 2.80 | 2.20 |
| King Safety | -3.20 | -0.70 |
| Threats | 5.40 | 11.80 |

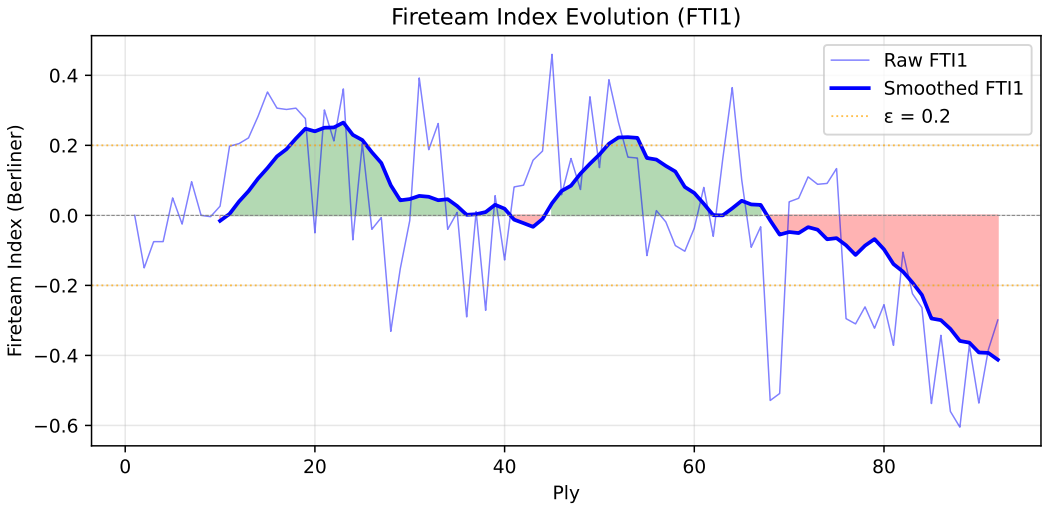
Instead of Qc1+: Qe1+ Best continuation: Qc2 Qg1 Kb2

## 14.6 Fireteam Index Prediction

Tracking: **Berliner** (White)

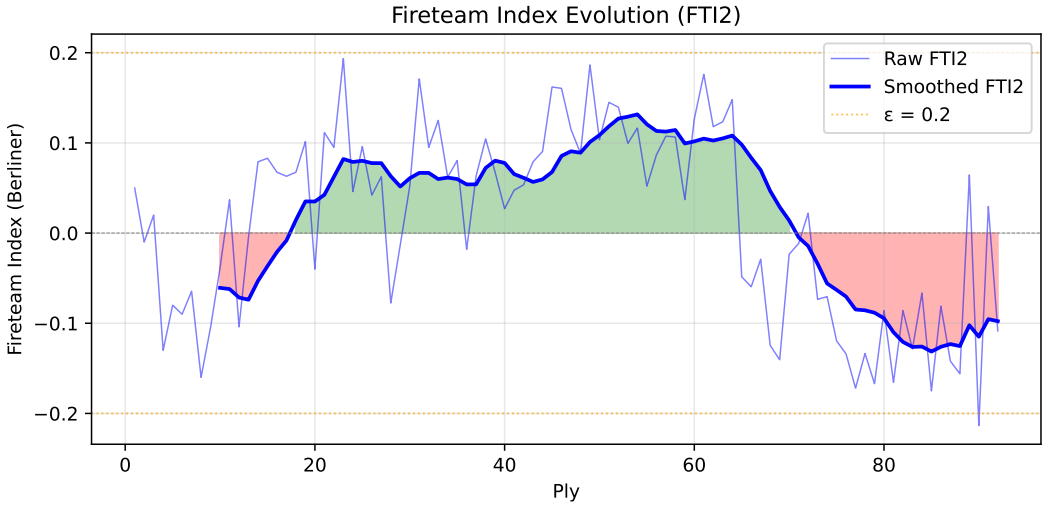
### FTI1: Harmonious ($\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}$)

| Algorithm | Prediction | Max Streak / Confidence |
|---|---|---|
| Per-ply | DRAW | 3 plies |
| Windowed | DRAW | 3 plies |
| Balance | DRAW | $B_n$=-0.09, C=0.06 |



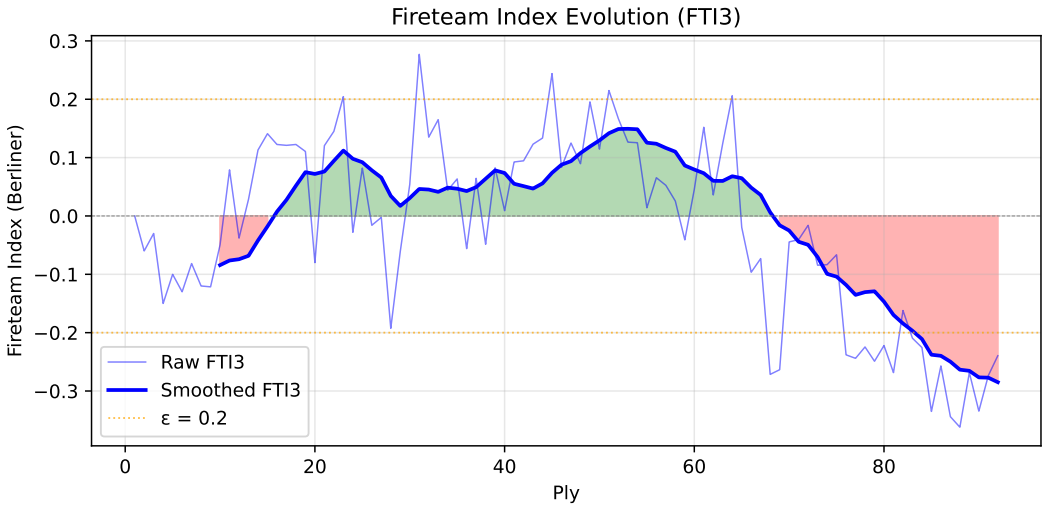

**FTI2: Tactical** $(0.6, 0.1, 0, 0.3)$

| Algorithm | Prediction | Max Streak / Confidence |
|---|---|---|
| Per-ply | DRAW | 0 plies |
| Windowed | DRAW | 0 plies |
| Balance | DRAW | $B_n$=-0.01, C=0.01 |



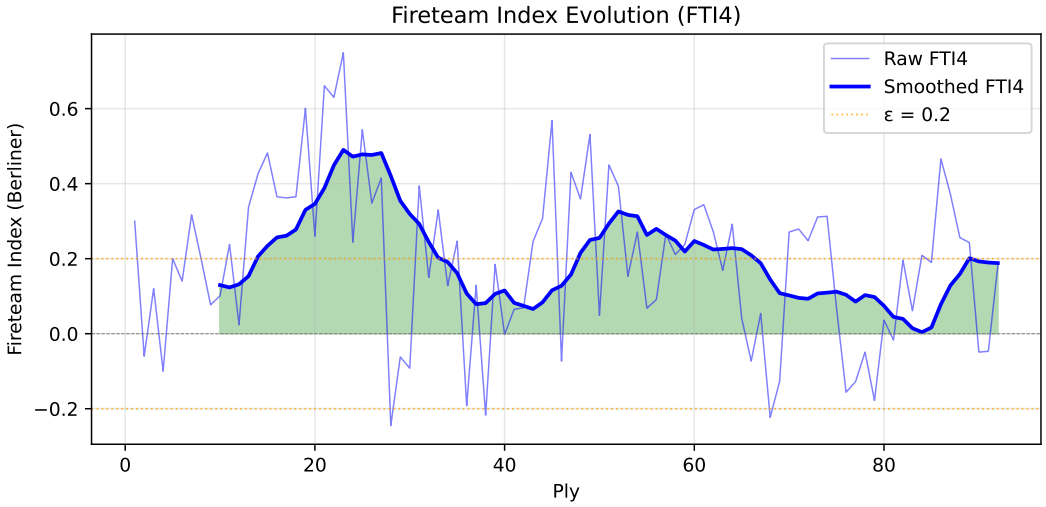

**FTI3: Strategic** $(0.7, 0.1, 0.1, 0.1)$

| Algorithm | Prediction | Max Streak / Confidence |
|---|---|---|
| Per-ply | DRAW | 1 plies |
| Windowed | DRAW | 0 plies |
| Balance | LOSS | $B_n$=-0.17, C=0.16 |



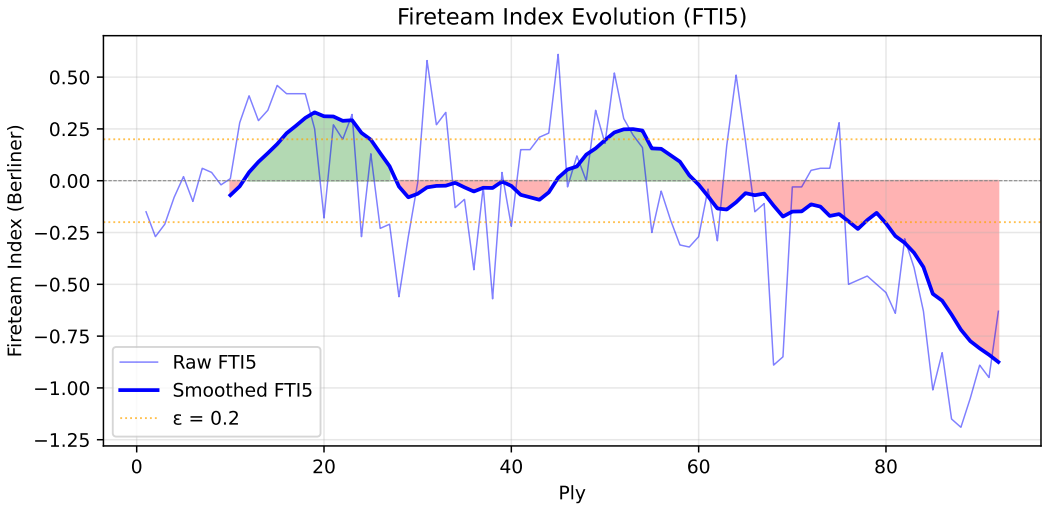

**FTI4: Dynamic** $(0.2, 0.6, 0, 0.2)$

| Algorithm | Prediction | Max Streak / Confidence |
|---|---|---|
| Per-ply | DRAW | 7 plies |
| Windowed | WIN | 15 plies |
| Balance | WIN | $B_n$=+0.49, C=0.29 |




**FTI5: Solid** $(0.3, 0.2, 0.5, 0)$

| Algorithm | Prediction | Max Streak / Confidence |
|---|---|---|
| Per-ply | DRAW | 3 plies |
| Windowed | DRAW | 7 plies |
| Balance | DRAW | $B_n$=-0.20, C=0.08 |




Actual result: 1-0 (Berliner's outcome: WIN)

FTI1: Per-ply × Windowed × Balance ×
FTI2: Per-ply × Windowed × Balance ×
FTI3: Per-ply × Windowed × Balance ×
FTI4: Per-ply × Windowed ✓ Balance ✓
FTI5: Per-ply × Windowed × Balance ×

## 14.7 Analysis Information

- Engine: Stockfish 16
- Depth: 20
- Analysis time: 222.8 seconds

# Chapter 15

# Werner Stern vs Hans Berliner

## 15.1 Game Information

| | |
|---|---|
| **Event** | 5th Correspondence World Championship |
| **Site** | ICCF |
| **Date** | 1965.07.01 |
| **Round** | 11 |
| **White** | Werner Stern |
| **Black** | Hans Berliner |
| **Result** | 0-1 |
| **Opening** | B04 – Unknown |

## 15.2 Player Statistics

### 15.2.1 Werner Stern (White)

- Total moves: 20
- Accuracy: 83.0%
- Average centipawn loss: 37.2
- Best/Excellent moves: 0 / 4
- Good moves: 7
- Inaccuracies: 6
- Mistakes: 1
- Blunders: 2

### 15.2.2 Hans Berliner (Black)

- Total moves: 20
- Accuracy: 94.2%
- Average centipawn loss: 11.8

- Best/Excellent moves: 9 / 3
- Good moves: 5
- Inaccuracies: 2
- Mistakes: 1
- Blunders: 0

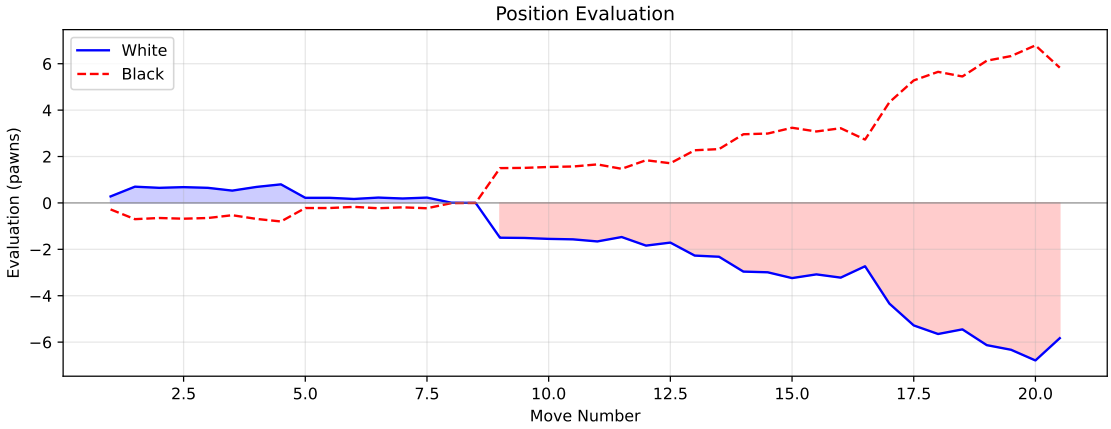
## 15.3 Positional Analysis

This section shows how key positional factors evolved throughout the game.

### 15.3.1 Position Evaluation Over Time

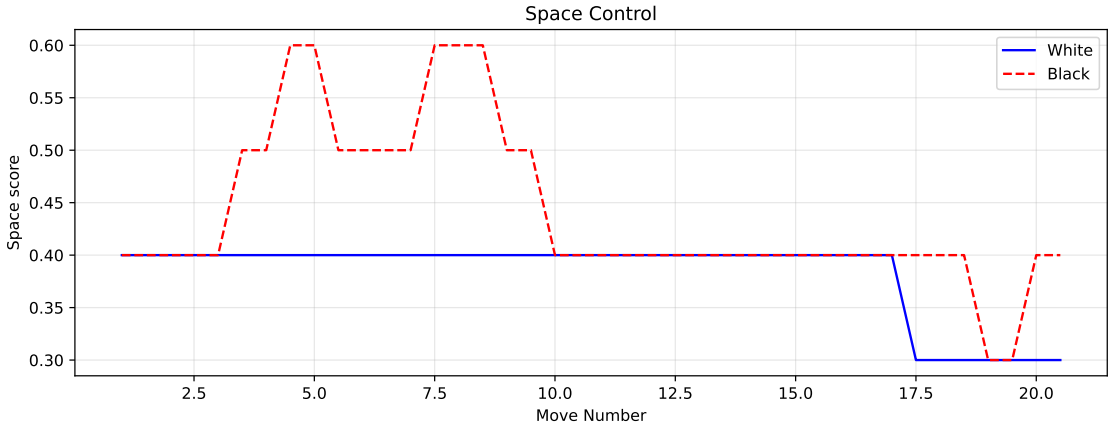
**Game Character Classification.** Evaluation range: minimum $m_1 = -6.79$, maximum $m_2 = +0.80$, spread $d = m_2 - m_1 = 7.59$.

This game is classified as **chaotic** (normal). A chaotic game with moderate swings.



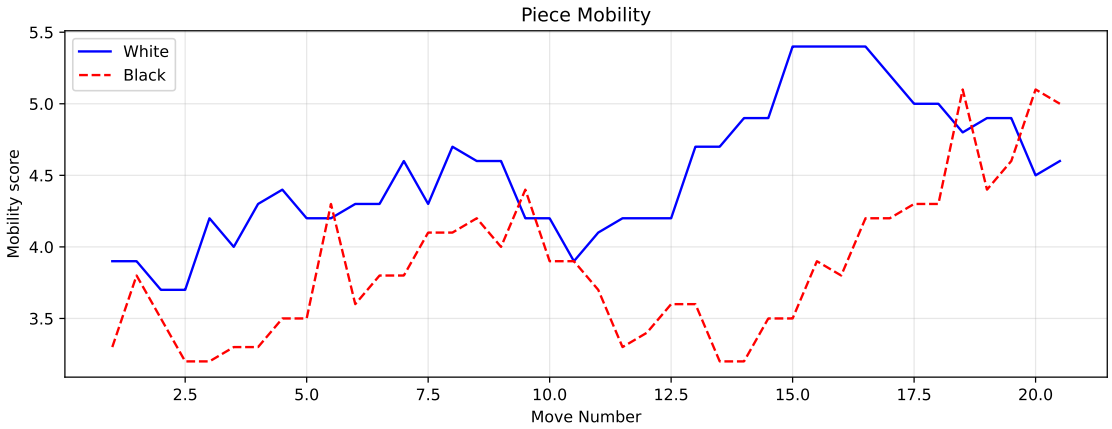

### 15.3.2 Space Control

- White average: 0.38 (range: 0.30 to 0.40)
- Black average: 0.44 (range: 0.30 to 0.60)



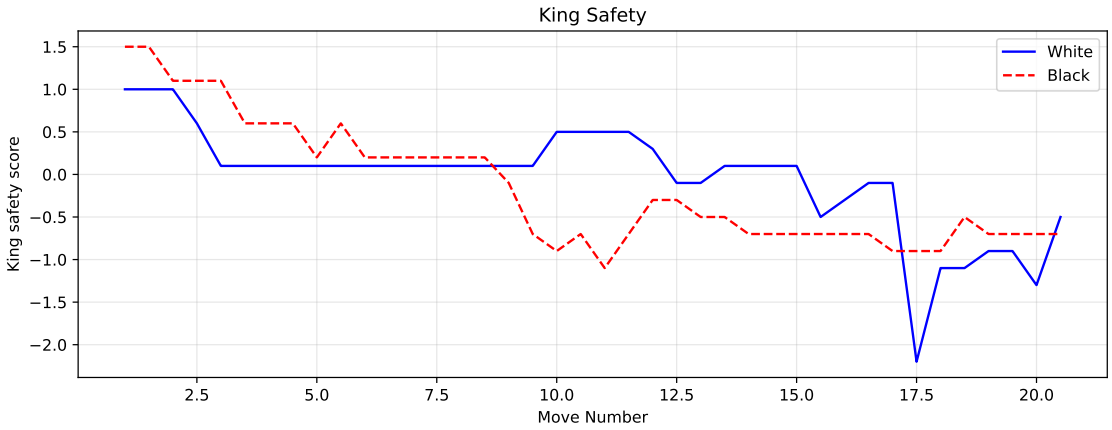
### 15.3.3 Mobility (Piece Activity)

- White average: 4.52 (range: 3.70 to 5.40)
- Black average: 3.87 (range: 3.20 to 5.10)



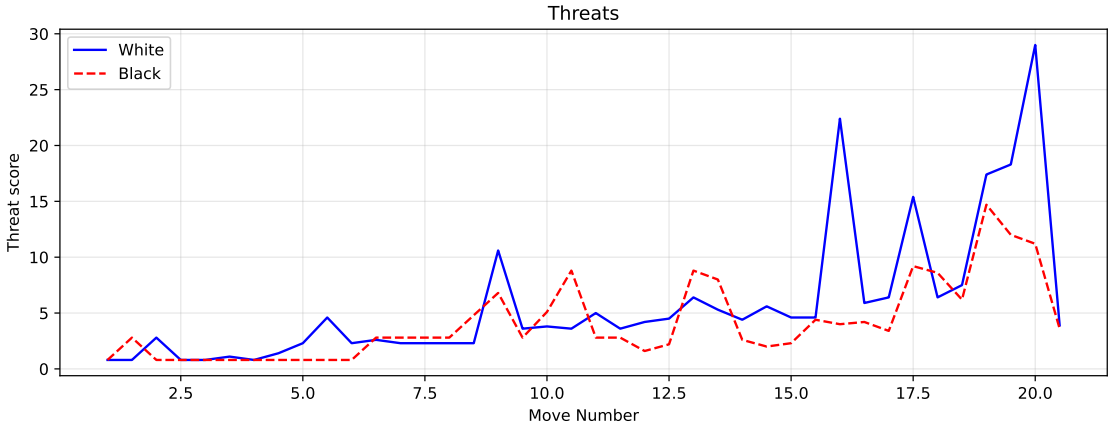

### 15.3.4 King Safety

- White average: -0.04
- Black average: -0.15




### 15.3.5 Threats

- White average: 5.82 (range: 0.80 to 29.00)
- Black average: 4.12 (range: 0.80 to 14.70)
- Average advantage: 1.69 (White maintained more threats)




## 15.4 Annotated Game

1. e4 Nf6?! 2. e5 Nd5 3. d4 d6 4. Nf3 g6 5. Ng5?! dxe5 6. dxe5 Bg7 7. Bc4 c6 8. Nc3 h6! 9. Nxf7?? *White loses 150cp. Consider instead: Nge4, Nf3.*

9...Kxf7 10. Nxd5 cxd5! 11. Bxd5+ Ke8 12. e6?! Rf8! 13. Qg4?! Rf6! 14. Qc4? *White loses 61cp. Consider instead: Qe4, Qa4+.*

14...Na6! 15. Be3?! Qa5+! 16. Bd2 Qb6?! 17. Bc3?? *White loses 162cp. Consider instead: O-O, O-O-O.*

17...Qxf2+! 18. Kd1 Bxe6 19. Bxf6?! Rd8! 20. Bh4?! Qb6? *Black loses 82cp. Consider instead: Qe3.*

0-1

## 15.5 Critical Positions

This section highlights critical moments where the evaluation shifted substantially—either due to errors or missed opportunities. Each diagram shows the position *after* the move was played.

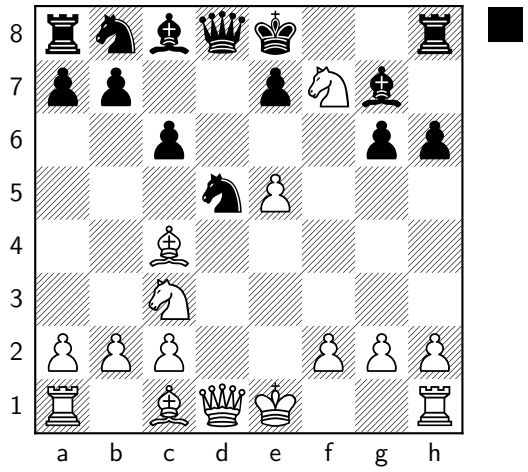
- **Instead of [move]**: The move(s) that should have been played instead.
- **Best continuation**: The optimal sequence of moves from the diagrammed position.

Positions marked with ★ represent the biggest evaluation swings in the game. The positional metrics table summarizes the spatial control, piece activity, king safety, and tactical threats for each side.

### Position 1: After 9. Nxf7

*White blunders (-150cp)*

Evaluation: -1.50



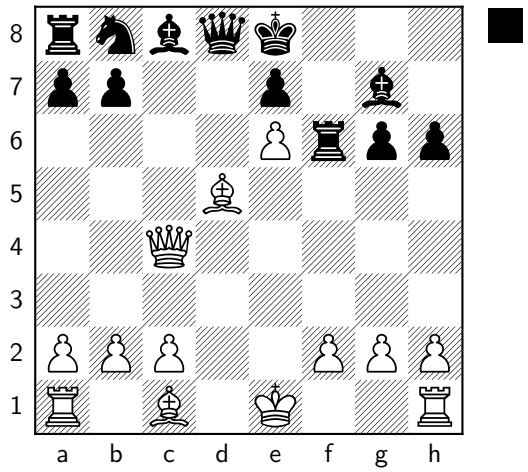
| Metric | White | Black |
|---|---|---|
| Space | 0.40 | 0.50 |
| Mobility | 4.60 | 4.00 |
| King Safety | 0.10 | -0.10 |
| Threats | 10.60 | 6.80 |

Instead of Nxf7: Nge4, Nf3 Best continuation: Kxf7 Nxd5 cxd5

## Position 2: After 14. Qc4 ★

*Biggest evaluation swing(s) (64cp)*

Evaluation: -2.96



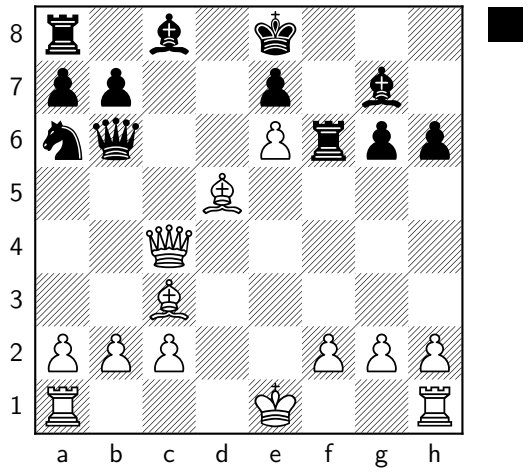
| Metric | White | Black |
|---|---|---|
| Space | 0.40 | 0.40 |
| Mobility | 4.90 | 3.20 |
| King Safety | 0.10 | -0.70 |
| Threats | 4.40 | 2.60 |

Instead of Qc4: Qe4, Qa4+ Best continuation: Na6 O-O Nc7

### Position 3: After 17. Bc3

*White blunders (-161cp)*

Evaluation: -4.34



| Metric | White | Black |
|---|---|---|
| Space | 0.40 | 0.40 |
| Mobility | 5.20 | 4.20 |
| King Safety | -0.10 | -0.90 |
| Threats | 6.40 | 3.40 |

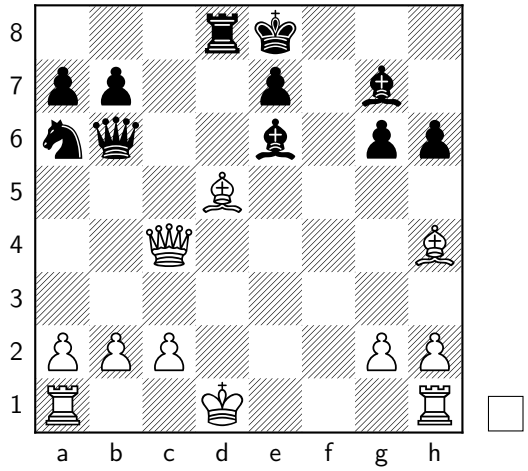
Instead of Bc3: O-O, O-O-O

Best continuation: Qxf2+ Kd1 Bxe6

### Position 4: After 20...Qb6 ★

*Biggest evaluation swing(s) (96cp)*

Evaluation: -5.83



| Metric | White | Black |
|---|---|---|
| Space | 0.30 | 0.40 |
| Mobility | 4.60 | 5.00 |
| King Safety | -0.50 | -0.70 |
| Threats | 3.90 | 3.60 |

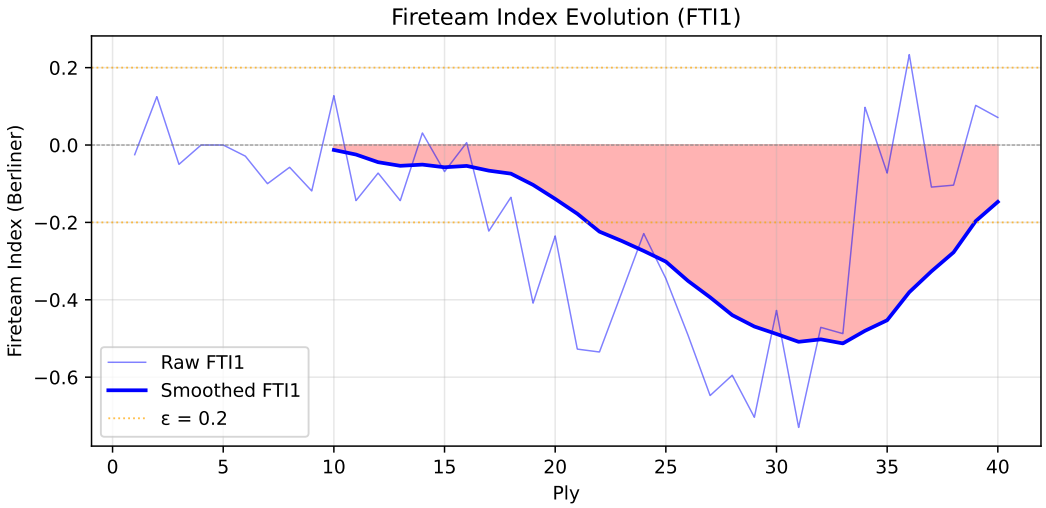
Instead of Qb6: Qe3 Best continuation: Qa4+ Rd7 Re1

## 15.6 Fireteam Index Prediction

Tracking: **Berliner** (Black)

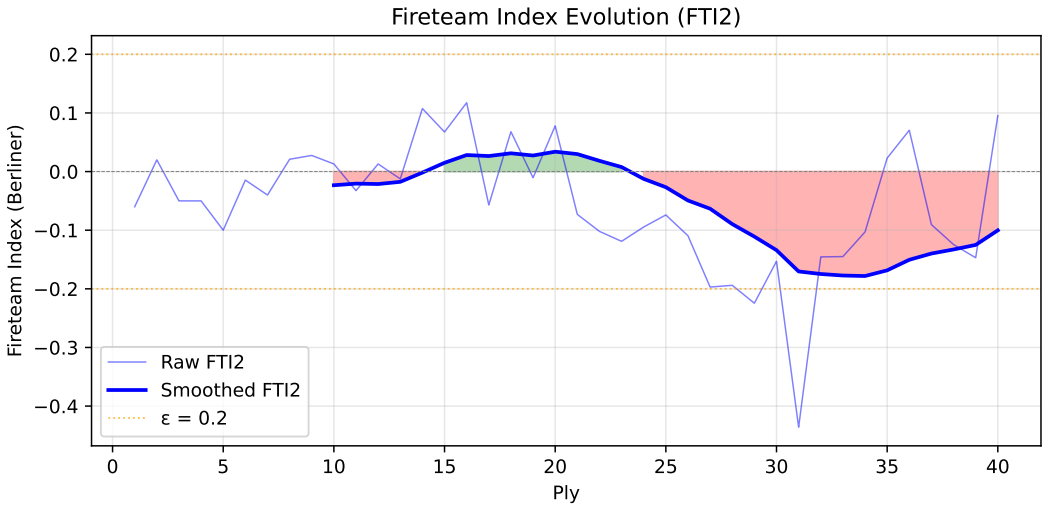
### FTI1: Harmonious $(\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4})$

| Algorithm | Prediction | Max Streak / Confidence |
|---|---|---|
| Per-ply | DRAW | 0 plies |
| Windowed | DRAW | 0 plies |
| Balance | LOSS | $B_n$=-0.62, C=0.49 |



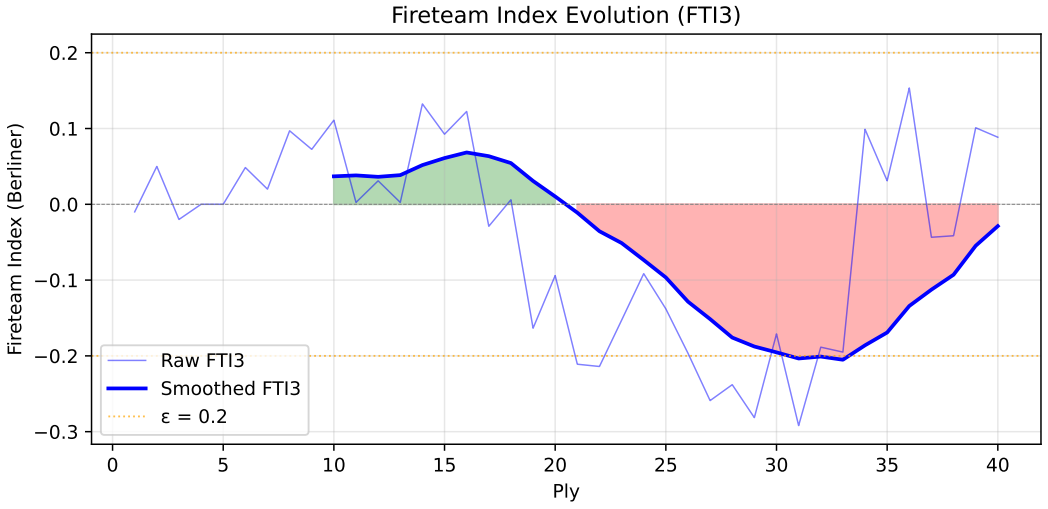

### FTI2: Tactical $(0.6, 0.1, 0, 0.3)$

| Algorithm | Prediction | Max Streak / Confidence |
|---|---|---|
| Per-ply | DRAW | 0 plies |
| Windowed | DRAW | 0 plies |
| Balance | DRAW | $B_n$=-0.08, C=0.08 |



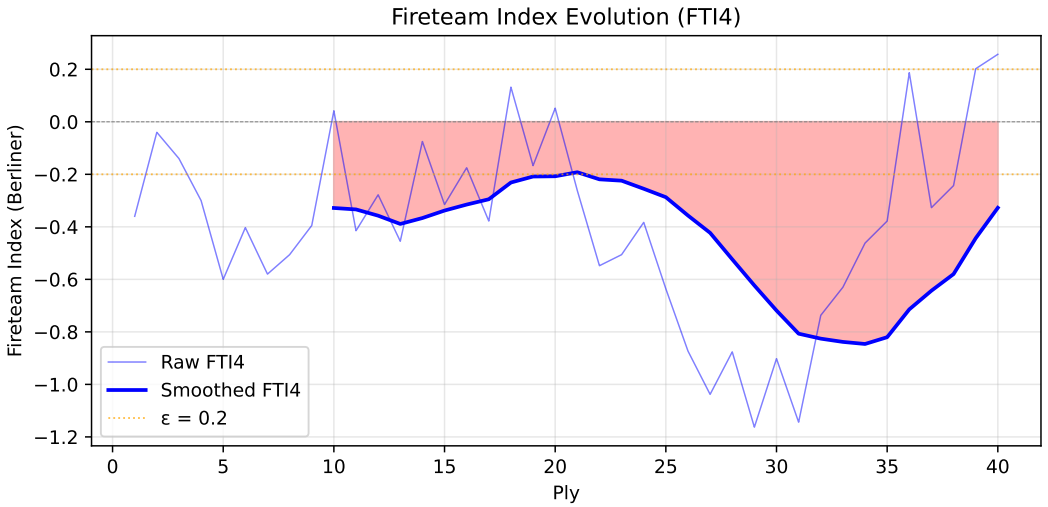

**FTI3: Strategic** $(0.7, 0.1, 0.1, 0.1)$

| Algorithm | Prediction | Max Streak / Confidence |
|---|---|---|
| Per-ply | DRAW | 0 plies |
| Windowed | DRAW | 0 plies |
| Balance | LOSS | $B_n$=-0.25, C=0.25 |



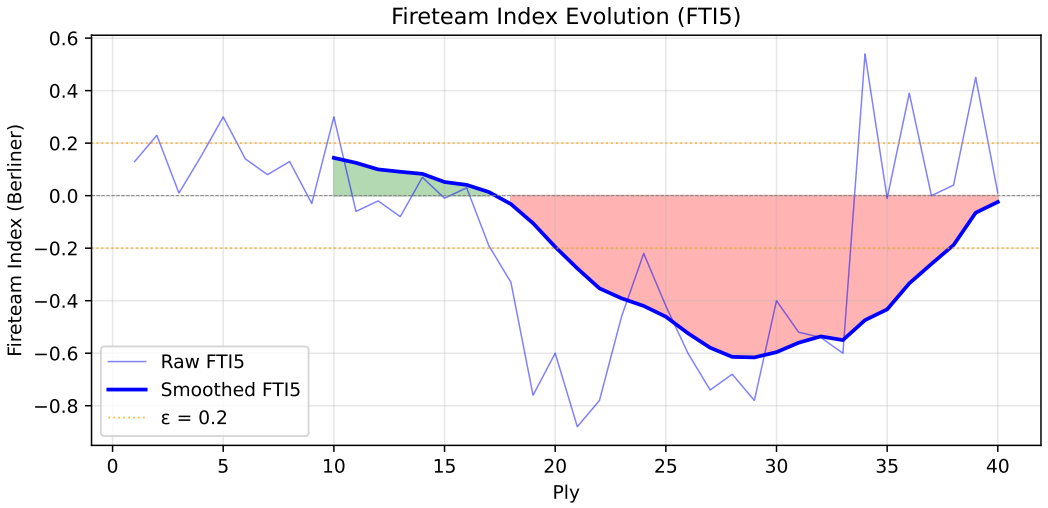

**FTI4: Dynamic** $(0.2, 0.6, 0, 0.2)$

| Algorithm | Prediction | Max Streak / Confidence |
|---|---|---|
| Per-ply | DRAW | 1 plies |
| Windowed | DRAW | 0 plies |
| Balance | LOSS | $B_n$=-0.67, C=0.53 |




**FTI5: Solid** $(0.3, 0.2, 0.5, 0)$

| Algorithm | Prediction | Max Streak / Confidence |
|---|---|---|
| Per-ply | DRAW | 1 plies |
| Windowed | DRAW | 0 plies |
| Balance | LOSS | $B_n$=-0.54, C=0.43 |




Actual result: 0-1 (Berliner's outcome: WIN)

FTI1: Per-ply × Windowed × Balance ×
FTI2: Per-ply × Windowed × Balance ×
FTI3: Per-ply × Windowed × Balance ×
FTI4: Per-ply × Windowed × Balance ×
FTI5: Per-ply × Windowed × Balance ×

## 15.7 Analysis Information

- Engine: Stockfish 16
- Depth: 20
- Analysis time: 110.3 seconds

# Chapter 16

# Hans Berliner vs Vladimir Zagorovsky

## 16.1 Game Information

| | |
|---|---|
| **Event** | 5th Correspondence World Championship |
| **Site** | ICCF |
| **Date** | 1965.07.01 |
| **Round** | 3 |
| **White** | Hans Berliner |
| **Black** | Vladimir Zagorovsky |
| **Result** | 1/2-1/2 |
| **Opening** | D35 – Unknown |

## 16.2 Player Statistics

### 16.2.1 Hans Berliner (White)

- Total moves: 38
- Accuracy: 94.0%
- Average centipawn loss: 12.3
- Best/Excellent moves: 15 / 6
- Good moves: 12
- Inaccuracies: 4
- Mistakes: 1
- Blunders: 0

### 16.2.2 Vladimir Zagorovsky (Black)

- Total moves: 38
- Accuracy: 95.0%
- Average centipawn loss: 10.2

- Best/Excellent moves: 22 / 4
- Good moves: 9
- Inaccuracies: 2
- Mistakes: 0
- Blunders: 1

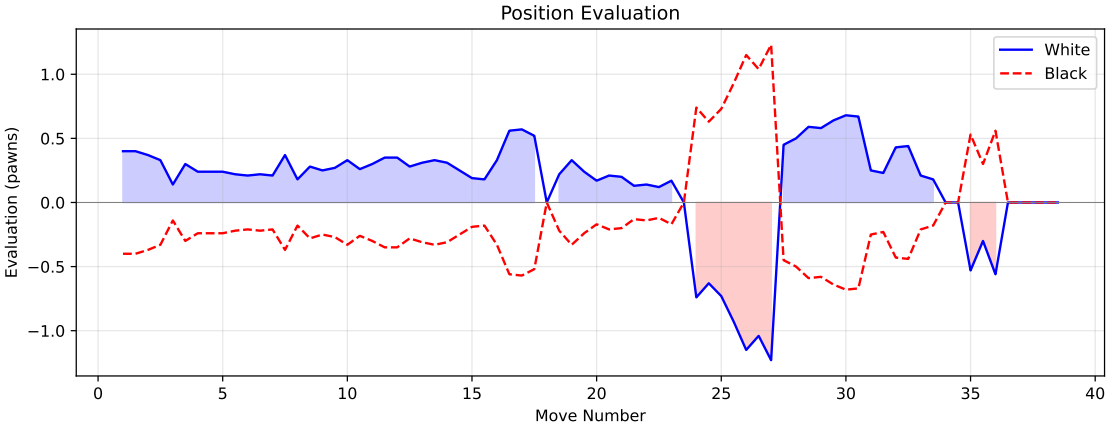
## 16.3 Positional Analysis

This section shows how key positional factors evolved throughout the game.

### 16.3.1 Position Evaluation Over Time

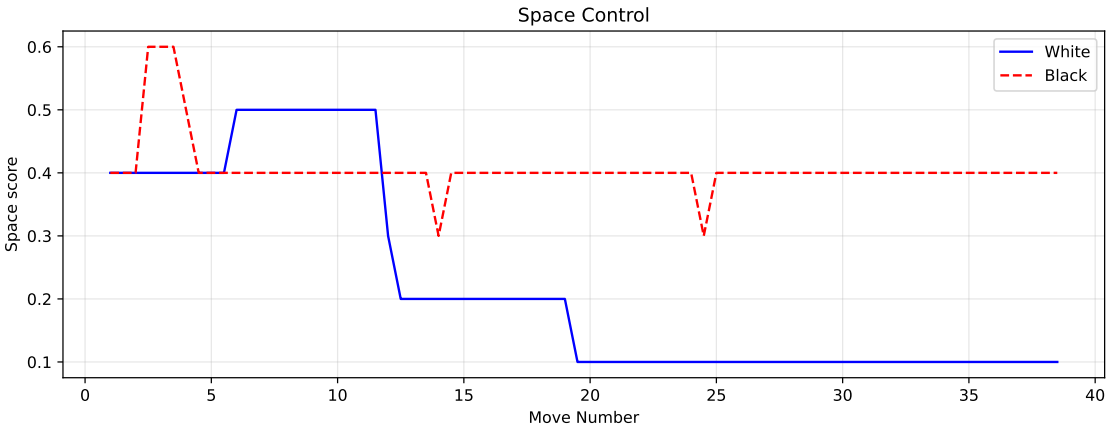
**Game Character Classification.** Evaluation range: minimum $m_1 = -1.23$, maximum $m_2 = +0.68$, spread $d = m_2 - m_1 = 1.91$.

This game is classified as **tense** (normal). A tense game with moderate swings.



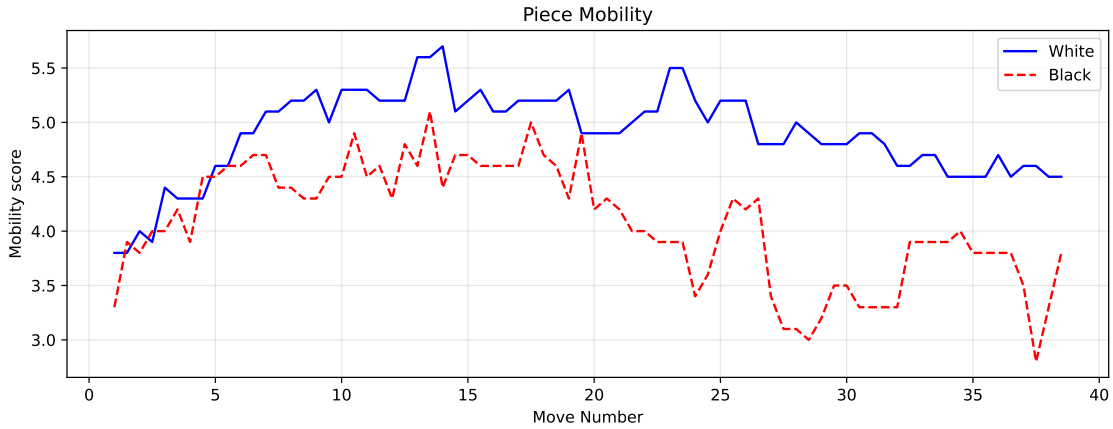

### 16.3.2 Space Control

- White average: 0.22 (range: 0.10 to 0.50)
- Black average: 0.41 (range: 0.30 to 0.60)



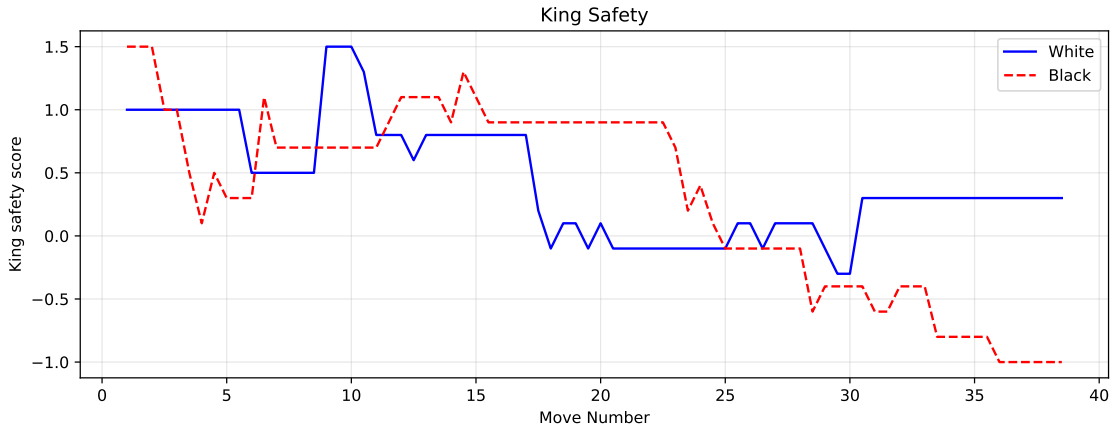
### 16.3.3 Mobility (Piece Activity)

- White average: 4.89 (range: 3.80 to 5.70)
- Black average: 4.08 (range: 2.80 to 5.10)



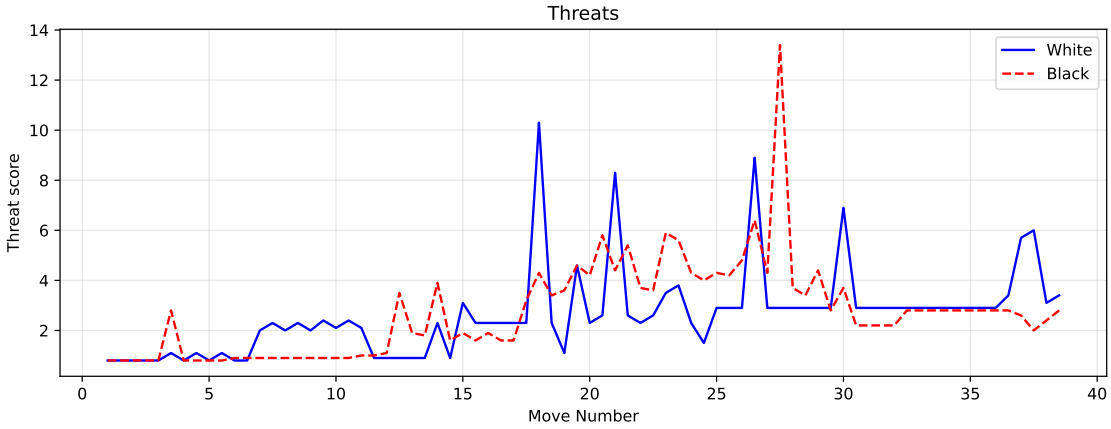

### 16.3.4 King Safety

- White average: 0.44
- Black average: 0.32




### 16.3.5 Threats

- White average: 2.64 (range: 0.80 to 10.30)
- Black average: 2.74 (range: 0.80 to 13.40)
- Average advantage: -0.10 (Black maintained more threats)




## 16.4 Annotated Game

1. d4 Nf6 2. c4 e6 3. Nc3 d5 4. cxd5 exd5 5. Bg5 Be7 6. e3 O-O 7. Bd3! Nbd7 8. Nge2 Re8 9. O-O! c6! 10. Qc2! Nf8! 11. f3 Ng6 12. e4! dxe4! 13. fxe4 Be6! 14. Bxf6 Bxf6! 15. e5! Be7! 16. Rad1 Bf8?! 17. Kh1! Qh4 18. g3?! Qh6 19. Nf4 Nxf4 20. gxf4 Bg4! 21. Be2 Bh3! 22. Rf3! Rad8! 23. Rg3 Kh8 24. f5? *White loses 77cp. Consider instead: Bc4, Bg4, Ne4.*

24...g6! 25. f6 Be6! 26. Bg4?! Bxg4 27. Rxg4! Qh5?? *Black loses 158cp. Consider instead: c5.*

28. Qe2 h6 29. Re4 Qf5! 30. Rf1 Qe6! 31. Qg2?! h5! 32. Qf3 Bh6 33. Rd1 Kh7! 34. Ree1 Rd7! 35. b3?! Red8! 36. Ne4 b6?! 37. Nd6! Bf8! 38. Ne4! Bh6!

1/2-1/2

## 16.5 Critical Positions

This section highlights critical moments where the evaluation shifted substantially—either due to errors or missed opportunities. Each diagram shows the position *after* the move was played.

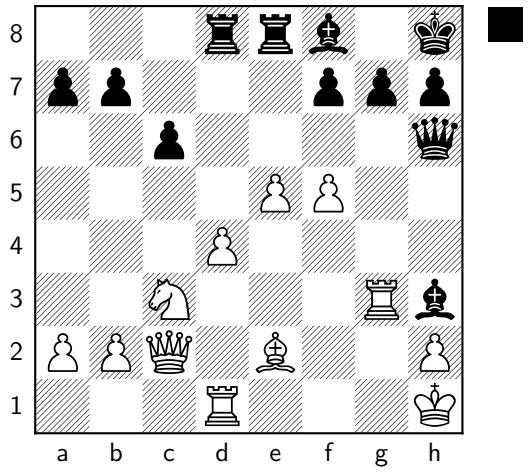
- **Instead of [move]**: The move(s) that should have been played instead.
- **Best continuation**: The optimal sequence of moves from the diagrammed position.

Positions marked with ★ represent the biggest evaluation swings in the game. The positional metrics table summarizes the spatial control, piece activity, king safety, and tactical threats for each side.

### Position 1: After 24. f5 ★

*Biggest evaluation swing(s) (74cp)*

Evaluation: -0.74



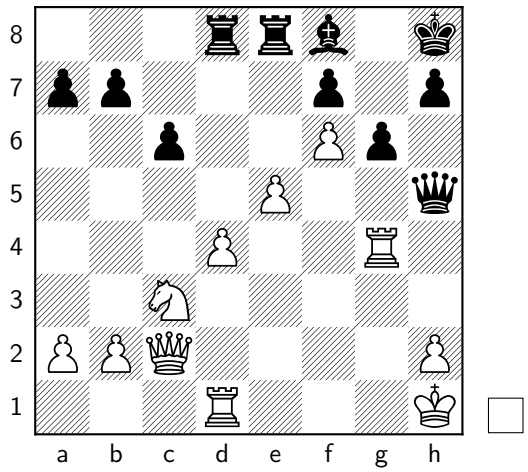
| Metric | White | Black |
|---|---|---|
| Space | 0.10 | 0.40 |
| Mobility | 5.20 | 3.40 |
| King Safety | -0.10 | 0.40 |
| Threats | 2.30 | 4.30 |

Instead of f5: Bc4, Bg4, Ne4 Best continuation: g6 f6 Bf5

### Position 2: After 27...Qh5

*Black blunders (+168cp)*

Evaluation: +0.45



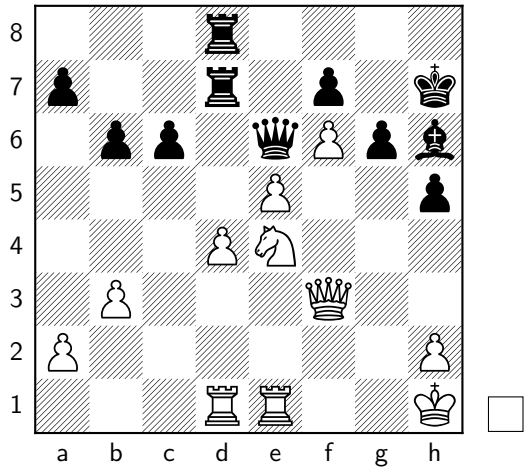
| Metric | White | Black |
|---|---|---|
| Space | 0.10 | 0.40 |
| Mobility | 4.80 | 3.10 |
| King Safety | 0.10 | -0.10 |
| Threats | 2.90 | 13.40 |

Instead of Qh5: c5 Best continuation: Qg2 c5 Rg5

### Position 3: After 36...b6 ★

*Biggest evaluation swing(s) (56cp)*

Evaluation: +0.00



| Metric | White | Black |
|---|---|---|
| Space | 0.10 | 0.40 |
| Mobility | 4.50 | 3.80 |
| King Safety | 0.30 | -1.00 |
| Threats | 3.40 | 2.80 |

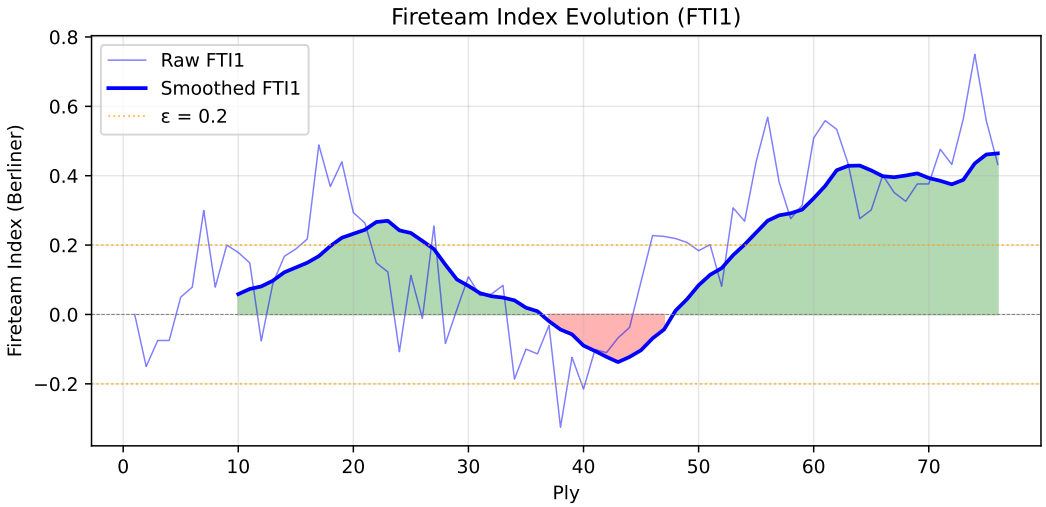
Instead of b6: Rxd4, Rd5 Best continuation: Nd6 Kg8 a4

## 16.6 Fireteam Index Prediction

Tracking: **Berliner** (White)

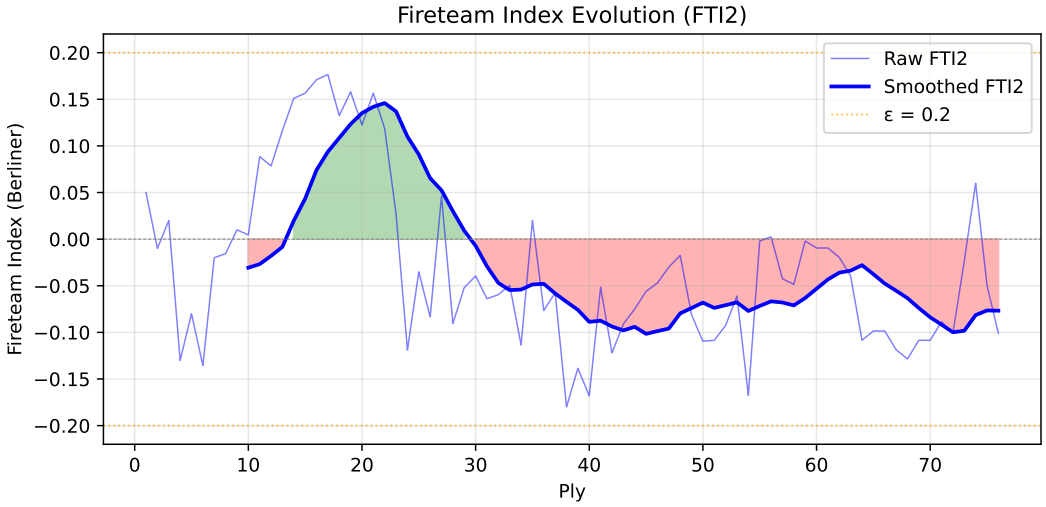
**FTI1: Harmonious** $(\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4})$

| Algorithm | Prediction | Max Streak / Confidence |
|---|---|---|
| Per-ply | WIN | 24 plies |
| Windowed | WIN | 21 plies |
| Balance | WIN | $B_n$=+0.55, C=0.46 |



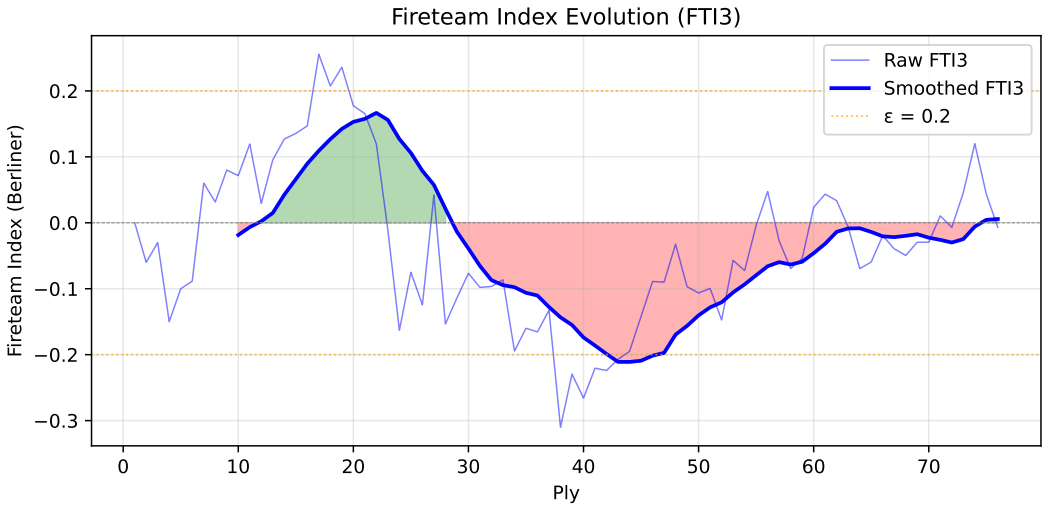

**FTI2: Tactical** $(0.6, 0.1, 0, 0.3)$

| Algorithm | Prediction | Max Streak / Confidence |
|---|---|---|
| Per-ply | DRAW | 0 plies |
| Windowed | DRAW | 0 plies |
| Balance | DRAW | $B_n$=+0.00, C=0.00 |



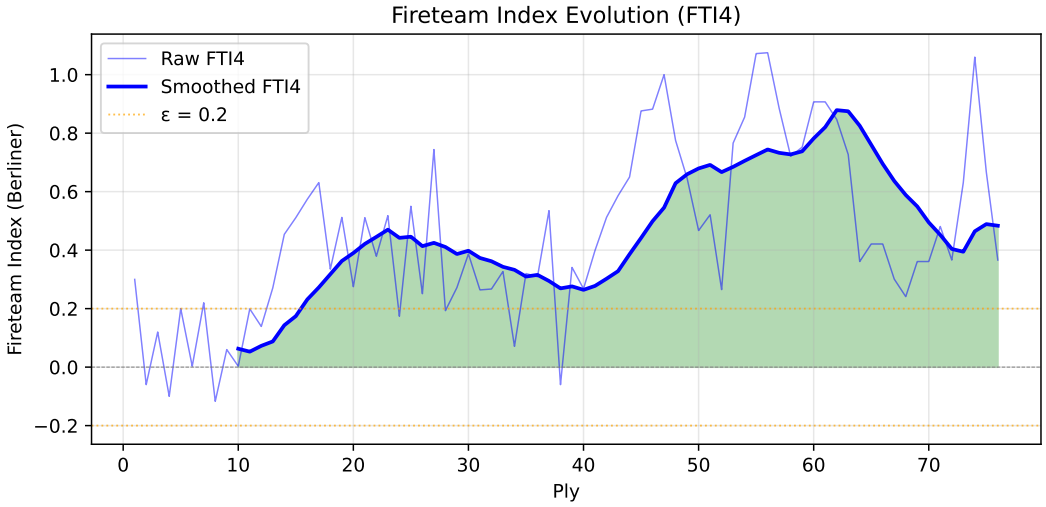

**FTI3: Strategic** $(0.7, 0.1, 0.1, 0.1)$

| Algorithm | Prediction | Max Streak / Confidence |
|---|---|---|
| Per-ply | DRAW | 1 plies |
| Windowed | DRAW | 0 plies |
| Balance | DRAW | $B_n$=-0.05, C=0.05 |



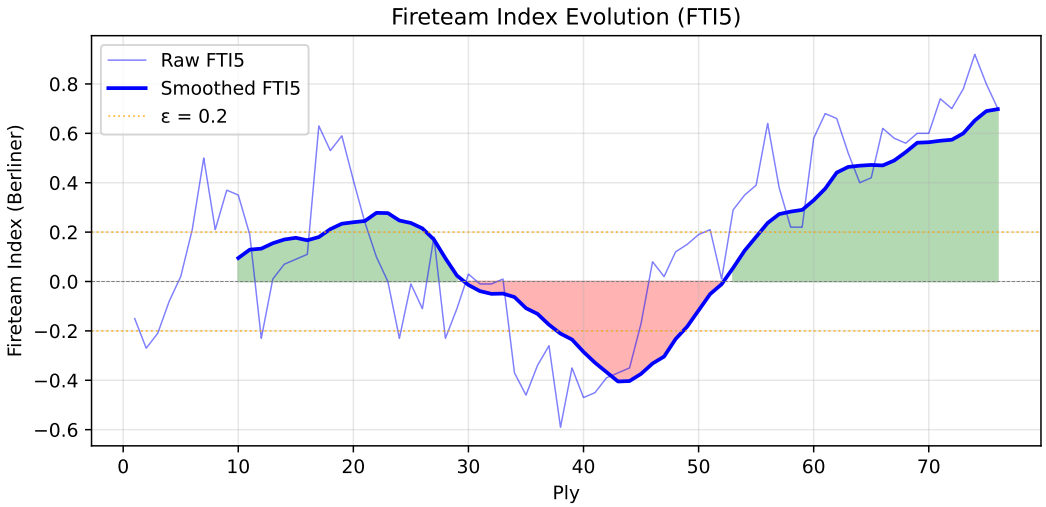

**FTI4: Dynamic** $(0.2, 0.6, 0, 0.2)$

| Algorithm | Prediction | Max Streak / Confidence |
|---|---|---|
| Per-ply | WIN | 29 plies |
| Windowed | WIN | 60 plies |
| Balance | WIN | $B_n$=+0.93, C=0.93 |




**FTI5: Solid** $(0.3, 0.2, 0.5, 0)$

| Algorithm | Prediction | Max Streak / Confidence |
|---|---|---|
| Per-ply | WIN | 17 plies |
| Windowed | WIN | 20 plies |
| Balance | WIN | $B_n$=+0.28, C=0.24 |




Actual result: 1/2-1/2 (Berliner's outcome: DRAW)
FTI1: Per-ply × Windowed × Balance ×
FTI2: Per-ply ✓ Windowed ✓ Balance ✓
FTI3: Per-ply ✓ Windowed ✓ Balance ✓
FTI4: Per-ply × Windowed × Balance ×
FTI5: Per-ply × Windowed × Balance ×

## 16.7 Analysis Information

- Engine: Stockfish 16
- Depth: 20
- Analysis time: 241.0 seconds

# Chapter 17

# Fireteam Index Prediction Summary

This chapter summarizes the accuracy of the Fireteam Index prediction algorithms across all games in this collection.

## 17.1 Overall Results

| Metric | Value |
|---|---|
| Total games | 16 |
| Decisive games | 0 |
| Draws (both sides analyzed) | 0 |
| Games with prediction | 16 |

### 17.1.1 FTI1: Harmonious $(\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4})$

| Algorithm | Correct | Accuracy |
|---|---|---|
| Per-ply (raw) | 7/16 | 43.8% |
| Windowed (smoothed) | 8/16 | 50.0% |
| Balance | 7/16 | 43.8% |

### 17.1.2 FTI2: Tactical $(0.6, 0.1, 0, 0.3)$

| Algorithm | Correct | Accuracy |
|---|---|---|
| Per-ply (raw) | 4/16 | 25.0% |
| Windowed (smoothed) | 5/16 | 31.2% |
| Balance | 7/16 | 43.8% |

### 17.1.3 FTI3: Strategic $(0.7, 0.1, 0.1, 0.1)$

| Algorithm | Correct | Accuracy |
|---|---|---|
| Per-ply (raw) | 7/16 | 43.8% |
| Windowed (smoothed) | 8/16 | 50.0% |
| Balance | 7/16 | 43.8% |

### 17.1.4 FTI4: Dynamic $(0.2, 0.6, 0, 0.2)$

| Algorithm | Correct | Accuracy |
|---|---|---|
| Per-ply (raw) | 8/16 | 50.0% |
| Windowed (smoothed) | 11/16 | 68.8% |
| Balance | 10/16 | 62.5% |

### 17.1.5 FTI5: Solid $(0.3, 0.2, 0.5, 0)$

| Algorithm | Correct | Accuracy |
|---|---|---|
| Per-ply (raw) | 7/16 | 43.8% |
| Windowed (smoothed) | 10/16 | 62.5% |
| Balance | 6/16 | 37.5% |

### 17.1.6 Comparison

**FTI4 (Dynamic)** shows the best predictive accuracy at 68.8%.

### 17.1.7 Held-out Tests of the Fireteam Index

Earlier versions of CGA printed "empirically optimal" FTI weights fitted to all of the games at once. Those weights and their $R^2$ values were not meaningful: the fit had no constant term and its weights were clipped and rescaled afterwards. This section instead tests the FTI the honest way: every prediction below is for a game the model was *not* fitted on (leave-one-game-out), and it is compared with simple baselines.

**1. Predicting results.** For each game, FTI weights (with a constant term) were fitted to the other games and used to predict the held-out game's result (win, draw or loss).

| Method | Correct | Accuracy |
|---|---|---|
| FTI with weights fitted to the other games | 13/16 | 81% |
| Always predict the most common result | 12/16 | 75% |
| Streak rule on Stockfish's evaluation | 15/16 | 94% |

The fitted FTI does no better than a simple baseline, so on this collection it adds no predictive value.

**2. Predicting the type of move a player chooses.** In positions where Stockfish rates two or more moves within 50 centipawns of its best, which one does the player pick? If Space, Mobility, King Safety and Threats captured a personal style, adding them to Stockfish's evaluation would predict the player's choices better, and a model fitted to the player's own moves would beat one fitted to the other players' moves.

| Player | Choices | Random | Eval only | Eval + metrics | Others' model |
|---|---|---|---|---|---|
| Berliner | 235 | 40% | 65% | 61% | 60% |

Each percentage is how often the model's most likely move was the move actually played, on games the model was not fitted on.

Adding the four metrics does not improve the predictions for any player, so on this collection they do not reveal a playing style. No player's own model predicts their choices better than the model fitted to the other players' moves.

## 17.2 Interpretation

These predictions tracked **Berliner** across all games where they appeared. Three prediction algorithms are compared:

- **Per-ply**: Predicts WIN if FTI $> \epsilon$ for 10 consecutive plies (raw values)
- **Windowed**: Same logic but applied to a 10-ply rolling average
- **Balance**: Uses normalized balance $B_n$ (plies ahead − plies behind) and volatility $V$ (sign changes), combined as confidence $C = |B_n| \cdot \max(0, 1 - V/V_{\text{thr}})$ with $V_{\text{thr}} = 0.2\times(\text{plies})$. Predicts DRAW if $C < 0.15$

The Balance algorithm explicitly models draws by detecting games where neither side maintains a consistent advantage, or where the advantage changes hands frequently.

The different FTI signatures may perform better or worse depending on the playing style of the players being analyzed. FTI2 (Tactical) tends to predict well for players who emphasize territorial control and tactical threats, while FTI3 (Strategic) may better capture the style of players who prioritize space and piece activity.

# Appendix A

# Methodology: How Metrics Are Computed

This appendix explains how the positional metrics are computed.

This section explains how Stockfish's classical evaluation computes each positional factor. Note that modern Stockfish primarily uses NNUE (neural network) evaluation, but the classical evaluation terms provide interpretable metrics for educational purposes.

## A.1 Space

Space evaluation counts squares in the center (files c-f, ranks 2-4 for White, 5-7 for Black) that are:

1. Attacked by at least one friendly pawn, AND
2. Not occupied by any enemy piece

The computation is weighted by the number of pieces behind the pawn chain—this rewards positions where you control territory AND have pieces to exploit that space. The space term is scaled by total piece count; it matters more in closed positions with many pieces.

## A.2 Mobility

Mobility counts legal moves for each piece type, with different weights:

- **Knights**: Safe squares attacked (excluding squares defended by enemy pawns)
- **Bishops**: Safe squares on diagonals (bonus for long, unobstructed diagonals)
- **Rooks**: Squares on files/ranks (bonus for open files, 7th rank penetration)
- **Queens**: Combination of bishop and rook mobility patterns

"Safe squares" exclude those defended by enemy pawns or where exchanges would lose material. Mobility in the opponent's territory is valued higher than mobility in your own.

## A.3 King Safety

King safety combines several factors:

- **Pawn shield**: Pawns on 2nd/3rd rank directly in front of the castled king
- **King tropism**: Distance of enemy pieces to your king (closer = more dangerous)
- **Attack units**: Count of enemy pieces attacking squares adjacent to your king
- **Safe checks**: Number of safe checking squares available to the opponent

## A.4 Threats

Threat evaluation considers:

- Hanging pieces (attacked but not adequately defended)
- Safe pawn advances that create threats
- Minor pieces attacking major pieces
- Weak squares in the pawn structure

## A.5 Middlegame vs Endgame (MG/EG)

Each term has separate middlegame (MG) and endgame (EG) values. Stockfish blends these based on remaining material (the "phase"). With all pieces on the board, MG dominates; as pieces trade, EG values become more important. For example, space matters greatly in the middlegame but is nearly irrelevant in endgames.

## A.6 Game Character Classification

Games are classified based on evaluation volatility and directionality. Let $m_2$ be the maximum evaluation for White during the game (in pawns) and $m_1$ be the minimum. The *spread* $d = m_2 - m_1$ measures the total evaluation range.

**Spread Classifications.** Based on the magnitude of $d$:

- **Balanced** ($d < 1$): Minimal advantage shifts; controlled, technical play with neither side gaining a significant edge.
- **Tense** ($1 \leq d < 3$): Normal competitive tension; typical of well-played games where small advantages are contested.
- **Tactical** ($3 \leq d < 6$): Significant evaluation swings; tactical complications, missed opportunities, or speculative play.
- **Chaotic** ($d \geq 6$): Wild swings in evaluation; likely includes major blunders, speculative sacrifices, or complex tactical melees.

**Directionality Classifications.** Based on whether the advantage changed hands:

- **One-sided**: Either $m_1 > -\frac{1}{2}$ (White always comfortable) or $m_2 < \frac{1}{2}$ (Black always comfortable). The leading side maintained control throughout.
- **Seesaw**: Both $m_1 < -1$ and $m_2 > 1$. The advantage genuinely changed hands, with each side holding a significant advantage at some point.
- **Normal**: Neither one-sided nor seesaw; moderate swings without complete reversals.

These classifications combine to describe the overall game character. For example, a "tactical seesaw battle" indicates significant evaluation swings ($3 \leq d < 6$) with the advantage changing hands multiple times.

## A.7 Fireteam Index (FTI) and Win Prediction

The Fireteam Index is a composite metric inspired by the ancient mathematical board game *Rithmomachia*, where a "progression victory" is achieved when a coordinated group of pieces (a "fireteam") establishes control deep in enemy territory. Similarly, in chess, sustained positional dominance across multiple factors often precedes victory.

The FTI weights can be thought of as *signatures* that capture different playing styles. By adjusting which positional factors receive emphasis, the FTI can be tuned to better predict outcomes for players with different strategic preferences.

**Threats Scaling.** The raw Threats metric ($T$) produces values typically ranging from 5–20, while Space, Mobility, and King Safety range from approximately 0.1–0.5. To make these comparable, Threats is divided by a scaling factor of 20 before weighting. The scaled value $T' = T/20$ brings Threats into a similar range as the other metrics.

**FTI1: Harmonious Formula.** The harmonious Fireteam Index weights all four positional metrics equally (with Threats pre-scaled):

$$\text{FTI}_1 = \frac{1}{4}\Delta S + \frac{1}{4}\Delta M + \frac{1}{4}\Delta K + \frac{1}{4}\Delta T' \tag{A.1}$$

where $\Delta S$ = Space, $\Delta M$ = Mobility, $\Delta K$ = King Safety, and $\Delta T' = T/20$ = scaled Threats, each computed as (Player's value − Opponent's value). Positive FTI indicates the player has overall positional superiority; negative indicates the opponent is better.

**FTI2: Tactical.** This variant emphasizes territorial control and tactical pressure through Space and Threats:

$$\text{FTI}_2 = 0.6\Delta S + 0.1\Delta M + 0.3\Delta T' \tag{A.2}$$

**FTI3: Strategic.** This variant emphasizes long-term spatial dominance:

$$\text{FTI}_3 = 0.7\Delta S + 0.1\Delta M + 0.1\Delta K + 0.1\Delta T' \tag{A.3}$$

**FTI4: Dynamic.** This variant emphasizes piece activity and maneuvering:

$$\text{FTI}_4 = 0.2\Delta S + 0.6\Delta M + 0.2\Delta T' \tag{A.4}$$

**FTI5: Solid.** This variant emphasizes king safety and defensive play:

$$\text{FTI}_5 = 0.3\Delta S + 0.2\Delta M + 0.5\Delta K \tag{A.5}$$

**Prediction Algorithms.** Three algorithms are used to predict game outcomes based on FTI (applied independently to each of the five variants):

1. **Per-Ply (Raw)**: Examines the raw FTI value at each ply after the opening phase (ply 16, approximately 8 moves). If FTI $> \epsilon$ (default $\epsilon = 0.25$) for 10 consecutive plies (5 full moves), a WIN is predicted for that player.

2. **Windowed (Smoothed)**: Computes a 10-ply rolling average of FTI before applying the same streak detection. This smooths out tactical noise and captures sustained trends rather than momentary fluctuations.

3. **Balance (Confidence-based)**: Predicts outcomes using a confidence score that combines balance and volatility:

   - *Normalized Balance* $B_n = \frac{(\text{plies with FTI}>\epsilon) - (\text{plies with FTI}<-\epsilon)}{\text{total plies}}$
   - *Volatility* $V =$ number of times FTI crosses zero (sign changes)
   - *Volatility threshold* $V_{\max} = 0.2 \times$ total plies (scales with game length)
   - *Confidence* $C = |B_n| \times \max(0, 1 - V/V_{\max})$

   The prediction is: WIN if $C > 0.15$ and $B_n > 0$; LOSS if $C > 0.15$ and $B_n < 0$; DRAW otherwise. Low confidence (due to balanced play or high volatility) predicts a draw.

The Per-Ply and Windowed algorithms predict DRAW only when no streak is achieved, whereas the Balance algorithm explicitly detects drawish characteristics (balanced plies, back-and-forth play).

**Interpretation.** The FTI prediction measures *positional* dominance, which is a necessary but not sufficient condition for winning. A player may dominate positionally yet fail to convert due to:

- Tactical errors (blunders, missed combinations)
- Time pressure in rapid/blitz games
- Tenacious defense by the opponent
- Drawing tendencies in certain structures (opposite-colored bishops, fortresses)

Conversely, a player may win without ever achieving sustained FTI dominance through a single tactical strike or opponent error. The FTI metric is most predictive in games where positional understanding—rather than pure calculation—determines the outcome.

**Choosing a Signature.** Different FTI signatures may perform better for different players or playing styles. FTI2 (Tactical) tends to predict well for players who build up territorial control and convert through direct threats. FTI3 (Strategic) may better capture the style of players who emphasize piece activity and spatial domination. Comparing prediction accuracy across signatures can reveal which positional factors most strongly correlate with a player's wins.

**For Draws.** When analyzing drawn games, the FTI is computed from both players' perspectives. This reveals whether:

- Neither side achieved dominance (a "true draw" with balanced play)
- One side dominated but failed to convert (a "missed win")
- Both sides dominated at different points (a "mutual chances draw")